



Molecular dynamics simulations in biomaterials science: Atomistic modeling across bioactive glasses and ceramics, polymers, metals and composites

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ABSTRACT

Over the last decade, molecular dynamics (MD) simulations have become an important tool in biomaterials research by linking atomic-scale interactions to macroscopic biological and mechanical performance. This review examines the application of MD across major biomaterial classes, including bioactive glasses and ceramics, polymers and hydrogels, metallic biomaterials, composites, and biological interfaces. For inorganic biomaterials, MD has clarified structure–property relationships, ion transport, hydration, and early dissolution processes. For polymers and hydrogels, it has provided insight into chain conformation, swelling, degradation, crosslinking, and responses to physiological environments. In metallic systems and composites, MD has helped explain interfacial adhesion, load transfer, deformation mechanisms, and corrosion-related behavior, while at biointerfaces it has revealed key aspects of protein adsorption and membrane interactions. This review summarizes the recent advancements in the methods for studying the properties of biomaterials through computer simulation and, specifically, highlights the latest developments in the field of reactive molecular dynamics (rMD). This class of methodologies has developed various schemes beyond the traditional view of materials as being represented by a fixed topology (the arrangement of the atoms), including the use of multiple surfaces within the same framework (multisurface reactions) and different schemes to accelerate the polymer crosslinking. Examples of these strategies encompass the use of template-based reactions and coarse-grained or multiscale approaches to create models, along with using machine-learning techniques to create potentials. All of these new methodologies increase the range of chemical, time, and length scales accessible to simulations and increase the ability to predict how a biomaterial will dissolve, mineralize, crosslink, degrade and respond dynamically at the interface with another one. Overall, these developments show that MD is evolving from a mainly interpretive technique into a predictive framework for the rational design and optimization of next-generation biomaterials.

1. Introduction

In the early 21st century, biomaterials science underwent a transformation from a subcategory of biomedical engineering to a major pillar of modern medicine and bioengineering. Today, there is hardly a field of advanced medicine that does not directly or indirectly depend on a smart biomaterial: from self-expanding vascular stents and artificial

heart valves to 3D-printed scaffolds for bone/cartilage regeneration, from cancer-targeted drug nanocarriers to drug-eluting contact lenses, from bioactive dental implants to self-healing artificial skin [1–9]. All of these advances became possible only when humans were able to create materials that were not only tolerable by the body, but also actively interacted with living tissue, responded to it, and even disappeared at the right time once having performed their therapeutic or diagnostic

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function [10,11].

However, designing such materials using purely experimental methods was slow, expensive, and largely based on a trial-and-error approach. This is where molecular dynamics (MD) simulations came into play, allowing researchers for the first time to see what was happening within a few angstroms of a biomaterial, in a fraction of a nanosecond, in contact with water, ions, proteins, or cell membranes [12–14]. This method not only explained the interactions but also predicted, optimized, and in many cases suggested entirely new routes for synthesis and design [15]. Bioactive glasses (BGs), born more than half a century ago with the discovery of Prof. Larry Hench [14], remain one of the most successful and widely used classes of biomaterials and, at the same time, one of the best testbeds for the power of MD simulations. These amorphous biomaterials have an amazing ability to form direct chemical bonds with living bone tissue [16–19]. The mechanism of this bonding, described by Prof. Larry Hench [16], involves, at the atomic level, the rapid migration of sodium and calcium ions, hydrolysis of silicate bonds, formation of a silica layer rich in silanol groups, adsorption of calcium and phosphate, and finally the crystallization of hydroxycarbonate apatite. As conventional all-atom MD is typically limited to picosecond-to-microsecond timescales, it can resolve the early atomistic events that govern processes observed experimentally over much longer periods. In bioactive glasses, for example, MD can capture initial hydration, ion migration, bond rearrangement, and early surface reactions, thereby providing mechanistic insight into phenomena whose full experimental evolution may occur over hours or days. Access to longer effective timescales generally requires reactive force fields, enhanced-sampling strategies, coarse-grained models, or coupling with higher-scale approaches. MD simulations have shown that even before contact with physiological body fluids, the lattice-modifying ions are not uniformly distributed within the glass and tend to migrate towards the surface [20]. This pre-migration phenomenon, which is difficult to demonstrate experimentally, is now recognized as one of the main reasons for the high initial reaction rate of BGs. It has also been shown that the ratio of bridging to non-bridging oxygens (BO/NBO) in the surface layers differs by up to 10–15 Å from that of the bulk glass, and this structural heterogeneity is key to understanding the differences in behavior of glasses with the same composition but different thermal cycles [21,22]. The distribution of Q^n structural species in the silicate lattice is another major achievement of MD simulations. In glass chemistry, the Q^n notation describes the connectivity of tetrahedral network-forming units (e.g., SiO_4 or PO_4), where n denotes the number of bridging oxygens (BOs) connected to a given tetrahedron. Thus, Q^0 corresponds to an isolated tetrahedron with no BOs, whereas Q^4 denotes a fully polymerized tetrahedron with four BOs; intermediate Q^1 – Q^3 species represent progressively higher network connectivity. In the classical 45S5 composition, the majority of the silicon tetrahedra are Q^2 , about 30–35% are Q^3 , and the rest are Q^1 and Q^4 [23]. Reducing the average lattice connectivity below 2.5 is the threshold for rapid bioactivity. The higher the percentage of Q^2 , the higher the solubility, but the lower the mechanical strength; therefore, optimal design requires a delicate balance that MD simulations can predict in the lattice connectivity [24,25].

In the field of bioactive ceramics, hydroxyapatite (HA) remains the most important biomaterial to be used in tissue engineering applications. The HA surface is highly hydrophilic and forms a structured layer [26]. MD simulations show that HA surface Ca^{2+} and phosphate groups strongly bind water, leading to a hydrated, charged surface. Experimental work then demonstrates that adhesion proteins bind efficiently to HA, consistent with a model in which the hydrated surface and double layer facilitate protein adsorption [27,28]. The final orientation of the arginylglycylaspartic acid domain in these proteins, which is critical for binding cellular integrins, is highly dependent on the precise surface chemistry (Ca/P ratio, presence of OH^- or ionic substitutions) and can now be predicted with sub-nanometer precision [29].

As for biodegradable polymers, poly(lactic acid) (PLA),

polycaprolactone (PCL), and poly(lactic-co-glycolic acid) PLGA copolymers all degrade via ester hydrolysis, but the rate and specific mechanism depend strongly on the spatial arrangement of the chains, crystallinity, and the presence of water in the amorphous phase [30]. Simulations have shown that the diffusion of water molecules into the amorphous phase of PLA is initially very slow, but after the first ester bonds are broken, terminal carboxyl and hydroxyl groups are formed that act as acid and base catalysts, driving the reaction in an autocatalytic manner [31,32]. This is precisely the phenomenon that causes the sudden degradation of PLA implants after a few months, and it can now be fully modeled and controlled [33,34]. Chitosan, as a naturally occurring cationic polymer, has a pH-dependent behavior that makes it ideal for drug delivery, gene therapy, and tissue engineering applications [35]. At low pH, the chains are fully protonated and stretched; at pH above 5.6, the amino groups are deprotonated, and the chains are coiled [36]. This deformation changes the diffusion coefficient of anionic drugs by several orders of magnitude. MD simulations have shown that even a 5% difference in the degree of deacetylation changes the free energy of interaction with DNA by up to 40 kJ/mol. Hydrogels, as materials that are constituted for more than 90% water but retain their shape, represent a huge computational challenge [37,38]. In poly(ethylene glycol) (PEG) hydrogels, the negative elastic energy (i.e., the tendency to swell) arises entirely from favorable water–PEG interactions. Simulations have shown that each ethylene glycol unit holds an average of 3–4 water molecules tightly in the first hydration layer, which prevents protein adsorption and immune activation [39–41]. Alginate, a naturally occurring polysaccharide composed of β -D-mannuronate (M) and α -L-guluronate (G) residues arranged in block sequences, is a classic example of ionic gelation. Long-term simulations have shown that each calcium ion coordinates precisely with eight oxygens of the four guluronic acid blocks, and this coordination increases the gel strength by up to 100 times compared to hydrogen bonds alone [42,43]. The ratio of G to M blocks is the main determinant of shear modulus: the higher the G, the stiffer and more brittle the gel; the higher the M, the softer and more elastic the gel [43–45]. In a new generation of self-healing double-network hydrogels, MD simulations have shown that reversible hydrogen bonds and coordinated ionic bonds increase the failure energy, while permanent covalent bonds ensure ultimate strength [46,47].

Organic–mineral composites are materials where MD simulations have truly gained a lot of attention and excelled in studying and experimenting with multiple formulations. In PLA/nanoHA composites, the addition of HA significantly increases the stiffness and modulus of the material due to the rigid filler effect and improved interfacial interactions between PLA chains and HA particles [48]. Some studies report that the presence of nanoHA can slightly modify the hydrolytic degradation of PLA, likely by influencing water diffusion and buffering acidic degradation products, although the exact molecular mechanism at the interface remains under investigation [49,50].

At the level of interaction with the cell membrane, simulations have shown that nanomaterials with a sharp radius of curvature (such as carbon nanotubes or gold nanoparticles) easily perforate the membrane, while smooth surfaces do not [51]. Many simulation and biophysical studies suggest that amphipathic peptides can insert into lipid bilayers at a tilt or oblique angle, and under suitable conditions nucleate barrel-stave or toroidal pores. Functional assays (leakage, conductance, dye release) provide indirect evidence for such pore formation. However, high-resolution structural confirmation (e.g., via cryo-electron microscopy) of stable peptide-lined pores remains elusive; no consensus exists on a universal ‘entry angle’ that guarantees pore formation [52,53]. Despite these advances, important limitations remain. Conventional non-reactive classical MD simulations rely on fixed-topology force fields and therefore cannot directly represent covalent bond breaking or formation. Chemically reactive phenomena relevant to biomaterials, such as hydrolysis, mineralization, redox processes, and bond exchange, instead require complementary approaches,

including reactive classical MD methods, ab initio MD, and QM/MM schemes, each with their own trade-offs in cost, transferability, and accessible system size [54,55]. The accessible timescales are still in the microsecond range, while many biological phenomena occur in seconds to years. Large systems (more than a million atoms) are still very time-consuming, and existing force fields for transition metals, complex oxides, and charged polymers do not possess the highest accuracy. Because the primary focus of this review is the role of MD as a methodology rather than an exhaustive treatment of any single biomaterial family, we intentionally compare representative biomaterial classes, including bioactive glasses and ceramics, polymers and hydrogels, composites, metals, and biointerfaces that display distinct bonding, hydration, degradation, and interfacial behaviors. This cross-class organization allows common MD strengths, limitations, and transferable design principles to be identified across different material systems. Accordingly, the following sections are organized by biomaterial class while maintaining emphasis on the molecular descriptors and simulation strategies that link atomistic behavior to functional performance.

In the field of reactive molecular dynamics (MD) new options have become available; all include useful complementary capabilities in addition to ReaxFF and first principles (Ab Initio) calculations (MD) for simulating and predicting reaction processes for materials. An example of an additional option is Multisurface Adiabatic Reactive Molecular Dynamics (MS-ARMD), which allows for the treatment of reaction processes through the coupling of reactive surfaces and when the reaction process involves known bond-rearrangement pathways. MS-ARMD is suitable when the relative positions of atoms and molecules involved in the reaction can be determined by experimental and theoretical means [56]. Another example of a new option in reactive MD for materials is the application of a Bond-Boost methodology combined with an Accelerated ReaxFF methodology: it is useful to address the issue of rare-event bottlenecks by facilitating MD simulations of polymer crosslinking and

other reactions involving low-temperature chemical processes [57,58]. Furthermore, template-based protocols, such as REACTER, allow for the re-creation of bonding topology during a MD (reactive) simulation based on the bond topology. This new methodology allows the creation of new bonding topologies to facilitate polymerization and network modification of polymeric materials as well as the dynamic behavior of polymeric materials in larger-scale heterogeneous systems. These newly emerging capabilities to conduct simulations of chemical topology changes allow obtaining correlations and even predictions for physical transport and mechanical properties, degradation rates, and overall biocompatibility of biomaterials, thereby playing an increasing role in the targeted design of new biomaterials [59].

From a general perspective, MD methodologies span a hierarchy of resolution from electronic-structure-based and reactive approaches to classical all-atom and coarse-grained (CG) models. No single simulation level is sufficient to address all questions relevant to biomaterials. While all-atom simulations are essential for resolving local coordination environments, hydrogen bonding, ion-specific interactions, and early interfacial events, many important biomaterials phenomena, such as fibrillar assembly, hydrogel network formation, mesoscale mechanics, collective protein adsorption, membrane remodeling, and long-term degradation, extend beyond atomistic length and time scales. In such cases, coarse-grained models provide an essential bridge by grouping atoms into effective interaction sites, thereby reducing the number of degrees of freedom and enabling access to larger systems and longer effective timescales. Rather than replacing atomistic MD, CG approaches complement it within a multiscale framework that links molecular descriptors to experimentally observable structure, transport, mechanics, and biofunction. These multiscale MD simulation methodologies are schematically summarized in Fig. 1. This review article is a comprehensive attempt to review the achievements of MD simulations in the field of biomaterials over the past two decades, analyze the remaining

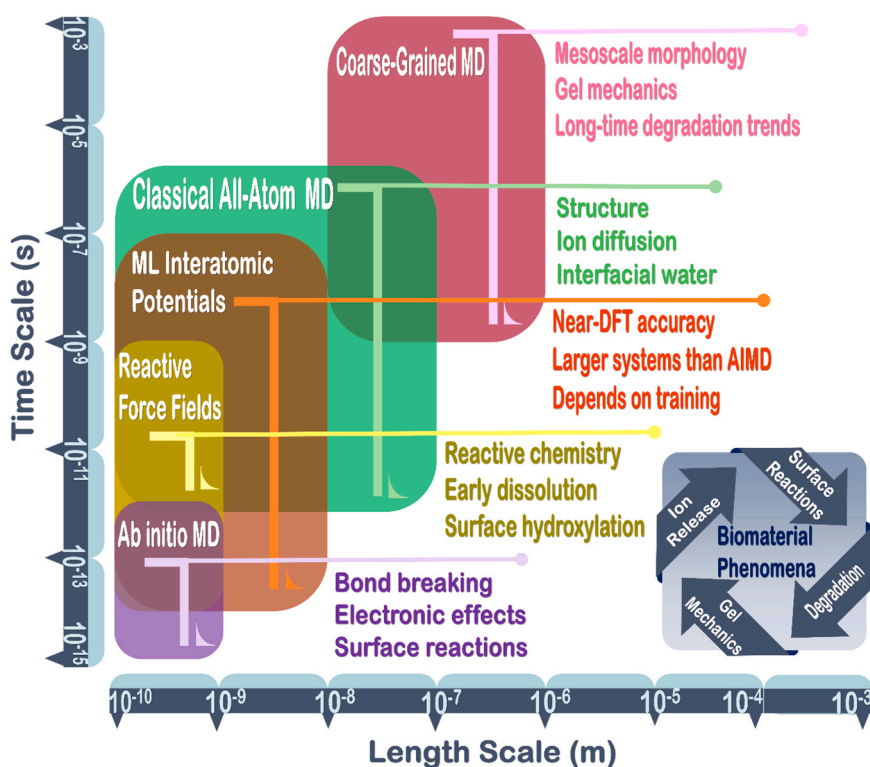


Fig. 1. Multiscale MD approaches for investigating biomaterials through different time and length scales. Classical all-atom MD simulations provide insights into structure, ion diffusion, and hydration effects, although being incapable of introducing chemical reactions, while reactive force fields and ab initio MD can simulate bond breaking, hydrolysis, and surface reactions at shorter times and smaller system sizes. ML interatomic potentials extend the accuracy of larger systems, while coarse-grained MD enables the simulation of morphology, mechanics, and long-term degradation processes. These approaches can address multiple aspects of biomaterials research and biological interactions in various study types.

challenges, and draw a clear roadmap for future studies.

2. Applications of MD across biomaterial classes

2.1. Bioactive glasses

In MD study of BGs, structural metrics such as radial distribution functions (RDFs), Q^n -species distributions, and coordination environments provide atomistic insight into network topology, interactions, and connectivity [12]. These metrics link atomic-scale arrangements to the macroscale physical and chemical properties of BGs. RDF measures the probability of finding an atom at a given distance from a reference point atom, providing a view of short-range structure in glasses. In BGs, RDF reveals the geometry of the atoms and allows the identification of BOs and NBOs. Q^n species extends the picture by providing network connectivity metrics using atomic positions. In a tetrahedral unit of a silicate-based glass, each Si atom is connected to 4 oxygen atoms [13]. If the said oxygens are connected to 2 Si atoms, making a Si-O-Si bond, the oxygen is considered a BO. Q^n species complement RDF analysis by describing network connectivity. In this notation, Q refers to a tetrahedral network-forming unit and n indicates the number of bridging oxygens (BOs) connected to that unit. Accordingly, Q^0 denotes an isolated tetrahedron, whereas Q^4 represents a fully polymerized tetrahedron; Q^1 – Q^3 correspond to intermediate degrees of connectivity. Because higher Q^n species reflect greater network polymerization, shifts in Q^n distributions help explain changes in dissolution behavior, ion release, and bioactivity. Studying Q^n species opens doors to understanding network connectivity and bioactivity of the BGs. For instance, increasing the modifying atom content in the glass reduces high-Q species and increases the Q^1 and Q^2 fractions, leading to a less integrated structure and easier dissolution of the glass [14]. Coordination environment analysis complements RDF by detailing how atoms interact in their coordination shell. Atoms like Si and P maintain a tetrahedral unit in the structure, thus having a coordination number of 4; however, dopants can have varying coordination numbers (5–9) [13]. These parameters together provide a comprehensive and unified framework for interpreting MD simulations across all BGs. This approach allows for rationally designing glass compositions with tailored structural features and predictable biological responses.

Using MD simulations helps studying multicomponent silicate-, phosphate- and borate-based BGs, with the ability to specifically observe the structural changes, bonding properties, atomic scale interactions, short- and medium-range structures [60–62]. Based on the studies by Du et al. [63–65], silicate BGs can be analyzed through various properties such as RDF, Q^n distributions, coordination number, and bond angles. This approach provides information about how the structure interacts when multiple dopants like Sr, Mg, Cu, Al, etc., are incorporated. In the glass structures, atomic presences are generally divided into 2 categories: network formers and network modifiers [66]. For instance, in a silicate-based BG, the Si atoms provide the majority of bonding and make Si-O-Si bonds, which possess high structural integrity and high network connectivity, making the glass able to interact only moderately with simulated body fluid (SBF). To overcome this phenomenon, modifier atoms break Si-O-Si bonds and make Si-O-M bonds, which possess lower bond strength, allowing the structure to have lower connectivity and, thus, better tendency to dissolution, while providing many added values through the local release of therapeutic ions [67,68].

Reactivity in aqueous media is faster in borate-based BGs compared to silicate ones. In this regard, Stevenson et al. [69] investigated multiple compositions of borate-based glasses ranging from binary $\text{Na}_2\text{O}-\text{B}_2\text{O}_3$ to complex $\text{Na}_2\text{O}-\text{CaO}-\text{B}_2\text{O}_3-\text{SiO}_2-\text{P}_2\text{O}_5$ systems. The model provided an accurate reproduction of BO_3/BO_4 structures similar to experimental evaluations of borosilicate glasses, as well as network connectivity and medium-range properties across diverse compositions. The results indicated that the boron existed in triangular and tetrahedral coordination, and the bonding (e.g., NBO, BO content) extremely

depended on the modifier atom types and concentrations. Meanwhile, this simulation provided valuable information on structural descriptors that correlate with the macroscopic glass properties, such as density, glass transition temperature, and ion release kinetics, thus providing insights into designing borosilicate BGs for biomedical applications. The study reported BO_4 fractions of 50–60%, based on $\text{Na}_2\text{O}:\text{B}_2\text{O}_3$ ratio, resulting in the majority of boron atoms being tetrahedrally connected under normal glass conditions. Additionally, the radial distribution function of borate glasses shows a peak at 1.40 Å associated with BO_3 and another at ≈ 1.50 Å associated with BO_4 tetrahedra; meanwhile, bond angles are 120° and 109.5° , respectively.

Phosphate glasses, exhibiting a quite wide range of dissolution kinetics in aqueous media from hours to months [70], are often described in terms of Q^n speciation (Q^2 , Q^3 , etc.), and many glasses exhibit a predominance of chain (Q^2) and branched (Q^3) phosphate units. The typical P–O bond lengths in such glasses lie around 1.5–1.6 Å, and P–P second-neighbor distances (across P–O–P bridges) are compatible with values near 3.0 Å. These structural features have been inferred from a combination of experimental (e.g., NMR, diffraction) and computational modeling studies [71,72]. Pedone et al. [73] further provided phosphate force fields and proved that the addition of alkali modifiers depolymerized the network. Meanwhile, in CaO- and MgO-containing phosphate glasses, divalent modifiers produced stronger and more localized cation-oxygen interactions, making the P–O–P angle wider. These findings provide the necessary data in the evaluation of glasses and help to better understand the effects of atomic-scale interaction and compositions on the properties of BGs, including biological effects on cells. In this regard, an interesting study on Mg-doped glasses was reported by Milborne et al. [74].

Fig. 2 depicts a single atomistic configuration of a phosphate-based bioactive glass created through Reactive Molecular Dynamics. The results obtained through the atomistic simulation and from experimental evidence support the conclusion that a strong relationship exists between local phosphate connectivity and cation environments, and allow estimating how these relationships affect the functional performance of phosphate-based bioactive glasses.

MD simulations have established a direct relationship between atomic-scale properties and macroscopic responses in physiological

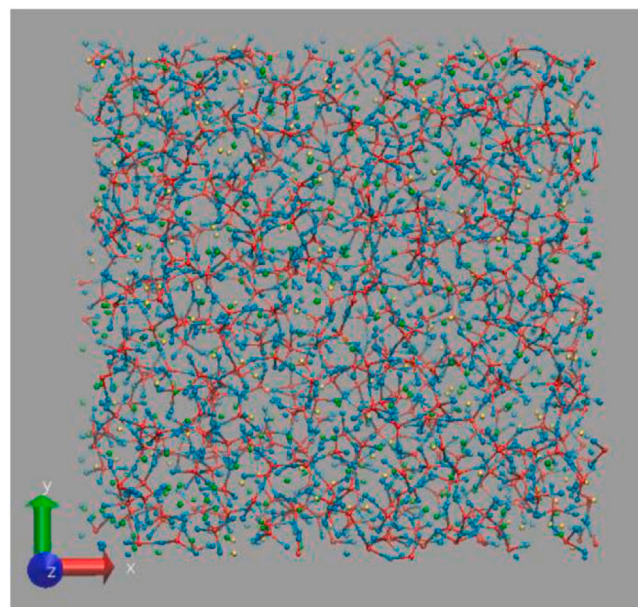


Fig. 2. Representative atomistic snapshot of a ternary phosphate-based bioactive glass (P45C30N25) obtained from reactive molecular dynamics (ReaxFF) simulation. Phosphorus, oxygen, sodium, and calcium atoms are shown in red, blue, yellow, and green, respectively [75].

body fluids [76,77]. Beyond describing local structure, MD simulations enable quantitative links between glass-network descriptors and three functionally important outcomes in bioactive glasses: ion transport, mechanical response, and interfacial reactivity. Network connectivity strongly influences ionic mobility in physiological environments. Classical MD studies show that depolymerized networks, characterized by lower connectivity and modifier-associated oxygen environments, exhibit higher diffusion coefficients and faster ion transport [78,79]. The addition of different modifier atoms leads to an altered local field strength of the network, and the depolymerization differs, leading to distinct ion transport even with similar network connectivity [80]. Importantly, modifier identity also matters by altering local field strength and cation–oxygen interactions, different dopants can produce distinct transport behavior even at comparable average connectivity. However, beyond short-range structure, MD has excelled in revealing the importance of medium-range structure, such as ring size distribution, ion clustering, and network connectivity, in the release patterns, dissolution, and even mechanical properties. Recent studies have reported that the high amounts of network former fraction in a composition lead to a significant rise in elastic moduli, while having partial network repolymerization and lower ion release [81–84]. Moreover, MD reveals that medium-range structural features are closely related to mechanical response. Descriptors such as ring-size distribution, ion clustering, and the degree of network polymerization correlate with elastic properties obtained from atomistic stress-strain simulations. In general, compositions enriched in network-forming species and more polymerized frameworks display higher Young's modulus, whereas increasing modifier content tends to reduce stiffness while often favoring ion release. Thus, MD helps explain the experimentally observed trade-off between mechanical integrity and glass reactivity.

Ab Initio MD are particularly useful to study the very early interfacial process involved in the dissolution of biomaterials, also providing information about subsequent apatite formation. The use of Ab Initio MD to study the water/bioactive glass interface has provided direct evidence for dissociative adsorption of water, hydrolysis of Si-O-Si and silanol formation on the interface, allowing the free energy barrier for the activation of the hydrolysis process to be estimated [85]. Reactive MD simulations based on reactive force fields have provided a more extensive representation of events leading to the time scales and durations associated with silicate glass/water interfaces and have indicated the occurrence of ion exchange, proton transport and water infiltration as well as the time-dependent rates for the formation and de-formation of silanols during the early stages of glass corrosion [86,87]. Although these chemically-explicit descriptors do not represent the entire time span for apatite formation, they can provide information about hydrated and ion exchanged surfaces that are key for subsequent calcium phosphate deposition. The recent development of ReaxFF for phosphate-based glasses provides continuing support for the development of this concept in a greater variety of biomaterial glass chemistries [75].

2.2. Ceramics

Ceramic-based biomaterials stand for a distinctive and strategically important position in the biomaterials landscape, since they span a broad spectrum of biological functions, from bioactive and bioresorbable calcium phosphate (CaP) ceramics that chemically resemble the mineral phase of bone and teeth, to bioinert oxide ceramics that offer exceptional wear resistance, chemical stability, and favorable tribological performance in demanding load-bearing environments [88]. Clinically established CaP formulations, including hydroxyapatite and tricalcium phosphate, are widely utilized as bone graft substitutes, coatings, and scaffolds, where their osteoconductivity and controlled dissolution (e.g., for α - and β -tricalcium phosphate) can promote mineralized tissue integration. On the other hand, oxide ceramics such as zirconia- and alumina-based systems have become prominent in

orthopedic and dental applications due to their mechanical reliability and corrosion resistance, with zirconia implants in particular gaining substantial clinical interest [89]. From an interfacial viewpoint, ceramics are considered chemically expressive materials as their surface terminations, hydroxylation states, defect populations, and ionic environments collectively define the hydration structure and electrostatic landscape that governs biological recognition. In an *in vivo* scenario, upon contact with physiological fluids, the surface of ceramics becomes rapidly hydrated and is subsequently conditioned by adsorption of proteins and other biomolecules, forming the earliest molecular layer that shapes downstream events such as immune signaling, cell adhesion, and ultimately osseointegration [90]. For ionic ceramics in particular, interfacial phenomena are strongly modulated by long-range electrostatic interactions, ion-specific adsorption, and surface charge compensation, which makes it difficult to infer binding mechanisms from macroscopic observations only. Therefore, atomistic modeling by MD has emerged as a critical tool for resolving how ceramic chemistry and topography translate into biological performance. Classical all-atom MD enables nanosecond-to-microsecond exploration of hydration layers, ion distributions and biomolecule orientations, often improved through enhanced sampling methods (e.g., steered MD, umbrella sampling, metadynamics) [91]. Quantifying rare events and binding free energies is crucial as they are key to cell adhesion, detachment, and mechanochemical response. At the same time, the ceramic-water interface can involve reactivity (hydroxylation, proton transfer, ligand exchange) that is not always captured by fixed-charge force fields, motivating the application of complementary strategies. This methodological diversity is particularly relevant for biomaterials-oriented questions, where surface state, pH, and electrolyte composition can be as determinative as bulk composition. Fig. 3 illustrates an example of a hydrated hydroxyapatite (HAp) system represented using MD techniques, specifically displaying the actual HAP-water interface.

Within bioceramics, CaP minerals represent the closest inorganic analogue to native hard tissue and therefore dominate atomistic studies aimed at understanding biomineral interfaces. Seminal MD and steered MD investigations have elucidated how key osteogenic and adhesive proteins engage hydroxyapatite (HAp) surfaces at the residue level. For example, Dong and co-workers combined MD/steered MD with electronic-structure analysis to resolve adsorption–desorption pathways of BMP-2 on HAp(001), identifying functional groups that mediate anchoring and release under load [93]. Shen et al. [27], systematically examined orientation-dependent adsorption/desorption of the fibronectin FN-III10 module on HAp(001), providing a molecular basis for how protein presentation and surface crystallography jointly tune adhesion strength and interfacial stability. Chen et al. [94] conducted complementary work on mineralization-related proteins and revealed motif-specific binding strategies, such as the carboxylate-driven “claw” interaction reported for the leucine-rich amelogenin protein (LRAP) on HAp(001), which highlights how Ca-rich ionic lattices can selectively stabilize particular side-chain geometries. Beyond individual proteins, quantitative mapping of small-molecule and amino-acid affinities has also become feasible. Hoff et al. [95] used a large ensemble of steered MD simulations to determine binding mechanisms and binding free energies of all 20 capped amino acids and citrate across multiple HAp facets, pH values, and electrolyte conditions, moving an important step toward predictive, chemistry-aware models of biomolecule–mineral recognition. Interfacial adsorption studies on HAp about the role of protein secondary structure and mineral binding were also reported by Zhang et al. [96].

Furthermore, other members of the CaP family were investigated; for instance, Zhao et al. [97] extended adsorption–desorption modeling to BMP-7 on β -tricalcium phosphate, demonstrating how surface termination (Ca-rich vs P-rich) can shift preferred interaction sites and alter epitope exposure, thereby suggesting atomistic routes to tuning osteoinductive signaling via ceramic surface design.

Oxide ceramics, on the other hand, including zirconia and alumina,

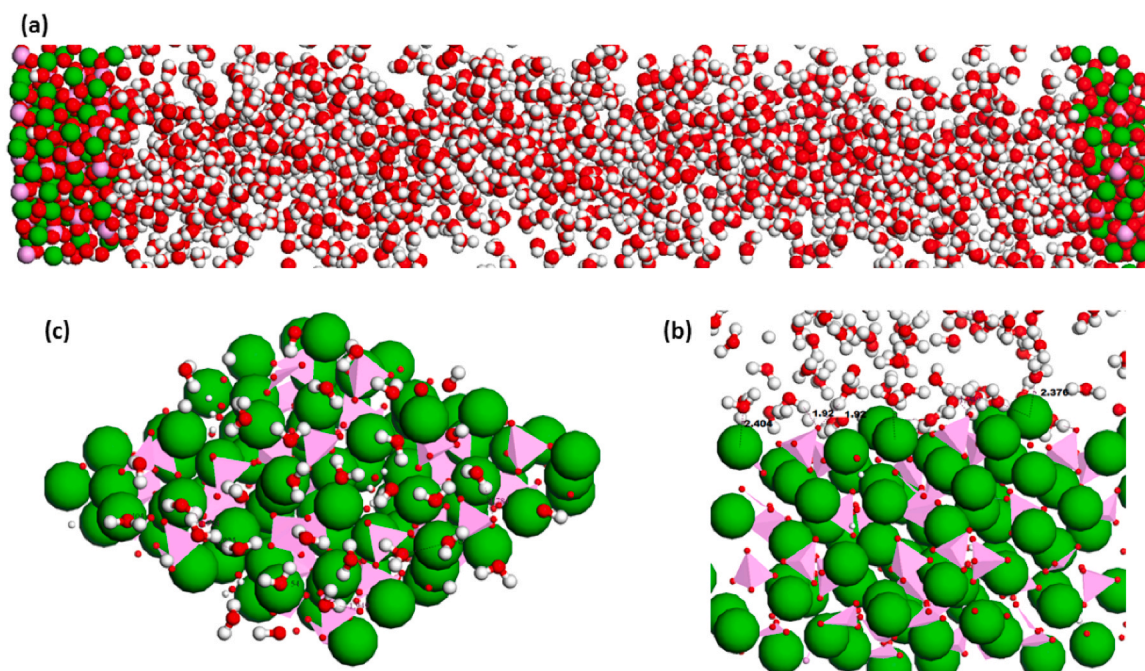


Fig. 3. MD model of a HAP–water system showing (a) the HAP–water nanopore system, (b) a parallel view of the HAP–water interface, and (c) a detailed view of the local surface environment [92].

provide model systems for understanding protein conditioning layers on implant-relevant surfaces and for designing biofunctional coatings. Zirconia is especially important in dentistry and orthopedics, where surface nanoengineering and chemistry strongly influence early biological interactions. Sabirianov and co-workers demonstrated via atomistic modeling that engineered nanostructuring of cubic zirconia can enhance initial protein fragment adsorption compared to flat surfaces, emphasizing that ceramic topography can amplify electrostatic and steric complementarity at the biointerface [98].

Related peptide-oxide studies employing long-timescale MD and umbrella sampling provide quantitative binding free energies for osteogenic sequences such as KRSR on anatase (TiO_2) surfaces, directly connecting peptide functionalization strategies to thermodynamic metrics of adsorption [99].

Beyond classical adsorption studies, *ab initio* MD has become essential for oxide ceramics where interfacial reactivity and proton-coupled processes shape the earliest stages of biointerface formation. In a representative example, Yang et al. [100], resolved how monoclinic ZrO_2 organizes interfacial water into alternating H-rich and O-rich layers and how this restructuring modifies hydronium solvation thermodynamics and accelerates proton hopping, providing mechanistic design cues for tuning oxide surface activity in aqueous (bio) environments.

Alumina is widely used for its hardness and chemical stability and has served as a platform for MD studies of catechol-based adhesion motifs relevant to bioinspired surface chemistries, clarifying how surface hydroxylation mediates adsorption modes and residence times [101]. Yeh et al. [102] utilized atomistically detailed MD simulations to elucidate how bioinspired oligomers adsorb onto a hydroxylated alumina surface, used as a model oxide ceramic. The oligomers were constructed from two repeat units, which were maleimide-bearing catechol motifs and methyl acrylate units, enabling systematic assessment of how sequence and composition modulate interfacial binding. Adsorption was examined under both anhydrous and aqueous environments to isolate the role of interfacial water in mediating binding competition at surface sites. The simulations revealed a strong environmental dependence: in anhydrous conditions, the oligomers exhibited robust adsorption to alumina, whereas in water the adsorption

interactions were substantially weakened due to competitive occupation of near-surface adsorption sites by water molecules. This study demonstrates that ceramic surface chemistry (hydroxylated alumina) and hydration competition jointly dictate the adsorption of biomolecules. It is a central research topic in ceramic biointerfaces where early-stage conditioning in aqueous media controls subsequent biological responses and functionalization outcomes. Ultimately, these developments position MD as an enabling framework for mechanistic and quantitative understanding of ceramic biomaterials: simulations can resolve how crystallographic facet, termination, hydration structure, and ionic milieu control the adsorption pathway and how surface engineering (chemistry and topography) can be translated into predictive design rules for improved biological integration.

2.3. Biopolymers and hydrogels

The design of advanced hydrogels for biomedical applications, ranging from tissue engineering scaffolds to drug delivery vehicles, fundamentally relies on a molecular-level understanding of their constituent polymers. MD simulations emerge as an indispensable computational methodology, offering atomistic resolution insights into the molecular conformation, hydration, self-assembly, and mechanical behavior that are often inaccessible to experimental techniques alone. The MD simulation of biopolymers and hydrogels presents unique computational challenges that distinguish these systems from conventional synthetic polymers, which will be analyzed in the next Section 2.3, because these biomacromolecules exhibit complex hierarchical structures spanning multiple length scales, from atomic-level conformational dynamics to mesoscopic assembly behaviors [103]. The intricate interplay between intramolecular and intermolecular interactions, coupled with the ubiquitous presence of water and ionic species, necessitates sophisticated modeling approaches that can accurately capture both the thermodynamic and kinetic aspects of these systems. The success of MD simulations in the field critically depends on the selection of appropriate force fields, the careful construction of initial configurations, and the implementation of suitable simulation protocols that can adequately sample the relevant conformational space [104]. Each biopolymer class presents distinct modeling considerations.

For instance, collagen's triple helix structure requires careful attention to hydrogen bonding networks and electrostatic interactions, chitosan's pH-dependent protonation states demand dynamic charge modeling, and gelatin's denatured collagen structure necessitates enhanced sampling techniques.

Among the many biopolymers used in biomaterials, we focus here on collagen, chitosan, gelatin, and alginate because they are both biologically important and methodologically representative for MD studies. From a biological perspective, these materials span key biomedical functions, where collagen is the principal structural protein of connective tissues and a major component of mineralized and soft-tissue matrices; chitosan is a bioactive, pH-responsive polysaccharide widely used in drug delivery, wound healing, and tissue engineering; gelatin is a denatured collagen derivative broadly employed in hydrogel, cell-encapsulation, and biofabrication platforms; and alginate is a widely used ionically crosslinked hydrogel for encapsulation and regenerative medicine [105–108]. From a modeling perspective, they also represent distinct classes of atomistic challenges: collagen requires accurate treatment of triple-helix stability, hierarchical assembly, and mineral/protein interactions; chitosan requires careful handling of protonation state, charge regulation, and polyelectrolyte behavior; gelatin presents disordered-chain conformational sampling, reversible association, and crosslinking complexity; and alginate provides a model system for ion-mediated gelation, M/G sequence effects, and cation-bridged network formation. Overall, these systems provide a representative cross-section of how MD can connect molecular organization with biological and mechanical function in hydrogel-based biomaterials.

2.3.1. Collagen

MD simulations have played a crucial role in unveiling the atomistic mechanisms underlying the structural and mechanical properties of collagen, the primary structural protein in connective tissues. Collagen represents one of the most structurally complex biopolymers encountered in biomaterials science and nature, with its characteristic triple helix configuration arising from the specific amino acid sequence motif Gly-X-Y, where X is frequently proline, and Y is often hydroxyproline [109]. MD simulations of collagen have evolved significantly from early studies focusing on isolated tropocollagen molecules to contemporary investigations of fibrillar assemblies and collagen-based composite materials. Collagen-based hydrogels are often modeled at the atomistic level by constructing triple-helix peptide fragments (e.g., Gly-X-Y repeats featuring proline and hydroxyproline) and building full triple-helical collagen segments using tools such as THeBuScr and CHARMM-GUI for system setup [110].

All-atom MD using CHARMM36m supports accurate modeling of hydrogen bonding, ionic interactions, and triple-helix stabilization important for fibrillar structure. Notably, simulations reveal that collagen peptides possess minimal bending resistance, and mechanical forces can induce helix disruption before covalent bond rupture, enabling insight into hierarchical mechanics relevant to biological function and biomaterial design [111]. Coarse-grained models, calibrated against reactive all-atom force fields, efficiently access longer time and length scales to study large-scale deformation and fibril interactions. Furthermore, water content and environmental conditions critically modulate collagen's structural stability and its interactions with mineral phases. These approaches are vital when evaluating crosslinking strategies (e.g., EDC/NHS, aromatic phenylenediacetic acid) and their impact on conformational integrity, swelling, and mechanical properties at the molecular scale [112]. These atomistic findings establish structure–function relationships underlying composite hydrogel mechanics and biological compatibility. Hartmann et al. [113], probed the atomistic mechanism underlying the propagation phase of collagen triple-helix folding, a rapid, biologically critical process that has so far eluded high-resolution experimental techniques.

The impact of disease-related central glycine mutations (Gly→Ala/Thr) on this propagation was also investigated. Simulations began with

unfolded (Gly–Pro–Pro) peptides, but with a C-terminal nucleus artificially stabilized via harmonic restraints on the backbone C α atoms of the last two residues in each chain. This mimicked the initiation of folding without modeling its nucleation step. Multiple independent MD trajectories (~10 per condition), extending up to ~1 μ s, were performed for both wild-type and Gly→Ala/Thr mutated peptides to map propagation dynamics [114]. Folding metrics included residue-triplet RMSD, native contact formation, dihedral transitions, folding times, and misfolding events (e.g., loop formation or strand shifts). Two-chain template formation was conducted by initial propagation involving two chains forming a short, transient near-native template, and the third chain folds sequentially, three residues at a time, onto this template. The mean propagation speed was found as ~75 ns per triplet unit. Misfolding occurred when starting from an unstable nucleus; loops spanning multiples of repeating units and strand shifts impeded correct propagation. Weaker harmonic restraints increased misfold frequency, suggesting nucleation stability is critical to productive folding. Gly→Ala or Gly→Thr mutations in the central region significantly reduced folding rates and lowered folding efficiency. Mutant chains showed sterically driven deformation that disrupted propagation, with multiple simultaneous substitutions preventing any complete fold in all simulations. Analyses of backbone dihedral angles revealed stepwise sampling toward native-like conformational regions as folding propagated from the C- to N-terminus. Regions further from the nucleus remained disordered until later simulation epochs.

2.3.2. Chitosan

Chitosan, a partially deacetylated derivative of chitin, presents unique modeling challenges due to its pH-dependent protonation behavior and the resulting changes in chain conformation and intermolecular interactions [115]. Atomistic MD simulations are invaluable for eliciting chitosan's conformational dynamics, protonation-dependent interactions, and performance as a biomaterial or drug carrier. Variations in the degree of acetylation (DA) and protonation (PD) significantly impact its solvation, folding, and ability to complex with molecules such as microRNA and peptides, influencing its functionality in gene delivery or complex formation [116].

Chitosan's biopolymeric chains (β -(1→4) linked glucosamine/N-acetylglucosamine) are modeled in atomistic MD simulations to quantify interactions with collagen segments (type II) as a function of pH and chitosan's degree of deacetylation (DD) [117]. These models reveal that higher DD yields stronger binding affinity (more hydrogen bonds, ionic contacts) with collagen frameworks and influences hydrogel co-assembly behavior. Simulations also provide an atomistic understanding of chitosan's adsorption onto surfaces (e.g., silica), interactions with model lipid bilayers (mimicking biological membranes), and thermoresponsive phase behavior. Key insights include how chitosan's structure remains stable in water, its inability to deeply penetrate lipid membranes, and the critical roles of van der Waals and hydrophobic interactions in its nanocomposite formation (Fig. 4).

Atomistic MD of chitosan alone or mixed with gelatin/collagen, sometimes in the presence of PEG or alginate, offers insights into solvation, charge screening, and cross-interaction networks [119]. Coarse-grained modeling approaches have also been developed to study chitosan assembly and gelation processes, which involve multiple chitosan chains and occur on time scales that are challenging for atomistic simulations. These models have provided insights into the molecular mechanisms underlying chitosan gel formation, including the role of hydrogen bonding, electrostatic interactions, and chain entanglement in determining gel properties. The understanding gained from these simulations has been very valuable in developing chitosan-based hydrogels with tailored mechanical and swelling properties.

Razmimanesh et al. [120], investigated all-atom MD to evaluate glycol chitosan as a biodegradable nanocarrier for the anticancer agent gemcitabine, focusing on how drug concentration and water presence influence loading efficiency and molecular interactions. Five chains of

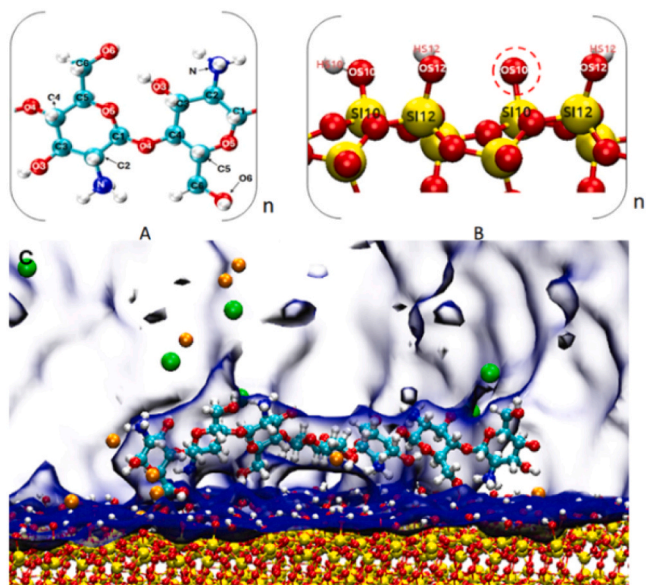


Fig. 4. (A) Chitosan molecules, (B) silica surface (the dashed red circle stands for deprotonated surface oxygen and (C) MD simulation of chitosan on silica surface [118].

glycol-chitosan (degree of polymerization ~ 20) with $\sim 20\%$ protonation to mimic physiological charge were used to form the carrier matrix. Drug concentrations of 10%, 40%, and 80% (w/w) were simulated. The hydration study was conducted by the simulation of both in the absence and presence of explicit water to probe solvation effects on drug-carrier interactions and loading efficacy. RDF, nonbonded energy terms (van der Waals and electrostatic), system energy, density, and structural equilibrium stability were all evaluated using VMD and MD output trajectories. Maximum drug encapsulation occurred at $\sim 40\%$ gemcitabine, where drug molecules were preferentially located closer to chitosan chains, maximizing nonbonded interactions and loading efficiency. At 10%, insufficient drug resulted in lower overall loading, whereas at 80% crowding, it likely reduced binding efficiency. In dry systems, gemcitabine molecules dwelt at shorter radial distances from chitosan functional groups (NH_2 , OH), yielding stronger interactions and greater loading. Introduction of water molecules weakens drug-chitosan interactions (via competition for hydrogen bonding and reduced van der Waals contacts), resulting in lowered loading efficiency despite solvation benefits. The simulations reached equilibrium with stable density and total energy profiles, indicating reliable sampling. At optimal 40% loading, the system displayed favorable energy configurations, including stable nonbonded energy minima, corroborating stronger drug-carrier affinity.

Shariatnia et al. [121], explored chitosan-based nanocomposite systems as potential carriers for the anticancer drug ifosfamide, using all-atom MD simulations to evaluate drug-polymer-nanofiller interactions, loading behavior, and temperature sensitivity. Nanocomposite systems were also considered, such as glycol-chitosan chains with various inorganic nanofillers (e.g., functionalized graphene derivatives, silica, etc.) to assess optimal drug-loading performance. They have simulated at different drug loading ratios, pH environments, and temperature conditions, reflecting physiological and storage-related variability; the resulting analyses included RDF, hydrogen bond quantification, mean square displacement (MSD), and binding energy calculations to characterize drug affinity and system stability. Ifosfamide preferentially localized near chitosan amino and hydroxyl groups, forming hydrogen bonds and significant electrostatic and van der Waals interactions, promoting stable adsorption within the polymer matrix. The inclusion of nanofillers enhanced drug loading by providing increased surface area and anchoring points, thereby increasing

interfacial adhesion and reducing drug mobility. Higher filler-polymer interaction strength yields more efficient encapsulation and slower release kinetics. Elevated temperatures (above physiological range) increased MSD and diffusion coefficients, facilitating faster drug mobility, while lowered temperatures correspondingly reduced diffusion and improved retention. Acidic pH enhanced the protonated state of chitosan, strengthening electrostatic attraction to ifosfamide and increasing hydrogen bond formation under drug loading conditions. Equilibrium MD trajectories showed stable system energy and density profiles, confirming robust sampling and reliable simulation conditions. Binding energy analysis revealed favorable nonbonded interaction minima, particularly for systems containing nanofillers, indicating a thermodynamically preferred drug-carrier state.

Moradi et al. [122], conducted a study to employ atomistic MD simulations to assess how chitosan, with varying degrees of deacetylation (DDA), influences the thermal stability of interferon αII , in comparison to synthetic polymers PEG and PLGA, under elevated temperature stress. Interferon αII was simulated alone and in complex with each polymer (chitosan, PEG, PLGA). Chitosan models differed in DDA to explore how polymer charge and solubility affect protein stabilization. MD simulations were conducted at 343 K and compared to baseline simulations at 300 K (physiological) to evaluate structural integrity under heat stress. At 343 K, low-DDA chitosan and PLGA effectively preserved the native conformation and secondary structure of interferon αII , showing minimal unfolding or backbone perturbation. In contrast, PEG penetrated the protein core and introduced instabilities, with increased backbone fluctuations and partial structural disruption.

Chitosan with lower deacetylation (more acetylated) offered superior stabilization compared to high-DDA variants. This suggests that optimal hydrogen-bonding and reduced excessive surface charge help buffer thermal denaturation. Between the three polymers studied, PLGA and low-DDA chitosan emerged as preferable carriers for maintaining interferon structure at elevated temperatures. High-DDA chitosan and PEG exhibited weaker protective effects, with PEG notably destabilizing the protein conformation via penetration and disruption. Chitosan and PLGA acted as nonpenetrating protective matrices, preserving structural integrity via surface adsorption or gentle encapsulation. PEG's higher tendency to infiltrate the protein structure may have disrupted hydrogen-bond networks and accelerated thermal instability.

2.3.3. Gelatin

Gelatin, a denatured derivative of collagen, represents a fascinating case study in protein denaturation and subsequent reassembly into functional materials, as it is widely used in hydrogel applications due to its biocompatibility and tunable physical properties [123]. Unlike native collagen's ordered triple helix structure, gelatin consists of denatured collagen chains that retain segments of their original amino acid sequence but lack the long-range structural organization. This structural complexity makes gelatin modeling particularly challenging, as simulations must capture both the random coil nature of individual chains and their propensity for local structure formation and intermolecular association. Recent advancements in atomistic MD modeling, based on modified FASTA codes and validated against experimental observables, offer predictive power for understanding gelatin's matrix structural evolution and suitability as a bioactive carrier [124]. Such simulations, often leveraging the INTERFACE force field, accurately capture the thermal and mechanical response of gelatin networks over physiologically relevant temperature ranges, providing crucial predictions for the design and optimization of functional polymeric matrices.

Gelatin is also modeled either as denatured collagen fragments or peptides retaining Gly-X-Y repeats and RGD motifs, which is important for cell adhesion within atomistic MD frameworks. In MD, gelatin-polysaccharide complexes such as alginate or carrageenan have been simulated to analyze binding energetics, hydrogen bonding, and conformational stability using CHARMM36m in GROMACS environments under NPT [125]. These simulations typically include explicit

water and physiological salts, with equilibration and production runs that capture the solvation structure and complexation behavior. MD simulation also elucidates details of enzymatic cross-linking and resultant network configurations necessary for hydrogel engineering. Central to many studies is the modeling of gelatin methacryloyl (GelMA) chains, which is one of the most utilized biomaterials in the field in atomistic simulations when crosslinked chemically, to probe network formation under UV-initiated polymerization and assess how methacryloyl functionalization affects stiffness and network connectivity [126]. Enhanced sampling techniques, such as replica exchange MD and metadynamics, have been employed to study the complex conformational landscape of gelatin chains and the free energy barriers associated with triple helix formation and dissociation. These advanced simulation methods have provided quantitative insights into the thermodynamics of gelatin gelation and have been instrumental in understanding how modifications to gelatin structure, such as chemical cross-linking or the incorporation of other polymers, affect gel properties.

Xiong et al. [127], investigated the molecular-level interaction mechanisms and mechanical behavior of lignin/gelatin composites via atomistic MD simulations, which stemmed from the need to reinforce gelatin, a biopolymer with inherently weak intermolecular binding, for potential textile fiber applications. Composite models were constructed at multiple lignin loadings: 0%, 5%, 10%, and 20% (w/w) within a gelatin matrix. All-atom MD simulations characterized hydrogen bonding patterns, density, and load responses at each composition. Simulation data were cross-validated against experimental measurements of mechanical properties to ensure consistency between computational predictions and real-world performance metrics. Introduction of lignin into the gelatin matrix significantly enhanced intermolecular hydrogen bonding, primarily between gelatin's amino/hydroxyl groups and lignin's phenolic hydroxyls. Hydrogen bond analysis demonstrated that 10% lignin content maximized interfacial interaction while preserving network homogeneity and compatibility. Simulated mechanical properties, including elastic modulus and tensile strength, reached their peak at 10% lignin loading, indicating an optimal balance of reinforcement and flexibility. Compared to pristine gelatin, the composite demonstrated markedly improved stiffness and predicted load-bearing capacity due to stronger gelatin–lignin interfacial coupling. Higher lignin fractions (20%) did not enhance performance further, likely due to phase separation or excessive filler disrupting gelatin continuity. The MD predictions were supported by experimental testing, showing consistent trends in mechanical enhancements at the 5–10% lignin range. This alignment validated the simulation framework as a predictive tool for designing lignin-reinforced gelatin materials. Zaupe et al. [128], aimed to predict the mechanical behavior of gelatin-based biomaterials using atomistic MD simulations. They investigated how physically interacting aromatic modifications, namely tyrosine-derived moieties, introduced into gelatin chains, affect crosslink density, swelling behavior, and elastic properties of amorphous gelatin systems. Amorphous bulk models of pure and functionalized gelatin were created at two representative hydration levels, which were 0.8 wt% and 25 wt% water content. Gelatin was chemically grafted with tyrosine-based groups to induce non-covalent crosslinks via aromatic interactions. Pure gelatin simulations were benchmarked against experimental data, density, backbone dihedral angle distributions, and X-ray scattering profiles, to ensure realism. Simulations revealed that increasing the density of aromatic (tyrosine-derived) groups leads to a higher number of physical netpoints as reversible junctions via π – π stacking or hydrogen bonding, within the amorphous gelatin network. By coupling simulation results with the Flory–Rehner polymer-network theory, the team predicted that functionalized gelatin exhibits reduced equilibrium swelling in water compared to unmodified gelatin due to its higher crosslink density. The functionalized materials showed lower swelling at equilibrium. MD-derived estimates of Young's modulus via static deformation methods for pure gelatin were consistent with experimental findings. Young's modulus was found to increase with functionalization,

linked directly to the formation of additional network junctions. Although the simulated modulus for unmodified gelatin (~ 7.8 GPa) was higher than the experimental one (~ 2.9 GPa), relative trends and values were deemed valid. The simulation of functionalized systems predicted higher stiffness and better mechanical stability.

Liang et al. [129], studied the molecular basis of antioxidant activity in donkey-hide gelatin peptides in order to understand how peptide structural features and peptide–protein interactions contribute to reactive oxygen species (ROS) scavenging. Utilizing enzymatic hydrolysis, structural spectroscopy, and molecular docking/dynamics, the work identified bioactive sequences and elucidated the mechanism via interactions with the Keap1 protein. Donkey-hide gelatin was hydrolyzed and fractionated into five molecular weight groups as > 30 kDa, 10–30 kDa, 3–10 kDa, 1–3 kDa, and < 1 kDa. Amino acid composition was quantified via FTIR and circular dichroism (CD) spectroscopy, characterized functional groups and secondary structure content across MW fractions. DPPH and ABTS radical scavenging assays were performed, revealing that peptides in the 1–3 kDa range exhibited the strongest antioxidant efficacy, which was shown in Fig. 5 as the docking interactions. Moreover, twenty-six pentapeptides/hexapeptides were sequenced, among which PGPAP was identified as the most active. PGPAP was docked to the Keap1 protein; the complex was subjected to ~ 50 ns MD simulation to assess conformational stability, binding interactions, and hydrogen-bond dynamics. The 1–3 kDa peptide fraction showed the highest DPPH ($\sim 32.3 \pm 2.2\%$) and ABTS ($\sim 94.1 \pm 0.4\%$) radical scavenging, indicating optimal bioactivity in this size range in MD studies. Interestingly, peptides < 1 kDa showed decreased activity, potentially due to excessive free amino acid content lacking structured peptide conformation. All fractions contained abundant Gly (G), Pro (P), Glu (E), Asn (N), Ala (A), and Arg (R), but specific proportions varied with MW. Notably, the 1–3 kDa fraction had higher Gly and Ile content, which may influence radical scavenging via structure and hydrophobicity. FTIR and CD measurements indicated differences in amide I/II band intensity and secondary structure proportions (e.g., β -sheet vs random coil) among MW classes, correlating with antioxidant potency. Docking predicted a binding energy of approximately -5.38 kcal/mol between PGPAP and Keap1. MD analysis (50 ns) demonstrated stable RMSD (~ 0.2 – 0.3 nm), compact radius of gyration, and persistent hydrogen bonding throughout the simulation. Key interacting residues identified as Arg415, Gly462, Phe478 and Tyr572. These interactions included hydrogen bonds, salt bridges, and π – π stacking, critical to stable peptide–Keap1 binding.

2.3.4. Alginate

Alginate, a naturally occurring polysaccharide composed of guluronic and mannuronic acid residues, presents unique modeling challenges due to its polyelectrolyte nature and its ability to form gels through ionic cross-linking with divalent cations [130]. The MD simulation of alginate systems requires careful treatment of electrostatic interactions, ionic atmospheres, and the conformational coupling between polymer structure and charge distribution. Alginate, especially in the form of calcium-crosslinked hydrogels, has been effectively studied via MD and steered MD (SMD) approaches [131]. MD explains the aggregation of alginate chains in the presence of divalent cations, notably Ca^{2+} , and identifies the dominant interactions, which are ionic bridging, hydrogen bonding, and van der Waals forces underlying adhesion strength and gel formation. Alginate modeling at the atomistic level typically represents blocks of β -D-mannuronate (M) and α -L-guluronate (G) units. Calcium ions are introduced to simulate gelation via physical cross-linking (“egg-box” junctions), capturing chelation geometry, ion-bridging, and polymer stiffness [132].

In MD, alginate chains are simulated in aqueous ionic environments, with physiological Ca^{2+} , Na^+ , Cl^- , and explicit water, to explore gel microstructure, diffusion pathways, and mechanical reinforcement in alginate or hybrid hydrogels. SMD simulations can quantify unbinding forces and structural deformation during gel detachment from interfaces

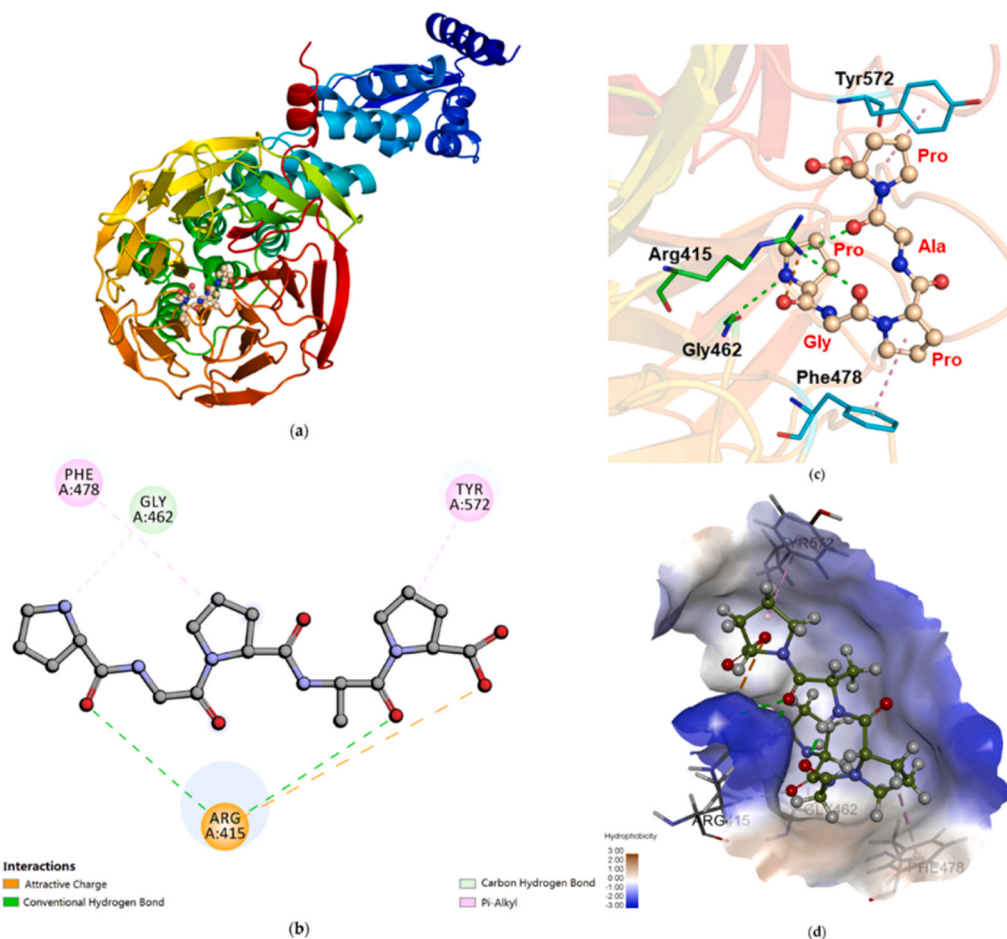


Fig. 5. PGPAP docking modeling with Keap1 as the crystal structure of PGPAP linked with (a). The two-dimensional structure of PGPAP interaction with Keap1 (b). The three-dimensional structure of PGPAP interaction with Keap1 (c) and hydrophobic interfaces of PGPAP binding to Keap1 (d) [129].

or other biomolecules, revealing extensive chain deformation and interaction specificity [133]. Such insights guide algorithmic optimization for designing resilient alginate-based biomaterials for encapsulation, tissue engineering, and membrane technologies. Coarse-grained modeling approaches have been developed to study larger-scale phenomena in alginate systems, such as the formation of gel networks and the mechanical response of alginate gels to external forces [134]. These models have enabled investigations of how the molecular-level interactions between alginate chains and cations translate into macroscopic gel properties, providing a bridge between atomistic simulations and experimental observations of bulk material behavior. These simulations help interpret alginate stability, swelling, and deformation under strain.

Liu et al. [135] studied the interactions, structural characteristics, and formation mechanisms of whey protein isolate (WPI) and sodium alginate (SA) complexes, focusing on how varying pH, WPI:SA ratio, and temperature influence assembly, physicochemical properties, and potential functional performance. WPI-SA mixtures were prepared at different mass ratios and temperatures, across pH values from acidic to neutral ($\sim$ pH 3–7) in order to generate protein-polysaccharide complexes. FTIR, fluorescence, circular dichroism, rheology, zeta potential, and particle size were carried out to reveal structural changes and complex stability; furthermore, molecular docking and MD simulations were performed at selected conditions to map key amino acid–alginate site interactions, hydrogen bonding patterns, and conformational adaptations under different protonation states. Complexation was favored near the isoelectric point (pH $\sim$ 4.5–5.0) of WPI, where electrostatic repulsion among protein molecules is minimized, thus promoting

WPI-SA interactions. At optimal pH, WPI undergoes partial unfolding, exposing hydrophobic and charged regions that facilitate interactions with SA via hydrogen bonding and electrostatics. Raman and FTIR spectra revealed increased α -helix content ($\sim$ 7.5% higher) and reorganization of secondary structure in WPI upon binding SA at pH 5.0. This indicates a more compact, ordered protein conformation in complexes. MD simulations identified key residue-specific interactions, including several acidic residues as Glu and Asp, in WPI form, persistent hydrogen bonds/salt bridges with SA's carboxyl functionalities, driving complex assembly. Complex binding energy and hydrogen bonding networks vary with pH, closely matching experimental observations of increased compactness around optimal pH conditions. Boukoussa et al. [136] investigated the development of a novel composite adsorbent combining mesoporous silica SBA-15 encapsulated within calcium alginate beads, and used both experimental adsorption tests and MD simulations to demonstrate the adsorption behavior and mechanism toward a model cationic dye. SBA-15 mesoporous silica was functionalized and encapsulated within calcium alginate beads to produce stable composite microspheres. Dye adsorption was performed in batch mode, varying initial dye concentrations and contact times. Isotherm and kinetic modeling were conducted using Langmuir, Freundlich, and pseudo-second-order frameworks, while SEM, nitrogen sorption, surface charge analysis, and thermal analysis were performed as part of SBA-15/Alg bead experimental characterization.

MD simulations modeled SBA-15 pores, alginate network, and dye molecules at atomic-level resolution to assess interaction energies, hydrogen bonding, and dye diffusion constraints within the composite. The performed simulations helped link interfacial energies and binding

modes to experimental adsorption capacity and kinetics. It was shown that the SBA-15@Alg beads exhibited high adsorption capacity for the cationic dye. Adsorption followed by the pseudo-second-order kinetics indicated chemisorption or electrostatic surface interactions as the rate-limiting. Equilibrium data fitted well to the Langmuir isotherm model, suggesting monolayer adsorption on homogeneous active sites. MD revealed strong electrostatic attraction and hydrogen bonding between the $-Si-O^-$ groups of SBA-15, carboxylate moieties of alginate, and the positive charges on the dye molecules. Confinement within alginate-SBA-15 matrices restricted dye diffusion but enhanced binding via multiple site interactions, effectively augmenting adsorption capacity and retention efficiency. The composite beads maintained the ordered mesoporous structure of SBA-15 even after encapsulation, as proved by textural analyses. The alginate matrix provided mechanical stability and prevented silica micro-particle leaching, while the high surface area of SBA-15 supplied abundant adsorption sites in this case.

2.3.5. Summary about the potential of MD applied to biopolymers and hydrogels

The elastic properties of biopolymer hydrogels emerge from the complex interplay between polymer chain flexibility, cross-linking density, and the thermodynamic interactions between polymer chains and the surrounding aqueous medium. Understanding these relationships at the molecular level requires detailed analysis of how local chain conformations and intermolecular interactions contribute to the macroscopic mechanical response of the hydrogel network. The theory of rubber elasticity provides a fundamental framework for understanding hydrogel mechanics, but its application to biopolymer systems requires careful consideration of the unique features that distinguish these materials from synthetic elastomers [137]. Unlike synthetic rubber networks, biopolymer hydrogels often exhibit significant deviations from ideal rubber elasticity due to factors such as chain entanglement, physical cross-links that can break and reform under stress, and the complex interactions between polymer chains and water molecules. MD simulations have provided detailed insights into the molecular mechanisms underlying hydrogel elasticity by enabling direct observation of how individual polymer chains respond to applied deformation. These simulations have revealed that the elastic response involves contributions from both conformational entropy changes of individual chains and intermolecular interactions that resist chain separation or compression [138]. The swelling behavior of biopolymer hydrogels, on the other hand, is a fundamental thermodynamic phenomenon that involves the competition between the elastic restoring forces of the polymer network and the osmotic pressure driving water uptake. Understanding this balance at the molecular level is crucial for predicting and controlling the swelling response of hydrogel materials under different environmental conditions. The thermodynamic framework for hydrogel swelling, originally developed by Flory and Rehner, provides a quantitative description of the equilibrium swelling ratio in terms of the elastic contribution from the polymer network and the mixing contribution from polymer-solvent interactions [139]. However, the application of this framework to biopolymer systems requires careful consideration of the unique features of these materials, including their tendency to exhibit non-ideal mixing behavior and the presence of specific intermolecular interactions. MD simulations have provided insights into the molecular mechanisms underlying hydrogel swelling by enabling direct observation of the structural changes that occur as water molecules penetrate the polymer network. The role of polymer-water interactions in determining swelling behavior has been extensively studied through simulations that examine how changes in the strength and specificity of these interactions affect the equilibrium swelling ratio [140].

For hydrophilic polymers such as PEG, the favorable interactions between polymer and water molecules drive extensive swelling, while for polymers with both hydrophilic and hydrophobic character, the swelling behavior reflects the balance between these competing

tendencies. For polyelectrolyte hydrogels such as chitosan and alginate, the swelling behavior is significantly influenced by the ionic atmosphere around the charged polymer chains. The Donnan equilibrium, which describes the distribution of mobile ions between the hydrogel and the external solution, creates an additional osmotic pressure that contributes to swelling [141]. MD simulations have enabled understanding how the ionic composition and concentration of the external solution affect the swelling equilibrium. When the mobility of polymer chains within hydrogel networks is considered, it has an important role in determining both the mechanical properties and the transport characteristics of these materials. Understanding chain dynamics at the molecular level requires detailed analysis of how local chain motions contribute to larger-scale relaxation processes and how these dynamics are influenced by the presence of cross-links and the surrounding aqueous environment. The relaxation dynamics of polymer chains in hydrogel networks exhibit a complex hierarchy of time scales, ranging from fast local motions involving bond rotations and small-amplitude bending modes to slow global motions involving large-scale chain rearrangements [142]. MD simulations have also led to an understanding of this hierarchy by enabling direct observation of chain motions across multiple time scales. The presence of cross-links in hydrogel networks significantly constrains polymer chain mobility, creating localized regions of high mobility between cross-link points and effectively immobilizing chain segments at cross-link locations. The dynamics of chains between cross-links have been extensively studied through simulations that examine how cross-link density and functionality affect local chain mobility. For physically cross-linked networks, such as those formed by hydrogen bonding or ionic interactions, the chain dynamics are further complicated by the transient nature of the cross-links, which can break and reform on time scales that are comparable to or shorter than the characteristic relaxation times of the polymer chains [143]. This creates a dynamic equilibrium between cross-linked and uncross-linked states that significantly influences the mechanical and transport properties of the hydrogel.

While atomistic MD simulations provide unparalleled detail about the molecular-level interactions that govern biopolymer behavior, they are fundamentally limited in their ability to address the length and time scales that are relevant for understanding macroscopic hydrogel properties. Typical atomistic simulations can access system sizes of tens of nanometers and time scales of microseconds, while the characteristic dimensions of hydrogel networks and their relaxation times often extend to much larger scales [144]. The development of coarse-grained models represents a crucial advance that enables the study of larger hydrogel systems while retaining the essential physics that determines network behavior. The key challenge in coarse-graining lies in identifying which molecular details are crucial for predicting macroscopic properties and developing effective interactions between coarse-grained units that accurately reproduce these essential features [145]. Coarse-grained models are particularly valuable for biopolymer hydrogels because they enable access to fibril assembly, network formation, swelling, and relaxation processes beyond atomistic simulation scales. In this context, MARTINI-type representations and related CG strategies have been used to study collagen fibrillogenesis, chitosan network formation, and polyelectrolyte-rich hydrogel behavior while retaining the key balance between chain flexibility, intermolecular association, and solvent-mediated interactions.

The application of MARTINI to collagen systems has enabled studies of collagen fibril formation and the mechanical properties of collagen networks that would be computationally prohibitive with atomistic models. These simulations have revealed how the hierarchical assembly of collagen molecules into fibrils depends on the balance between intermolecular attractions and the entropic cost of chain ordering, providing insights into the conditions that favor fibril formation [146]. For chitosan systems, MARTINI-based simulations have been used to study the formation of physical hydrogel networks through hydrogen bonding and electrostatic interactions. These simulations have

demonstrated how the pH-dependent protonation of chitosan affects network formation and have provided insights into the relationship between molecular-level interactions and macroscopic gel properties. The treatment of polyelectrolyte effects in coarse-grained models presents particular challenges, as the long-range nature of electrostatic interactions must be accurately represented while maintaining computational efficiency [147]. Recent developments in the MARTINI force field have incorporated improved treatment of charged species, enabling more accurate simulations of polyelectrolyte hydrogels such as alginate networks. Table 1 shows the general MD methods in terms of structural focusing and model levels.

2.4. Metallic biomaterials

Metallic biomaterials represent a crucial application area of modern biomedical applications, especially in orthopedic implants, cardiovascular devices, and dental prosthetics. Pure metals (e.g., gold) and alloys (e.g., steel, cobalt-chromium alloys) have been used for decades in the manufacturing of permanent dental restorations and orthopedic implants. Among the diverse types of metallic biomaterials, titanium (Ti), magnesium (Mg), and zinc (Zn) alloys have recently emerged as highly appealing candidates due to their unique combination of biocompatibility, mechanical properties, and – to some extent – degradation characteristics [149]. The atomic-scale understanding of these materials through MD simulations has become increasingly vital for optimizing their performance in biological environments. The complexity of metallic biomaterial behavior arises from the intricate interplay between atomic structure, surface chemistry, mechanical deformation, and interactions with the biological environment. The atomistic modeling of metallic biomaterials begins with the selection and validation of appropriate interatomic potentials that accurately capture the essential physics governing atomic interactions [150]. For titanium alloys, the embedded atom method (EAM) and modified embedded atom method (MEAM) potential have proven particularly effective in describing the complex electronic environments characteristic of transition metals. The EAM formalism treats each atom as embedded in an electron density field created by neighboring atoms, with the total energy expressed as the sum of pair interactions and embedding energies.

Recent developments in machine learning potential, particularly neural network potentials and Gaussian approximation potentials, have shown remarkable promise for capturing the subtle electronic effects that govern Ti alloy behavior. These advanced potentials excel in describing phase transformations between α (hexagonal close-packed) and β (body-centered cubic) phases, which are crucial for understanding the mechanical and biocompatible properties of Ti-6Al-4V and other clinically relevant alloys [151].

For magnesium alloys, modeling challenges are related to the high chemical reactivity of the metal and the need to accurately represent Mg-based intermetallic phases. The development of reliable interatomic potentials for Mg alloys has focused on capturing the anisotropic mechanical properties inherent to the hexagonal close-packed structure [152]. Second-nearest-neighbor modified embedded atom method (2NN-MEAM) has demonstrated superior performance in reproducing experimental elastic constants, stacking fault energies, and twin

formation mechanisms that are critical for understanding Mg alloy deformation behavior.

Zinc alloys present unique modeling challenges due to their intermediate position between transition and post-transition metals. The electronic structure of Zn, with its filled d-shell, requires careful consideration of charge transfer effects and the potential for covalent bonding contributions. Recent advances in charge-optimized many-body (COMB) potential have shown promise for accurately describing Zn-based systems, particularly in environments where oxidation and charge redistribution play significant roles.

2.4.1. Microstructure

The atomistic modeling of crystal structures in metallic biomaterials extends beyond perfect crystal lattices to encompass the full spectrum of defects that govern material properties. Point defects, including vacancies, interstitials, and substitutional atoms, play crucial roles in diffusion processes, mechanical properties, and corrosion resistance. MD simulations have revealed that vacancy formation energies in Ti alloys are highly dependent on local chemical environment, with β -stabilizing elements like V and Mo significantly altering defect energetics [153]. Dislocation dynamics represent another critical aspect of atomistic modeling, particularly for understanding plastic deformation mechanisms. In titanium alloys, the competition between basal, prismatic, and pyramidal slip systems determines the overall mechanical response. Large-scale MD simulations with millions of atoms have elucidated the role of solute atoms in controlling dislocation motion, revealing how Al atoms in Ti-6Al-4V create local distortions that impede dislocation glide and enhance strength. The modeling of twinning mechanisms in Mg alloys has provided fundamental insights into the anisotropic mechanical behavior that limits their structural applications. MD simulations have demonstrated that extension twins nucleate through the coordinated motion of partial dislocations on consecutive basal planes, with the activation stress highly dependent on grain orientation and local chemical composition [154]. These atomistic insights have proven invaluable for developing texture-modified Mg alloys with improved ductility. Modern atomistic modeling approaches have revolutionized alloy design by enabling the systematic exploration of compositional effects on material properties. High-throughput MD calculations, combined with thermodynamic databases, allow for the rapid screening of potential alloy compositions before expensive experimental validation. For Ti alloys, this approach has identified novel β -stabilizing element combinations that maintain biocompatibility while enhancing mechanical properties. The development of biodegradable Mg alloys has particularly benefited from atomistic modeling approaches. MD simulations have revealed how alloying elements like Ca, Zn, and rare earth elements alter the local atomic environment and influence degradation kinetics. These studies have shown that the formation of intermetallic phases generates galvanic couples that accelerate corrosion, providing crucial guidance for optimizing alloy compositions for controlled degradation rates. For Zn alloys, atomistic modeling has explored the potential for developing alloys with tailored mechanical properties and degradation characteristics.

Table 1
General overview of biopolymers and hydrogels in MD methods [137,133,148].

Biopolymer	Model Level	Key Structural Focus	Key Insights from MD	MD Methods / Force Field
Collagen	Atomistic, CG, Triple-helix peptide	Hydrogen-bonding, helix stability, and crosslinks	Triple-helix folding, microunfold, length/time scaling, solvent/mineral interactions	CHARMM36m, GROMACS, THeBuScr scripts
Chitosan	Atomistic, Oligomers/polymers	Electrostatics, DD effects, charge binding	DA/PD effects, stability, membrane interactions, nanoscale complexation	All-atom MD, binding free energy
Gelatin	Atomistic, Peptide fragments, or GelMA	Confirmation, RGD motifs, crosslink density	Density and Tg prediction, enzyme crosslinking, bioactive matrix design	CHARMM36m simulations, functionalized chains
Alginate	Atomistic, SMD, M/G blocks + Ca ²⁺	Ion-bridged gelation, stiffness, swelling	Ca ²⁺ aggregation, unbinding forces, structural deformation, and crosslinking mechanisms	Atomistic MD with explicit ions

2.4.2. Surface properties and interactions with biomolecules

The interaction between metallic biomaterial surfaces and biological molecules represents a complex interplay of chemical, physical, and biological factors that ultimately determine biocompatibility and biological response. MD simulations open a pathway into protein adsorption mechanisms, cell adhesion processes, and the role of surface chemistry in modulating biological interactions. The TiO₂ surfaces characteristic of Ti implants exhibit unique properties that promote favorable biological responses through specific protein-surface interactions. Albumin adsorption on TiO₂ surfaces has been extensively studied through large-scale MD simulations, revealing the preferential binding sites and conformational changes that occur upon protein contact with the oxide surface [155]. These studies have demonstrated that the amphiphilic nature of albumin leads to partial unfolding and reorientation to maximize favorable electrostatic interactions with the TiO₂ surface. The resulting protein layer serves as a conditioning film that influences subsequent cellular interactions. The role of surface hydroxylation in promoting biocompatibility has been elucidated through MD simulations of water and biomolecule interactions with metal oxide surfaces. Hydroxyl groups on TiO₂ surfaces create hydrogen bonding networks that facilitate protein adsorption and maintain hydration layers essential for biological compatibility. The density and distribution of surface hydroxyl groups can be controlled through surface treatments, providing a means to optimize biointerface properties. For Mg and Zn alloys intended for biodegradable applications, the surface chemistry evolution during degradation significantly impacts the biological response [156]. MD simulations have shown that the release of metal ions creates local chemical environments that can either promote or inhibit cellular activities depending on ion concentration and speciation. Understanding these dynamic surface chemistry changes has been crucial for optimizing degradation rates and biological responses. Wang et al. [157] investigated how secondary phases in biodegradable magnesium alloys, such as those containing Y, Ce, Nd, or Zn, influence protein adsorption behavior, combining *in vitro* experiments and MD simulations to understand impacts on biocompatibility. Various Mg alloys featuring distinct second-phase compositions were immersed in plasma. Protein adsorption (specifically fibrinogen) was quantified using the BCA assay and mass spectrometry (MS). Biological impact was assessed through cell adhesion and viability assays using MG63 osteoblast-like cells. MD models simulated fibrinogen adsorption on representative second-phase surfaces versus α -Mg matrix regions. Simulations analyzed surface charge distribution, protein-surface interaction energy, and local solvent structuring (water distribution) around target surfaces. Fibrinogen adsorption was significantly enhanced on second phases containing rare-earth elements (Y, Ce, Nd). In contrast, Zn-containing second phases exhibited reduced protein adsorption, suggesting inhibitory surface properties. Higher surface-protein interaction energies on RE-rich phases led to stronger binding, largely driven by favorable charge distributions and surface polarity. On Zn-rich second-phase surfaces, weaker interactions and altered water layering hindered protein docking. Water density near the surface also varied: more structured water hydration layers inhibited fibrinogen contact on Zn phases, whereas RE-rich surfaces supported hydration shells conducive to adsorption. Sahoo et al. [158] investigated the influence of forging temperature (FT623, FT723, FT823) on the microstructure evolution, mechanical behavior, corrosion resistance, antibacterial response, and osteogenic differentiation performance of Mg–0.2Zr–0.1Sr–0.5Ce alloy. *Ab initio* MD (AIMD) simulations were utilized to elucidate further the atomic-scale stability and evolution of Mg₁₂Ce precipitates in the alloy. Microstructural characterizations by SEM, TEM, and X-ray phase analysis confirm the presence and stability of Mg₁₂Ce precipitates. Thermomechanical processing was conducted on alloys forged at 623 K, 723 K, and 823 K (FT623, FT723, FT823) to control precipitate fraction and basal texture intensity. *In vitro* corrosion tests and antibacterial assays were performed against *E. coli*, *S. aureus*, and *P. aeruginosa* bacteria for both biocorrosion and antibacterial

testing. Cellular compatibility was evaluated by measuring MC3T3-E1 osteoblastic cell proliferation and alkaline phosphatase (ALP) activity. While an *in vivo* evaluation was performed on rabbit femur implantation for bone regeneration assessment, AIMD simulations were conducted by quantum-level modeling of Mg₁₂Ce precipitate stability and interaction within Mg matrix. As a result, FT723 treatment yielded the strongest basal texture, the highest fraction of dynamic recrystallization-assisted grains, abundant, stable Mg₁₂Ce precipitates, balanced mechanical performance, and high strength with retained ductility. Moreover, FT723 exhibited significantly enhanced corrosion resistance, attributed to optimized texture, lower dislocation density, and stable precipitate distribution acting as micro-galvanic moderation sites. All alloys demonstrated 100% inhibition of Gram-negative *E. coli*, Gram-positive *S. aureus*, and *P. aeruginosa* within 24 h, suggestive of release of biocidal alloying ions (Ce, Sr) or surface-mediated reactive effects. The FT723 alloy produced the highest osteoblast proliferation and ALP activity in MC3T3-E1 cells, reflective of its low corrosion and favorable ion release profile. Finally, *in vivo* rabbit femur implantation showed no tissue infection, and greater bone regeneration around FT723 alloy after 60 days, confirmed via SEM and fluorescence labeling. Kaviani et al. conducted a study by combining MD and quantum mechanics to reveal the interaction of Fe³⁺ and human serum albumin to understand the iron overload disease [159]. The authors utilized MD simulation to predict two binding sites of human serum albumin (HSA) for complex formation, including Glu-Arg-Asn-Glu-Cys (BS1) and Lys-Glu-Cys-Cys-Glu-Lys (BS2) amino acids, to investigate the interaction of the corresponding binding sites with Fe³⁺. Moreover, the quantum chemistry calculations were performed at M06–2X/6–31 G(d) level in water: the density functional theory calculations showed that binding constant of Fe³⁺ to human serum albumin in BS1 (log K = 13.33) was lower than that of BS2 (log K = 18.99), demonstrating higher thermodynamic stability of the HAS single bond Fe³⁺ complex in BS2.

2.4.3. Mechanical properties

The mechanical behavior of metallic biomaterials under physiological loading conditions is a critical aspect for their long-term performance and clinical success. MD simulations under applied stress and strain have provided fundamental insights into the atomic-scale mechanisms governing plastic deformation, work hardening, and failure processes. The application of mechanical loads in MD simulations requires careful consideration of boundary conditions, loading rates, and system size effects to ensure physically meaningful results. Uniaxial tensile loading simulations of Ti alloys have revealed the sequence of events leading from elastic deformation to plastic flow and eventual failure [160]. The initial elastic response is characterized by reversible bond stretching and angle bending, with the elastic moduli determined by the second derivatives of the interatomic potential energy surface. As the applied stress approaches the theoretical yield strength, dislocation nucleation occurs preferentially at sites of high local stress concentration, including surface steps, grain boundaries, and pre-existing defects. The transition from elastic to plastic deformation involves the nucleation and propagation of dislocations on specific crystallographic slip systems. MD simulations have shown that the critical resolved shear stress for dislocation nucleation varies significantly between different slip systems, with basal slip typically exhibiting the lowest activation stress. The work hardening behavior observed in metallic biomaterials arises from the interaction between mobile dislocations and various obstacles, including other dislocations, grain boundaries, and second-phase particles. The evolution of dislocation density and arrangement during continued deformation determines the overall stress-strain response and mechanical properties.

Gowthaman et al. [161], employed atomistic MD to investigate the tensile mechanical behavior of Ti–5Mo–5Cu alloy under varying temperatures and strain rates, to elucidate how these factors influence yield stress, strain, dislocation formation, and structural deformation mechanisms in a biomaterial context. The alloy was modeled at the atomic

scale using classical LAMMPS, applying uniaxial tensile loading across a range of temperatures and strain rates. Strain rate varied typically from low ($\sim 1\text{e}^{-5}/\text{ps}$) to high ($\sim 2.5\text{e}^{-3}/\text{ps}$), while temperature conditions spanned ambient to elevated values. Structural variables analyzed included shear strain distribution, defect nucleation, and radial distribution functions to clarify atomic coordination shifts under load. Higher strain rates significantly increased yield stress and fracture strain across the simulation conditions, consistent with strain-hardening behavior and kinetically constrained deformation. The positive impact of strain rate on mechanical strength was attributed to atomic-level restraining effects between neighboring atoms, delaying dislocation formation and propagation. In contrast, increasing ambient temperature led to a pronounced reduction in tensile properties with reduced yield stress and fracture strain due to enhanced atomic motion, bond weakening, and increased dislocation density. RDF analysis confirmed that elevated temperatures promote increased atomic disorder and lattice strain, undermining mechanical integrity. Under high-temperature conditions, MD outputs showed the formation of multiple dislocations, altering shear strain fields and driving deformation. The structural change mediated by dislocations correlates with the observed degradation in tensile behavior at elevated temperatures. Sampath et al. [162] utilized all-atom MD augmented with well-tempered metadynamics to explore the binding mechanisms and specificity of amino acids and a biomimetic peptide (TBP) on various rutile TiO_2 surfaces differentiated by hydroxylation states. MD simulations were conducted for three amino acids, which were arginine (R), lysine (K), and aspartate (D), on four distinct TiO_2 rutile (110) surfaces varying in density and type of surface hydroxyl groups. The binding free energy was estimated via metadynamics to map adsorption landscapes. Subsequent simulations included the TBP hexamer (RKLPGA) and dodecamer (RKLPGA₂) to evaluate sequence-level binding and correspondence to single-residue energetics. Arginine (R) consistently binds strongly across all TiO_2 surface chemistries, driven by favorable electrostatic and hydrogen-bond interactions. Lysine (K) and aspartate (D) displayed binding affinities that depend on surface hydroxylation: K prefers surfaces with exposed hydroxyls, whereas D binds more strongly on less hydroxylated surfaces. The TBP sequences exhibited binding free energies in close alignment with the dominant residues, whose binding strength correlates with sequences rich in R and D, producing stronger adhesion than K-only dominated motifs. The dodecamer shows cooperative adsorption effects, calling for consideration of context beyond individual residues. Variation in TiO_2 surface chemistry (neutral vs. hydroxylated, charged vs. uncharged) significantly impacts binding affinity. Representative peptide-binding conformations on hydroxylated TiO_2 obtained from enhanced MD are shown in Fig. 6.

Sadhukhan et al. [163] investigated Ti–Nb binary alloys (5–25 wt% Nb) using MD simulations and experimental validation to identify optimal compositions that infer low elastic modulus and high hardness, aligning mechanical behavior with human bone and reducing stress-shielding risk in orthopedic implants. Spherical indenter simulations were performed on single-crystal Ti–Nb systems across Nb composition range. Modified embedded atom method (MEAM) potentials were employed to estimate mechanical properties. Young's modulus, hardness, and dislocation activity (via Dislocation Extraction Algorithm) were quantified under nanoindentation to gauge deformation mechanisms. Ti–12 wt% Nb alloy was fabricated via vacuum arc remelting and characterized to cross-validate simulation predictions. The Ti–20 wt% Nb alloy exhibited a Young's modulus of 55.8 ± 2.7 GPa and hardness of 2.34 ± 0.05 GPa, closely resembling that of human cortical bone (~ 10 – 30 GPa), and thereby minimizing stress-shielding potential. Alloys with lower or higher Nb content (5 wt% or >20 wt%) showed higher stiffness, less bone compatibility, and different hardness profiles. Analysis via DXA revealed that dislocation density increased with indentation depth. The alloy with 20 wt% Nb showed appropriately distributed dislocations supporting balanced ductility and hardness, essential for implant durability under mechanical loading. Physical

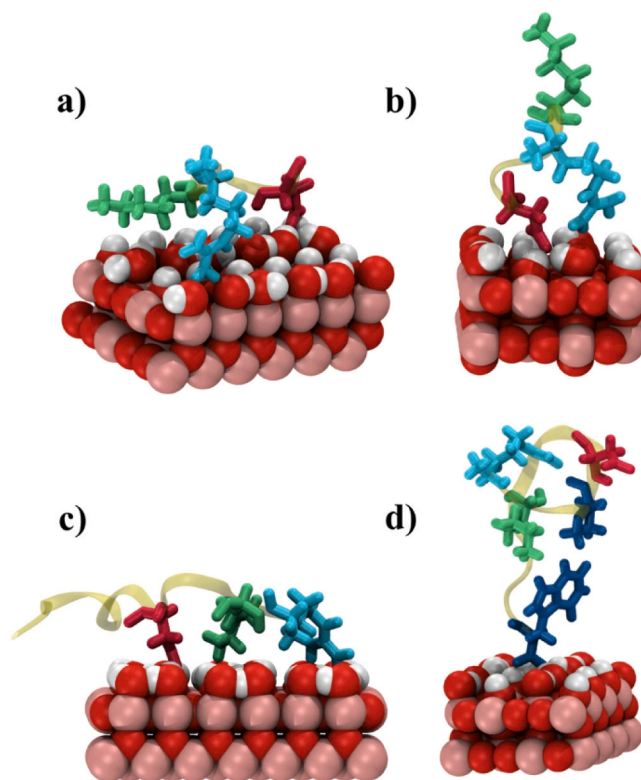


Fig. 6. Representative weighted peptide clusters of titanium-binding peptides on the hydroxylated (NH) TiO_2 surface obtained from metadynamics-enhanced molecular dynamics simulations. Backbone atoms are shown in transparent yellow; arginine, lysine, and aspartate residues are shown in blue, green, and red, respectively [162].

samples of Ti–12 wt% Nb alloy confirmed low elastic modulus behavior consistent with MD predictions. Wear testing indicated improved mechanical resilience compared with standard Ti–6Al–4 V and Co–Cr alloys, supporting suitability for orthopedic implantation. Zhang et al. [164] investigated the molecular mechanisms driving adsorption of bovine serum albumin (BSA) on anodized magnesium (AZ31) alloy surfaces, combining MD simulations with experimental insights to understand how surface charging and protein conformation govern initial biocompatibility interactions. BSA was simulated at physiological pH (~ 7.4), where it carries a net negative charge (BSA pI ~ 4.7), exposing $-\text{COO}^-$ moieties on its exterior. MD simulations modeled adsorption on two surface types, which were the bare AZ31 Mg alloy surface and micro-arc oxidized (MAO) coating that possesses a significant positive charge (pI ~ 12.4) at pH 7.4. Simulations tracked adsorption energetics, protein conformational change, ionic layering, and surface charge redistribution during interaction. Negatively charged BSA experienced strong electrostatic attraction toward the positively charged MAO-coated surface, leading to rapid, stable adsorption. On uncoated AZ31, initial interaction was weaker; corrosion-driven Mg^{2+} ion release contributes secondary adsorption sites. Upon adsorption on the MAO surface, BSA undergoes partial structural rearrangement, notably unfolding of domains that expose additional charged residues, enhancing surface contact area. Less conformational distortion was observed on the bare AZ31 surface, despite some bending near the interface. Pre-adsorption of BSA on MAO-coated surfaces slowed corrosion initiation by acting as a diffusion barrier, mitigating direct ionic exchange with chloride ions. In contrast, adsorption on bare Mg tended to coincide with localized Mg dissolution, promoting early-stage corrosion. Table 2 shows the atomistic modelling, surface oxidation/passivation, grain boundary dynamics/mechanical performance, and MD under stress/strain/corrosion simulations comparison among Ti, Mg, and Zn alloys.

Table 2
Overall MD-based comparison between Ti, Mg, and Zn alloys [165–167].

Alloy System	Atomistic Modelling	Surface Oxidation & Passivation	Grain Boundary Dynamics & Mechanical Performance	MD under Stress/Strain & Corrosion Simulations
Titanium (Ti)	Reactive MD (ReaxFF), DFT-based MD of Ti(0001) oxidation reveal oxygen adsorption, diffusion kinetics, compressive stress in oxide film, and oxide-substrate interface adhesion	Ti forms a stable native TiO ₂ layer within seconds; MD shows initial oxide growth self-limits due to built-up compressive stress in the oxide	Grain-boundary modeling is limited; MD of machining shows oxide fracture and defect nucleation at GBs under load. No extensive GB diffusion studies yet. MD simulations of oxide-coated surfaces on cutting highlight depth-dependent fracture behavior. Reactive MD rarely applied to Ti corrosion due to strong passivation barrier.	MD under constant stress reveals dislocation nucleation and GB-mediated creep behavior. ReaxFF-MD captures hydroxylation, Mg ²⁺ release, radical formation, and pitting initiation in corrosion models
Magnesium (Mg)	Machine-learning-assisted MD plus DFT elucidated Mg dissolution mechanisms, identifying MgOH as a solid-phase intermediate that grows inward and evolves into Mg(OH) ₂	In water Mg forms Mg(OH) ₂ , but this film rapidly dissolves in chloride environments. ML-MD captures MgOH / MgO film nucleation and growth dynamics (	MD creep studies of nanocrystalline Mg and Mg–Y show GB diffusion dominates deformation below ~423 K; Y addition reduces GB sliding energy, enhancing high-temperature creep resistance	MD of mechanical behavior and corrosion-fatigue coupling studied at the continuum scale for Zn stents. MD-level corrosion studies are sparse but imply GB segregation affects localized corrosion susceptibility.
Zinc (Zn)	Limited atomistic MD studies focused on solute segregation in Zn alloys and predictions of atomic-level mechanical behavior in Zn-based BMGs.	Zn forms Zn(OH) ₂ or basic carbonate passive films; grain refinement and Cu/Zn segregation aid uniform film formation and suppress galvanic corrosion zones	Few MD GB studies exist; simulations of Zn-based BMGs show atomic-level structure; continuum modeling used for fatigue-corrosion coupling in stents	

3. Protein and bio-interface modeling

Proteins are fundamental biological macromolecules whose interactions at material interfaces critically govern the performance and biocompatibility of biomaterials. The molecular-level understanding of protein adsorption, conformational dynamics, and bioactivity at interfaces is essential to the rational design of advanced biomaterials used in implants, tissue engineering scaffolds, and biosensor technologies. Protein–surface interactions constitute the molecular gateway at which biomaterials engage with biological systems, shaping processes such as adsorption, cell adhesion, immune recognition, and tissue integration. In the case of *in vivo*, cells typically interact with implant surfaces via intermediary adsorbed protein layers, most notably serum and extracellular-matrix proteins like albumin, fibronectin, and fibrinogen, making the underlying adsorption events central to biocompatibility and clinical outcome. Bio-interfaces, defined as the interface between a biological environment and synthetic or natural materials, present complex physicochemical and biochemical landscapes that modulate protein structure-function relationships and subsequent cellular responses. MD simulations, whether atomistic or coarse-grained, are uniquely capable of resolving these early-stage events. MD enables time-resolved observation of protein approach, surface docking, conformational rearrangement, and solvent structuring at the bio-interface. The theoretical foundation of protein adsorption involves a delicate interplay of thermodynamic driving forces, including van der Waals interactions, electrostatic forces, hydrogen bonding, and hydrophobic effects. The Derjaguin-Landau-Verwey-Overbeek (DLVO) theory, while initially developed for colloidal systems, provides a fundamental framework for understanding long-range protein-surface interactions, though it requires significant modifications to account for the complex geometry and charge distribution of proteins. Additionally, the Vroman effect, describing the sequential displacement of smaller, more mobile proteins by larger, higher-affinity proteins over time, adds temporal complexity to the adsorption process that can be elegantly captured through extended MD simulations. These atomic-level insights complement experimental proteomics and surface assays, which provide kinetic or compositional data yet cannot capture sub-nanometer events or transient states. Precise modeling of these phenomena enables the demonstration of mechanisms driving protein adsorption/desorption kinetics, conformational rearrangements upon surface binding, and specific binding interactions with cell membrane receptors or extracellular matrix components, with realistic ionic strength and pH conditions,

providing more accurate representations of physiological environments. Besides, the development of reactive force fields allows for the simulation of covalent bond formation between proteins and chemically active surfaces, opening new avenues for understanding bioconjugation and surface-mediated protein cross-linking. The predictive capability of MD simulations in protein bio-interface modeling has significant implications for biomaterial design and optimization. By establishing structure-property relationships at the molecular level, these simulations can guide the development of surfaces with tailored protein adsorption characteristics. For instance, understanding the molecular basis of protein resistance on polyethylene glycol-modified surfaces or the selective adsorption of specific proteins on engineered peptide sequences can inform the design of next-generation biomedical devices. Furthermore, the ability to predict protein denaturation and aggregation at material interfaces is crucial for developing strategies to minimize adverse biological responses such as thrombosis and inflammation. The integration of MD simulations with experimental techniques such as quartz crystal microbalance with dissipation monitoring (QCM-D), surface plasmon resonance (SPR), and atomic force microscopy (AFM) provides powerful validation of computational predictions and enables the development of more accurate models. This synergistic approach between computation and experiment is essential for advancing our understanding of protein-biointerface phenomena and translating molecular-level insights into practical biomaterial applications.

3.1. Protein adsorption

All-atom MD simulations offer atomic-level insight into the adsorption mechanisms of fibronectin and albumin by explicitly modeling every atom of the protein and surrounding solvent. These simulations capture crucial details such as protein unfolding/refolding events, secondary structure rearrangements, and specific sidechain-surface interactions under physiological conditions. For instance, all-atom MD has revealed how fibronectin's cell-binding domains adapt their conformations upon adsorption, impacting biological recognition, while albumin exhibits conformational plasticity that can lead to partial unfolding or altered surface coverage states. For longer adsorption times, multi-protein competition, and mesoscale organization, the coarse-grained strategies introduced earlier, particularly MARTINI-type models, become especially useful. In this context, CG simulations allow describing that fibronectin domain rearrangement, competitive albumin displacement, and collective adsorption behavior at biomaterial

interfaces. For fibronectin, MARTINI-based simulations have been particularly valuable for studying the protein's behavior at interfaces over microsecond timescales, revealing the kinetics of domain unfolding and the cooperative effects of multiple protein molecules during adsorption. These studies have demonstrated that fibronectin exhibits complex adsorption kinetics, with an initial rapid binding phase followed by slower conformational rearrangements that can extend over microseconds. On the other hand, the coarse-grained approach has proven especially powerful for investigating albumin's role in competitive adsorption scenarios, where multiple protein species compete for surface binding sites. MARTINI simulations have exhibited the molecular basis of the Vroman effect, showing how the relatively weak surface affinity of albumin allows it to be displaced by proteins with higher binding energies, such as fibrinogen or immunoglobulins. These studies have also revealed the importance of protein concentration gradients and surface heterogeneity in determining the final composition of adsorbed protein layers. These approaches simplify protein and surface representations by grouping atoms into larger beads, enabling simulations over longer timescales and larger biomolecular assemblies while preserving essential interaction motifs. Coarse-grained MD has been effectively employed to study fibronectin network formation on polymeric hydrogels and albumin layering on metallic implants, providing complementary insights into adsorption kinetics and mesoscale aggregation processes. The physicochemical nature of biomaterial surfaces profoundly influences protein adsorption. Glass and ceramic surfaces, often characterized by hydrophilic, charged, or hydroxylated moieties, engage in electrostatic and hydrogen-bonding interactions that stabilize adsorbed fibronectin and albumin configurations.

Fibronectin, a large glycoprotein (molecular weight ~ 440 kDa) composed of two nearly identical subunits connected by disulfide bonds, plays a pivotal role in cell adhesion, migration, and tissue repair processes. Its modular structure consists of repeating type I, II, and III fibronectin domains, with the type III domains containing the renowned RGD (Arg-Gly-Asp) sequence that serves as a primary recognition site for integrin receptors. The protein's inherent flexibility and multiple functional domains make it particularly susceptible to conformational changes upon surface adsorption, which can either enhance or diminish its biological activity depending on the nature of the substrate and adsorption conditions. In contrast, human serum albumin (HSA), the most abundant protein in blood plasma ($\sim 60\%$ of total protein content), presents a more compact globular structure with a molecular weight of approximately 66.5 kDa. Comprising three homologous domains (I, II, and III), each containing two subdomains, albumin exhibits remarkable conformational stability under physiological conditions. Its primary functions include maintaining oncotic pressure, serving as a transport vehicle for various ligands, and acting as an antioxidant. The protein's tendency to undergo the Vroman effect, whereby it initially adsorbs rapidly but is subsequently displaced by higher-affinity proteins, makes it a crucial component in understanding the dynamics of protein layer formation on biomaterial surfaces.

The interaction of fibronectin and albumin with different classes of biomaterial surfaces reveals the fundamental principles governing protein adsorption and highlights the critical role of surface chemistry, topography, and electronic properties in determining biological responses. Each surface category presents unique challenges and opportunities for controlling protein adsorption behavior through molecular-level design strategies. Glass and ceramic surfaces, characterized by their predominantly oxide compositions and hydrophilic nature, present complex interaction landscapes for protein adsorption. Silica-based glasses, the most extensively studied systems, offer hydroxyl-rich surfaces that can engage in extensive hydrogen bonding with protein residues. MD simulations of fibronectin adsorption onto silica surfaces have revealed that the protein's initial binding is mediated primarily by basic amino acid residues, particularly lysine and arginine, which form strong electrostatic interactions with the negatively charged silicate groups at physiological pH. The conformational behavior of fibronectin on glass

surfaces is particularly intriguing, as the protein tends to adopt extended conformations that maximize surface contact while preserving the accessibility of key functional domains. Simulations have shown that the type III domains, which contain the cell-binding RGD sequence, can maintain their structural integrity on appropriately designed glass surfaces, suggesting that controlled surface chemistry can preserve biological activity. The flexibility of the fibronectin molecule allows it to adapt to surface irregularities and heterogeneities, leading to higher binding affinities compared to more rigid proteins. Albumin interaction with glass surfaces follows distinctly different patterns, reflecting the protein's compact globular structure and high conformational stability. The protein tends to bind through a limited number of contact points, primarily involving surface-exposed hydrophobic patches and charged residues. Importantly, albumin often retains much of its native structure upon adsorption to glass surfaces, which explains its relatively weak binding and susceptibility to displacement by other proteins. This characteristic makes albumin-coated glass surfaces potentially valuable for applications requiring minimal protein fouling. Ceramic surfaces, including alumina, zirconia, and hydroxyapatite, present unique adsorption environments due to their distinct surface chemistry and crystallographic structures. Hydroxyapatite, the mineral component of bone, has received particular attention due to its biocompatibility and osteoconductive properties. MD studies have shown that fibronectin exhibits strong affinity for hydroxyapatite surfaces through interactions with both calcium and phosphate sites. The protein's modular structure allows different domains to engage with different surface features simultaneously, leading to highly stable adsorption complexes. The adsorption of albumin onto ceramic surfaces is strongly influenced by the surface's isoelectric point and the resulting electrostatic interactions. On positively charged ceramic surfaces such as alumina at physiological pH, albumin binding is enhanced due to favorable electrostatic interactions with the protein's net negative charge. However, the protein's tendency to maintain its compact structure often results in lower surface coverage compared to more flexible proteins like fibronectin. Polymer surfaces offer unparalleled versatility in controlling protein adsorption through molecular design of surface chemistry, chain architecture, and physical properties. The interaction of fibronectin and albumin with polymeric biomaterials encompasses a broad spectrum of behaviors, from strong adsorption to near-complete protein resistance, depending on the polymer's chemical composition and surface modification strategies. Hydrophobic polymer surfaces, such as those based on polystyrene or polyethylene, typically promote strong protein adsorption through hydrophobic interactions and van der Waals forces. MD simulations of fibronectin adsorption onto hydrophobic surfaces have revealed extensive protein unfolding and spreading, driven by the thermodynamic favorability of burying hydrophobic amino acid residues in contact with the surface. While this leads to high binding affinity, extensive denaturation often results in loss of biological activity, as critical functional domains become inaccessible or structurally compromised. Albumin behavior on hydrophobic polymer surfaces is somewhat different, as the protein's structure is more resistant to unfolding. However, the protein still undergoes significant conformational changes, particularly in domains I and II, which contain numerous hydrophobic residues. The binding is often accompanied by partial unfolding that increases the contact area with the surface, leading to stronger adsorption than typically observed on hydrophilic surfaces. Hydrophilic polymer surfaces, particularly those based on poly(ethylene glycol) (PEG) or other water-soluble polymers, have been extensively studied for their protein-resistant properties. The molecular basis of protein resistance involves the creation of a hydrated layer at the polymer-water interface that presents an energetic barrier to protein adsorption. MD simulations have shown that both fibronectin and albumin exhibit reduced binding affinities to well-hydrated PEG surfaces, with the proteins often maintaining greater distances from the surface and preserving more of their native structures. The development of stimuli-responsive polymer surfaces has opened new possibilities for

controlling protein adsorption behavior. Thermoresponsive polymers such as poly (N-isopropylacrylamide) (PNIPAM) undergo conformational transitions in response to temperature changes, allowing for reversible control of protein binding. Simulations have demonstrated that fibronectin can be captured at temperatures above the polymer's lower critical solution temperature and subsequently released upon cooling, providing a mechanism for controlled protein delivery. Metal surfaces present unique environments for protein adsorption due to their electronic properties, oxidation behavior, and ability to form coordination bonds with protein residues. Titanium and its alloys, widely used in orthopedic and dental implants, have been the subject of extensive computational studies aimed at understanding their excellent biocompatibility and osseointegration properties. The interaction of fibronectin with titanium surfaces involves multiple binding mechanisms, including electrostatic interactions with the oxide layer, coordination bonding through histidine and methionine residues, and van der Waals interactions. MD simulations have revealed that fibronectin can form stable complexes with titanium oxide surfaces while maintaining much of its structural integrity, particularly in the cell-binding domains. The protein's ability to bridge between the material surface and cellular integrins makes it a crucial mediator of osseointegration processes. Albumin's interaction with titanium surfaces is characterized by relatively weak binding, which contributes to the metal's biocompatibility by reducing non-specific protein fouling. The protein tends to bind through coordination interactions involving carboxylate groups from aspartic and glutamic acid residues, but the binding is generally reversible and does not lead to significant structural perturbation. Stainless steel surfaces, commonly used in temporary implants and surgical instruments, present more complex interaction environments due to their multi-component nature and variable oxide layer composition. The presence of chromium, nickel, and iron oxides creates a heterogeneous surface that can interact with proteins through multiple mechanisms simultaneously. Fibronectin adsorption onto stainless steel surfaces often results in heterogeneous binding modes, with different protein domains interacting preferentially with specific surface components. Gold surfaces, while not commonly used in bulk implant applications, have provided valuable model systems for fundamental studies of protein adsorption mechanisms. The chemical inertness of gold allows for precise control of surface chemistry through self-assembled monolayer formation, enabling systematic investigation of structure-activity relationships. MD studies of fibronectin and albumin adsorption onto functionalized gold surfaces have provided crucial insights into the role of surface charge, hydrophobicity, and chemical functionality in determining protein binding behavior. The influence of surface topography on protein adsorption has emerged as an important consideration in metal surface design. Rough or nanostructured metal surfaces can significantly enhance protein binding through increased surface area and the creation of binding sites with different local environments. Simulations have shown that fibronectin can adapt to surface irregularities by adopting extended conformations that maximize contact with topographical features, while albumin's more rigid structure limits its ability to conform to surface topology.

Liamas et al. [168] employed fully atomistic MD simulations to investigate how surface chemistry influences the adsorption behavior of the fibronectin type-III fragment FNIII 8–10 onto model self-assembled monolayers (SAMs) terminated with different functional groups (methyl, hydroxyl, amine, carboxyl). The work targeted anchoring kinetics, binding specificity, and exposure of the cell-binding domain (RGD/PHSRN), thereby informing biomaterial design for improved cell adhesion. Simulations used NAMD 2.6 with CHARMM27 force field, modeling FNIII 8–10 (~274 residues, ~4184 atoms) in explicit TIP3P water, Na^+/Cl^- ions at physiological ionic strength (~0.05 M), with a total of ~100,000 atoms per system. Adsorption trajectories ran for up to 100 ns on SAMs with methyl (neutral), hydroxyl (neutral/polar), amine (positive), and carboxyl (negative) end groups. Multiple initial orientations (FNa–FNd) were tested, with protein–surface distances

varied to probe anchoring orientation and kinetics. Charged surfaces facilitated rapid, site-specific adsorption, propelled by long-range electrostatic attraction. On these surfaces, FNIII 8–10 aligns its dipole moment with the electric field and anchors via residues such as Ser1261, Asp1263, Glu1312, and Thr1509, all located in the FNIII 8 or FNIII 10 modules. Anchoring occurs within ~3–5 ns for amine and silica-mimetic surfaces, as shown in Fig. 7A. A strongly bound hydration and ion layer near charged surfaces reduced adsorption efficiency compared to modestly charged ones. Methyl-terminated SAMs led to slower, non-specific adsorption dominated by van der Waals interactions. Anchoring primarily involved polar uncharged residues like Thr and Asn, often in loop regions. FNIII 8–10 retained high mobility on hydrophobic SAMs and exhibited multiple adsorption conformations ("head-on", "beta-on", "side-on") depending on initial orientation. Upon adsorption to positively charged surfaces, the fragment oriented such that the RGD/PHSRN region remained exposed, preserving its integrin-binding activity as demonstrated in Fig. 7B. Adsorption on negatively charged surfaces occluded that region. On neutral surfaces, fragment orientation was ambiguous, and cell-binding region availability was unclear.

Kubiak et al. [169] studied the atomistic adsorption mechanisms of the FN(III)9 fibronectin module as the integrin-binding synergy region onto three prototypical solid surfaces at physiological pH. Full all-atom MD simulations were conducted to compare the effects of surface charge, electrostatic field, and chemical composition on adsorption kinetics, orientation, and conformational behavior. FNIII9 (~40 Å × 30 Å ellipsoidal β -sandwich fold) with net $-1e$ charge at pH 7, containing seven Arg/Lys and nine Glu/Asp residues located primarily in flexible loop regions. The models were selected as the Mica model, a negatively charged ionic surface without a long-range electrostatic field (periodic boundary conditions suppress the field), the Silica (SiO_2) model, as negatively charged with a realistic long-range electric field, and the Au (111) model, an uncharged, non-polar surface. Adsorption monitored by tracking center-of-mass approach, dipole orientation change, anchor residues docking (<6 Å), and mobility on the surface. In the results, the silica model demonstrated rapid and highly specific adsorption driven by long-range electrostatics. Protein dipole reoriented perpendicular to the surface, and anchoring residues with opposite charges (Lys85 and Arg56) mediate binding within ~5–30 ns, leading to immobilization once multiple residues contact. The Mica model exhibited that, despite being negatively charged, adsorption occurred without needing a long-range field. Electrostatic attraction still enables docking, though orientation may differ due to dipole alignment. Anchorage proceeded via similar positively charged residues, followed by mobility reduction once 2–3 anchors establish contact. The Au (111) surface showed adsorption as less frequent, slower, and non-specific; multiple regions of the protein could make contact. Diffusion on the surface was prolonged, and orientations varied. Despite low directional specificity, adsorption occurs without major conformational changes. On charged surfaces, FNIII9 aligns its dipole for favorable interaction and anchors primarily via positively charged residues in loops (Lys85, Arg56, Arg51, Arg61, and Ser84). Immobilization occurred when multiple anchor residues docked at <6 Å. Protein mobility was high initially but decreased upon anchoring. Final orientation may have been end-on, side-on, or intermediate, depending on anchoring residues and surface geometry. Hao et al. [170] investigated at the molecular scale how surface chemical functionality modulates fibronectin adsorption and consequent cellular responses. The study integrated experimental characterization as SPR, XPS, and cell assays, with Monte Carlo (MC) and all-atom MD simulations to provide mechanistic insight into fibronectin orientation, conformation, and bioactivity on tailor-made self-assembled monolayers (SAMs). Four SAMs were fabricated on gold as $-\text{CH}_3$, $-\text{NH}_2$, $-\text{COOH}$, and $-\text{OH}$ terminated surfaces. Fibronectin adsorption was quantified via surface plasmon resonance to derive kinetics and binding capacity. XPS confirmed fibronectin presence, while hBMSC assays assessed biological responses as adhesion, proliferation, integrin β_1

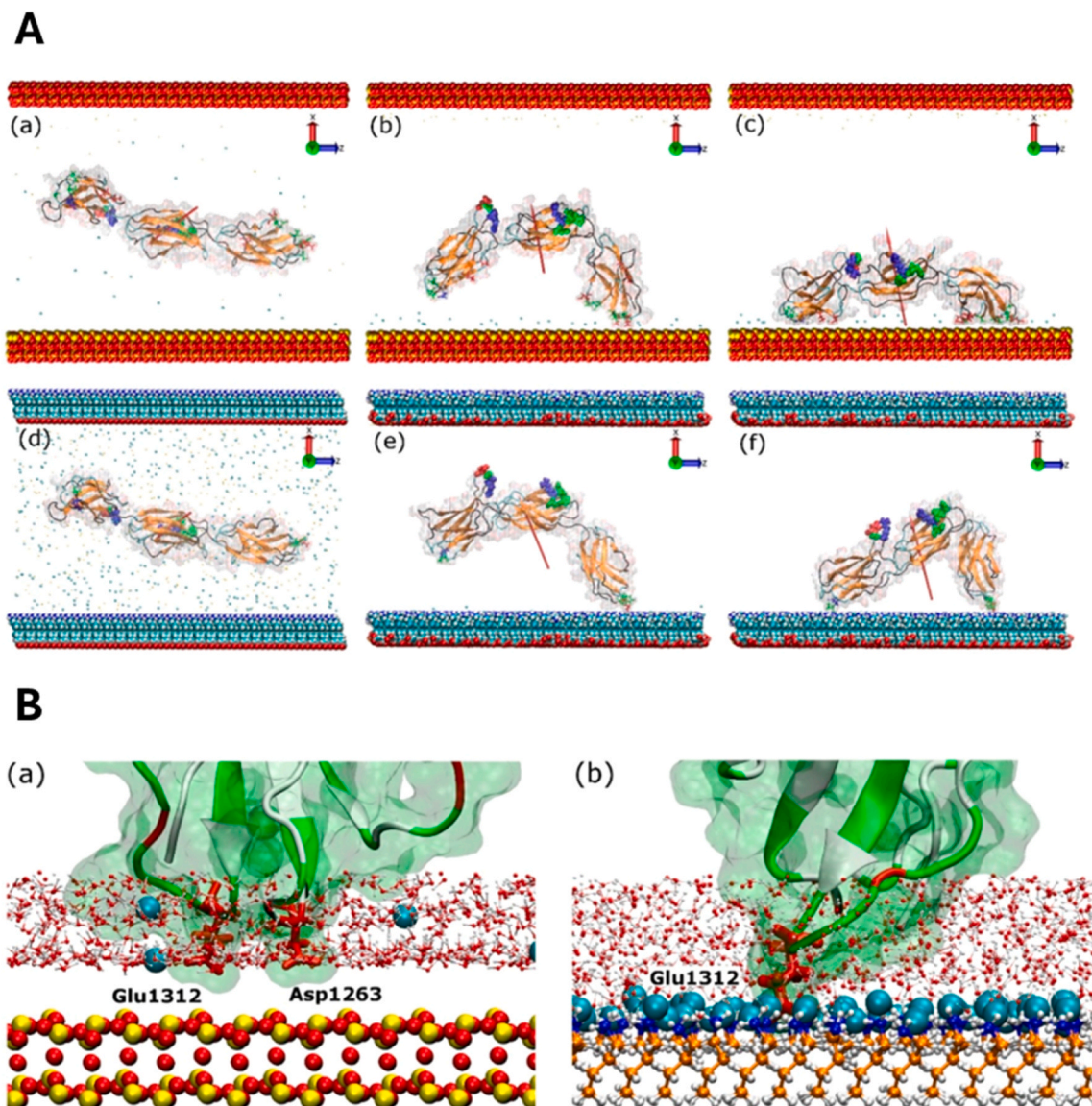


Fig. 7. (A) FNIH8–10 adsorption action on the model as the silica top and the amine the bottom, at 0 ns (a and d), anchoring event (b and e), end of simulation (c and f). (B) Demonstration of the anchoring site of the FNIH8 module on the (a) silica, (b) amine surfaces [168].

expression, and osteogenic differentiation. Monte Carlo orientation optimization preceded long (300 ns) MD simulations of the Fn-III₇₋₁₀ fragment on each SAM. Simulations examined orientation distributions, binding energetics (electrostatic, vdW), hydrogen bonding/salt-bridges, and solvent-accessible surface area (SASA) of RGD/PHSRN motifs. SAM-CH₃ displayed the highest fibronectin adsorption mass via hydrophobic interactions, but the poorest bioactivity due to the low SASA of RGD/PHSRN motifs and subtle conformational deformation. SAM-NH₂ and SAM-COOH enabled strong adsorption through combined van der Waals, electrostatic interactions, hydrogen bonding, and salt bridge formation. Fibronectin on SAM-NH₂ showed the greatest cell activity, driven by high exposure of RGD/PHSRN motifs; SAM-COOH also supported bioactivity, likely via activation of alternative binding motifs. SAM-OH discouraged fibronectin adsorption due to a persistent water film; yet the small amount of adsorbed fibronectin remained bioactive due to high exposure of PHSRN and substantial conformational flexibility. MC-optimized orientations (O₁–O₄) were used to initialize MD; after equilibration, fibronectin adsorbed in “side-on” or “end-on” conformations depending on surface chemistry. On SAM-NH₂, Fn-III₇₋₁₀ bound tightly via modules 7–8 while modules 9–10 remained

exposed away from the surface, preserving cell-binding function. On SAM-COOH, fibronectin assumed “end-on” or tilted orientations with moderate SASA coverage, enabling partial functional exposure. On SAM-CH₃ and SAM-OH, orientation was more variable, and binding was less specific.

Castellanos et al. [171], employed all-atom MD simulations over ~70 ns to assess the structural and dynamical role of native disulfide bonds in human serum albumin by comparing simulations in the presence and absence of each or all disulfide linkages. Full-length HSA (~585 residues, ~67 kDa) was simulated in explicit water with physiological ions using standard MD force fields for 70 ns. Multiple systems were generated as Native—all 17 intramolecular disulfide bonds intact and reduced—variants with either specific disulfide bonds removed (Cys168, Cys177, or Cys278–Cys289) or all bonds removed. Structural analyses included root-mean-square deviation (RMSD), secondary structure content, hydrogen-bond patterns, B-factors (atomic fluctuation), and dynamic cross-correlation mapping. Removal of all disulfide bonds led to significant conformational deviation after ~35 ns, with RMSD rising from ~2 Å (native) to ~4.5 Å in fully reduced structures, indicating loss of tertiary integrity. Selective removal identified specific

disulfide bridges, such as Cys168–Cys177 and Cys278–Cys289, as critical for stabilizing secondary and tertiary structure, while others (Cys200, Cys246, Cys461, and Cys477) had little local effect. Loss of key bonds disrupted specific inter-residue hydrogen bonds and correlated motions, particularly among adjacent domains, increasing overall flexibility. Primary ligand-binding sites showed partial structural perturbation when certain bonds were removed, implying potential functional impact. In the native simulation, the sole free cysteine became transiently solvent-exposed, information not evident from crystal structures, indicating a likely site for intermolecular disulfide formation and aggregation. Kooravand et al. [172] investigated, in combination with molecular docking, all-atom MD, and multi-spectroscopic methods to illuminate the binding mechanism, structural effects, and dynamic behavior of malachite green oxalate (MGO) interacting with human serum albumin. Molecular docking was performed to identify potential binding sites of MGO on HSA and estimate binding free energy. All-atom MD simulations were run to assess the stability of the docked complex, monitor changes in protein conformation, compactness, and flexibility. Spectroscopic experiments, such as fluorescence quenching, circular dichroism, and UV-Vis, validated and complemented computational findings, characterizing quenching behavior and secondary structural changes. MGO was predicted to bind specifically to HSA with a docking free energy of approximately 32.93 kJ/mol, indicating spontaneous interaction. MD simulations confirmed stable binding of MGO within the HSA binding pocket throughout the trajectory. HSA has undergone increased flexibility and a measurable decrease in compaction, suggesting loosening of tertiary structure. Fluorescence quenching studies showed that at an MGO concentration of 0.0005 mM, HSA intrinsic fluorescence was quenched by ~16.9%, supporting a static binding mechanism. Circular dichroism analysis revealed a gain in α -helical content and a loss in β -sheet content, reflecting conformational rearrangements upon ligand binding. Abdelaziz et al. [173], explored the binding interaction between the antidiabetic agent dapagliflozin (DAPA) and bovine serum albumin, using a suite of spectroscopic techniques, including fluorescence titration, UV-Vis absorption, synchronous fluorescence, thermodynamic analysis, molecular docking, and MD simulation to clarify the nature, mechanism, and dynamics of complex formation. Fluorescence spectroscopy at multiple temperatures assessed quenching effects and allowed calculation of binding constants and quenching mechanisms. UV-Visible absorption and fluorescence resonance energy transfer (FRET) corroborated the spectroscopic findings and helped estimate donor–acceptor distances. Site-specific displacement assays using known BSA marker ligands defined the specific binding site (Sudlow's site I). Thermodynamic parameters (ΔH , ΔS , ΔG) were determined to deduce the driving forces behind the interaction. Molecular docking and MD simulation provided atomistic insights into binding orientation, contact residues, structural stability, and binding energy landscapes. DAPA induced a combination of static and dynamic quenching of BSA fluorescence, indicating both complex formation and collisional deactivation. Analysis across different temperatures yielded a 1:1 binding stoichiometry between BSA and DAPA. ΔG was negative, which was a spontaneous binding, while positive ΔH and ΔS values point to hydrophobic forces as the dominant contributor to binding affinity. Competitive displacement experiments combined with docking place DAPA at Sudlow's site I (sub-domain IIA), near Trp213, consistent with synchronous fluorescence shifts that implicate proximity to the tryptophan residue. Docking confirmed a favorable orientation of DAPA within site I, aligning hydrophobic moieties into the cavity and supporting stable interactions during MD trajectories. MD simulations further confirmed the dynamic stability of the DAPA–BSA complex, with persistent contacts and minimal structural distortion over time.

3.2. Membrane and cell-level modeling

The intersection of biomaterials science and cellular biology represents one of the most challenging frontiers in MD simulations.

Understanding how synthetic materials interact with biological membranes and cellular structures requires sophisticated computational approaches that can bridge multiple length and time scales while maintaining atomistic precision. The complexity of membrane and cell-level modeling stems from the hierarchical organization of biological systems, where molecular-scale interactions between biomaterials and membrane components cascade into cellular-level phenomena such as adhesion, activation, and biocompatibility responses. The fundamental challenge in modeling biomaterial-membrane interactions lies in accurately representing the complex, dynamic nature of lipid bilayers while simultaneously capturing the specific chemical and physical properties of synthetic surfaces [174]. Lipid bilayers, composed primarily of phospholipids, cholesterol, and membrane proteins, exhibit remarkable structural diversity and dynamic behavior that varies significantly with composition, temperature, and environmental conditions. When biomaterials come into contact with these membranes, a cascade of molecular-level events occurs that ultimately determines biocompatibility, cellular uptake, and biological response. At the membrane scale, atomistic simulations resolve local contact chemistry and short-range bilayer perturbations, whereas the coarse-grained framework introduced earlier is typically required to capture slower membrane remodeling, larger curvature changes, and longer-timescale biomaterial–bilayer interactions. The simulation of surface-induced membrane perturbations requires careful consideration of several key phenomena. First, the approach of a biomaterial surface to a lipid bilayer typically induces local membrane deformation, characterized by changes in lipid tilt angles, area per lipid, and membrane thickness. These structural perturbations can propagate laterally through the membrane, creating long-range effects that influence membrane protein function and lipid domain organization. Advanced sampling techniques, such as umbrella sampling and metadynamics, have proven essential for calculating the free energy landscapes associated with these deformation processes [175]. Second, the chemical nature of the biomaterial surface profoundly influences the mode and extent of membrane interaction. Hydrophilic surfaces, such as oxidized silicon or titanium dioxide, tend to interact primarily through hydrogen bonding and electrostatic interactions with lipid headgroups, often inducing the formation of water bridges that mediate surface-membrane adhesion. In contrast, hydrophobic surfaces, including many polymer materials and carbon-based nanomaterials, can penetrate the hydrophobic core of the bilayer, resulting in significant membrane disruption and potentially leading to pore formation or membrane rupture.

3.2.1. Bacterial adhesion

The computational modeling of bacterial adhesion to biomaterial surfaces represents a critical challenge in developing infection-resistant medical devices and implants. Bacterial adhesion is a complex, multi-step process that begins with initial, reversible attachment mediated by physical forces, followed by irreversible adhesion involving specific molecular interactions, and ultimately culminating in biofilm formation. MD simulations have emerged as powerful tools for understanding these processes at the molecular level, providing insights that complement experimental studies and guide the design of antibacterial biomaterials. The initial stages of bacterial adhesion are governed by a combination of van der Waals forces, electrostatic interactions, and hydrophobic effects, collectively described by DLVO (Derjaguin-Landau-Verwey-Overbeek) theory [176]. However, the complex surface topology and chemical heterogeneity of both bacterial cells and biomaterial surfaces necessitate computational approaches that go beyond continuum theories. MD simulations enable explicit modeling of surface roughness, chemical functionality, and molecular-scale interactions that determine adhesion strength and specificity. Bacterial cell surfaces present unique challenges for MD modeling due to their complex, multi-layered architecture. Gram-positive bacteria possess thick peptidoglycan cell walls decorated with teichoic acids and surface proteins, while Gram-negative bacteria feature outer membranes containing lipopolysaccharides and

porins [177]. These surface structures exhibit significant conformational flexibility and can undergo dynamic reorganization in response to environmental conditions, including contact with biomaterial surfaces. Recent advances in coarse-grained modeling have enabled simulations of complete bacterial cell envelopes, capturing the interplay between different structural components during surface adhesion. These models typically represent peptidoglycan as a cross-linked polymer network, lipid membranes using established coarse-grained force fields, and surface proteins as flexible, multi-domain structures. The integration of these components into unified models has revealed how mechanical deformation of the cell envelope influences adhesion strength and can trigger adaptive responses in adherent bacteria [178]. The role of bacterial surface appendages, particularly pili and fimbriae, in mediating adhesion has been extensively studied using MD simulations. These proteinaceous structures act as molecular springs, capable of extending several micrometers from the bacterial surface and binding to specific receptors or surface features on biomaterials. Type IV pili, found in many pathogenic bacteria, exhibit remarkable mechanical properties, including the ability to generate forces exceeding 100 piconewtons during retraction [179]. MD simulations of pilus-mediated adhesion have revealed the molecular basis of this extraordinary mechanical strength. The helical structure of pilin subunits creates a supercoiled arrangement that can undergo reversible unfolding under tension, providing a mechanism for energy dissipation during mechanical loading. Steered MD simulations have characterized the force-extension relationships of individual pili, revealed multiple unfolding intermediates, and provided quantitative data for mesoscale models of bacterial adhesion.

An additional complexity in clinically relevant bacterial adhesion arises from the rapid formation of conditioning films composed of host proteins, polysaccharides, and ions on biomaterial surfaces before bacterial contact. Consequently, the relevant interface is often not the pristine material itself, but a dynamically restructured and hydrated interphase. MD simulations are particularly useful in resolving this pre-adhesion environment because they can quantify interfacial water residence, ion-mediated bridging, local electrostatic heterogeneity, and the free-energy landscape of bacterial surface motifs approaching the conditioned surface. These atomistic descriptors help explain why surfaces with similar bulk chemistry may display markedly different colonization tendencies once exposed to serum, saliva, or wound fluid, and they provide mechanistic guidance for designing anti-adhesive coatings that stabilize hydration layers or inhibit specific anchoring interactions.

Beyond the initial attachment event, MD also provides insight into the molecular transition from reversible adhesion to mechanically stabilized residence at the surface. Once adhesins, pili, extracellular polymers, and multivalent cations act cooperatively, the contact zone can evolve into an interfacial network with increased rupture force and longer binding lifetime, thereby promoting early microcolony formation. In this context, atomistic and steered MD can quantify adhesion energies, force-extension responses, and detachment pathways of specific surface-bound complexes, while coarse-grained models can extend these observations to larger bacterial envelope deformations and collective adhesion behavior. This multiscale perspective is especially valuable for biomaterial design because it shifts the focus from merely reducing initial attraction to interrupting the specific molecular events that stabilize colonization and precede biofilm development.

3.2.2. Cytocompatibility

The development of cytocompatibility models represents another critical application of MD in biomaterials science. Cytocompatibility encompasses the ability of materials to interact with mammalian cells without causing adverse biological responses, including cytotoxicity, inflammation, or unwanted cellular activation [180]. Unlike bacterial adhesion, which is generally undesirable, cytocompatibility requires a delicate balance that promotes beneficial cellular interactions while avoiding harmful responses. Modern approaches to cytocompatibility

modeling integrate MD simulations of material-cell membrane interactions with phenomenological models of cellular signaling pathways. This multiscale approach recognizes that molecular-level events at the material-cell interface can trigger cascades of intracellular processes that ultimately determine cellular fate and function. For example, the interaction of biomaterial surfaces with integrin receptors can activate focal adhesion kinase pathways, leading to changes in cell morphology, proliferation, and differentiation.

The simulation of protein adsorption to biomaterial surfaces represents a crucial component of cytocompatibility modeling. When biomaterials are introduced into biological environments, they are rapidly coated with a layer of adsorbed proteins that mediates subsequent cellular interactions. The composition and conformation of this protein layer, often termed the “protein corona”, determines whether cells recognize the surface as foreign or compatible [181]. MD simulations have demonstrated the mechanisms of protein adsorption, revealing how surface chemistry, topology, and hydrophobicity influence protein binding kinetics and conformational stability. Fibronectin, a key extracellular matrix protein, has been extensively studied as a model system for understanding how protein adsorption affects cell adhesion. Simulations have shown that fibronectin undergoes significant conformational changes upon adsorption to certain surfaces, exposing cryptic binding sites that enhance integrin-mediated cell adhesion [182]. The modeling of inflammatory responses to biomaterials represents an emerging frontier in cytocompatibility simulation. MD simulations have begun to elucidate how biomaterial surfaces can mimic pathogen-associated molecular patterns (PAMPs) or damage-associated molecular patterns (DAMPs), triggering inflammatory cascades. The development of predictive models for long-term biocompatibility represents a significant challenge in biomaterials science. While MD simulations excel at capturing short-term molecular interactions, biocompatibility is ultimately determined by processes occurring over weeks, months, or years. These multiscale approaches typically employ MD to parameterize coarse-grained models of cellular behavior, which can then be integrated into tissue-scale simulations of implant integration, wound healing, or device performance.

Building on the protein-adsorption mechanisms discussed in the previous section, cytocompatibility depends critically on whether the adsorbed interfacial layer presents receptor-accessible motifs in geometries compatible with productive cell binding. MD simulations are particularly valuable in this context because they can quantify descriptors such as solvent-accessible exposure of RGD-like motifs, orientation of adhesive domains, interfacial hydration, local charge density, and the nanoscale spacing between surface-bound ligands. These quantities influence whether cell membranes remain separated by a protective hydration barrier or establish stronger receptor-mediated contacts that nucleate focal adhesions. In this way, MD provides a mechanistic bridge between surface chemistry and downstream cellular observables such as spreading, morphology, proliferation, and differentiation.

Cytocompatibility is also affected by the degree to which a material perturbs membrane structure and inflammatory signaling. Simulations can resolve lipid extraction, local membrane thinning, water defect formation, ion accumulation, and transient pore-like events, thereby distinguishing benign interfacial contact from sub-lethal membrane stress. Likewise, the conformational fate of adsorbed proteins is highly relevant: surfaces that preserve near-native fibronectin or albumin structures are more likely to support constructive cell interactions, whereas surfaces that induce substantial unfolding may expose non-native epitopes or damage-associated patterns that bias inflammatory activation. These considerations are particularly important for degradable biomaterials, whose surface chemistry continuously evolves due to corrosion, hydrolysis, or ion release.

From a predictive standpoint, the major challenge is to connect the short-timescale molecular events accessible to MD with long-term cellular outcomes. A promising strategy is to use MD-derived

descriptors, binding free energies, motif exposure, hydration structure, membrane perturbation metrics, and local ion distributions as inputs for coarse-grained, kinetic, or tissue-scale models of adhesion, macrophage activation, wound healing, and implant integration. Such multiscale workflows do not replace in vitro or in vivo studies, but they can substantially improve early-stage biomaterial screening by identifying surfaces that are unlikely to trigger cytotoxicity or chronic inflammatory responses.

3.2.3. Conformational aspects

Zhang et al. [183], investigated the combination of atomistic MD with multiscale continuum concepts to elucidate the dynamics of pore closure (sealing) in lipid bilayer membranes. The study was conducted to demonstrate how bilayer structure and mechanics respond during the critical resealing of pores formed under mechanical or chemical perturbation. The MD systems modeled POPC lipid bilayers containing pre-formed pores of varied diameters (5 Å to 120 Å), using LAMMPS-compatible force fields. Simulations tracked the time evolution of pore radius, bilayer deformation, lipid tilt, and order parameters, as well as mechanical stress fields during spontaneous closing sequences. Results were compared against continuum mechanics models to validate MD-derived metrics such as edge line tension and membrane bending energy. In the results, smaller pores ($< \sim 20$ Å) closed rapidly (< 10 ns) via lipid migration and bilayer lap-over. Larger pores displayed slower resealing dynamics, with intermediate stages where bilayer edges bent inward before closure. Closure profiles exhibited a power-law decay in pore radius over time, qualitatively aligning with predictions from continuum mechanics. Lipids at pore edges exhibited elevated tilt angles, lower nematic order, and enhanced lateral mobility. These metrics gradually relaxed upon closure. The rim region displayed localized curvature and lateral sliding, consistent with edge-line tension minimization. The study derived line tension values ($\sim 10^{-10}$ N) and bending rigidity estimates by fitting MD trajectories to theoretical membrane models. Edge energy per unit length matches magnitudes predicted by continuum shell theory, validating atomistic mechanics bridging to continuum behavior. Grasso et al. [184], utilized all-atom MD simulations to investigate how embedding cell-penetrating peptides (CPPs) into lipid bilayers (specifically dioleoyl phosphatidylcholine (DOPC) membranes) modulates their mechanical properties, such as bending modulus, and how this relates to peptide hydrophobicity and membrane water permeability. Simulations embedded different CPPs (with varying hydrophobic/hydrophilic character) into dioleoyl phosphatidylcholine bilayers. Mechanical properties, including membrane bending modulus, were compared between peptide-free and peptide-containing bilayers. Analysis included quantifying water penetration, lipid order, and local bilayer disorder induced by peptide insertion. CPPs lower the bending modulus of DOPC membranes in proportion to their hydrophobicity, and more hydrophilic peptides promote enhanced water penetration into the core, resulting in greater local disorder and overall softening of the bilayer. A strong positive correlation was seen between CPP hydrophobicity and the number of water molecules transported inside the bilayer. Increased water infiltration weakened lipid packing, consistent with measured reductions in bending rigidity. These findings supported the hypothesis that CPPs modulate cell membrane mechanics by facilitating water infiltration, causing local lipid chaos, and reducing the energy barrier for bilayer deformation/insertion. Mirshahi et al. [185], conducted an interesting research to quantify the induced transmembrane voltage (ITV) generated during electroporation of a spherical MCF-7 breast cancer cell, using finite element method (FEM) modeling and complementary MD simulations. The goal was to determine threshold voltages for pore formation at both cellular and membrane (lipid bilayer) levels. The model was a spherical MCF-7 cell model, which was subjected to external electric fields (up to 1500 V/cm) to compute the steady-state ITV distribution. Surface conductivity representing the lipid membrane was explicitly incorporated to capture realistic boundary

conditions. All-atom MD simulations (GROMACS, ArgusLab, VMD) of POPC bilayers under varying field strengths were performed to observe spontaneous pore nucleation and evaluate the correlation between applied field and the fraction of membrane area permeabilized. A predictive machine learning model correlated pulse parameters with expected cell viability after four consecutive electroporation pulses, integrating FEM-derived ITV values. Both MD and FEM approaches produced comparable estimates of total electroporated cell surface area under 1500 V/cm: $\sim 66\%$ from MD and $\sim 68\%$ from FEM. Critical ITV thresholds for membrane permeabilization were calculated at 700 mV (MD, pore formation on POPC bilayer) and 739 mV (FEM, at cellular scale). MD simulations showed the formation of hydrophilic pores with an average radius of ~ 1.33 nm, which corresponded to local structural rearrangements in the POPC bilayer upon voltage exposure. The fraction of the bilayer surface involved in pore formation scaled with field magnitude, matching FEM predictions of macroscopic electroporation behavior. The integration of MD pore dynamics with FEM-generated ITV distributions allowed cross-validation of threshold voltages, and the consistency (~ 700 – 739 mV) across both methods indicates robustness of the multiscale modeling approach. Kaviani et al. developed a synergistic framework by combining MD simulations and machine learning techniques to predict, interpret, and rationalize the ion transport and mechanical properties of ionic liquid-polyvinylidene fluoride (IL/PVDF)-based gel polymer electrolytes (GPEs) for lithium-ion batteries [186]. In the study, a comprehensive analysis of MD-derived descriptors, including the lithium-ion diffusion coefficient, density, Pugh's ratio, and elastic moduli, was conducted in order to reveal the underlying structure-property relationships. Benchmarking five ML regressors on log-transformed conductivity data identified Extreme Gradient Boosting (XGB) as the most accurate, generalizable, and robust predictor. SHAP (SHAPley Additive exPlanations) analysis provided interpretability, highlighting the dominant roles of ion diffusion and matrix flexibility in conductivity. Khajavian et al. conducted experimental and density functional theory investigations for single and binary heavy metal adsorption using alginate acid structure [187]. The study demonstrated to address single and binary systems by analyzing various parameters, including adsorption energy, thermodynamics, quantum chemical, natural bond orbital, quantum theory of atoms, electron-localization-function, non-covalent interaction, UV-vis, and restrained electrostatic potential (RESP), which were calculated using a DFT-based method to assess the adsorption of heavy metals (Cd^{2+} and Co^{2+}) onto AA in both single and binary systems. The DFT calculations confirmed that Co^{2+} adsorbs more strongly on AA than on Cd^{2+} . Furthermore, the determination of the charge distribution on the adsorbent atoms, accomplished through the RESP method, revealed that oxygen atoms within AA allocate more negative charge for Co^{2+} adsorption than Cd^{2+} .

4. Advanced approaches and trends

4.1. Reactive MD and accelerated reactive schemes

A recent important development in methodological research is that reactive MD has gone beyond the simple association of ReaxFF with reactive MD in biomaterials to form a greater set of reactive classical approaches and different balances of chemical fidelity, transferability and attainable scale provided by each of these methods. The assumption of a multisurface adiabatic reactive MD (MS-ARMD) model requires the coupling of force field surfaces to represent predefined reactive channels, where the representation of the (classical) trajectories follows bond rearrangement while maintaining an explicit barrier description [56]. This contribution is particularly well-suited for situations where the reaction pathways have been defined, but all of the fully ab-initio method is too expensive.

For mechanistically biased events, accelerated or bond-boost ReaxFF schemes represent an alternative approach, with the goal to eliminate

the rarified phenomena of chemically-relevant reactions that occur on the time scale of standard MD runs. These methods provide an opportunity to promote existing close-proximity configuration (the state of which is very close to the chemical reaction) so that the process of polymer crosslinking can be represented accurately [58]. Furthermore, the accelerated or bond-boost approach has also been extended to the hydrolytic degradation of citrate-based polymeric scaffolds, providing a very consistent estimation biomaterial resorption and degradation [57]. It is expected to see a similar progress on the inorganic bioaterials side, too, with the recent development of dedicated ReaxFF parameterization of phosphate-based bioactive glasses, allowing the simulation of composition-dependent reaction dissolution reactions and surface reconstruction [75].

Another approach relies on the development of template-based reactive protocols. REACTER, which has been integrated into the LAMMPS software environment, uses user-defined templates to modify the bonding topology of an already-existing classical MD simulation. REACTER enables modifications of the bonding topology as a result of bond changes during the running of the reaction simulation and has been applied to photopolymers, composites, membranes, and other heterogeneous large-scale systems [59]. Such methods are very useful for resolving biomaterials problems and dealing with polymerization, dynamic network re-organization, transferring bonds, and even self-healing action as well as mechanical changes that occur in association with degradation. The future of reactive MD methods will continue to expand beyond a single approach into a growing number of options in which MD has evolved from purely post-hoc interpretation to chemical prediction for biomaterials [56,58,59].

4.2. Machine learning with MD

Machine learning has recently revolutionized MD simulations by providing interatomic potentials with near-*ab initio* accuracy at noticeably reduced cost. Frameworks such as the Gaussian Approximation Potential (GAP) [188], Deep Potential MD (DeepMD) [189], SchNet [190], Neural Equivariant Interatomic Potentials (NequIP) [191], and Moment Tensor Potentials (MTP) [192], learn the complex mapping between atomic configurations and possible energy surfaces from DFT datasets.

For biomaterials, machine learning-based force fields have assisted accurate modeling of BGs, ceramics, and polymer composites, where traditional empirical potentials regularly fail to capture mixed ionic-covalent bonding or structural disorder [193,194]. For example, Bartók et al. [188] used GAP to reproduce amorphous silica energetics, while Zhang et al. [189] demonstrated DeepMD scalability to large oxide systems. Recently, machine-learning interatomic potentials and ML-assisted surrogate models have been increasingly applied to disordered inorganic and polymeric systems. In such studies, model performance is typically benchmarked against DFT, MD, or experimental reference data using explicit quantitative measures—such as MAE/RMSE for energies and forces, and R^2 or percent deviation for predicted mechanical properties—rather than qualitative descriptions alone [195,196].

Beyond atomistic simulations, machine learning is progressively used to learn composition–structure–property relationships directly from simulation and experimental datasets [197,198]. Neural-network and Gaussian-process regressors can link descriptors such as atomic coordination or network connectivity to macroscopic outputs like stiffness, dissolution rate, and bioactivity [199]. These data-driven correlations enable reverse design, wherein target mechanical or biological responses guide optimal composition selection. The convergence of machine learning and MD thus enables predictive and design-oriented biomaterials engineering.

4.3. Multiscale and hybrid methods

Many phenomena in biomaterials (i.e., crack propagation, degradation, and ion diffusion) occur across length and time scales beyond the reach of classical MD. Multiscale and hybrid frameworks address this by coupling atomistic and continuum approaches. At the smallest scale, DFT describes bond formation and defect energetics; its output parameterizes MD force fields. MD then informs finite-element (FE) or coarse-grained models describing mesoscale mechanics and transport [200–202].

Accelerated sampling techniques, including metadynamics [203], hyperdynamics [204], and adaptive MD [205], further extend accessible timescales, enabling the study of rare events such as nucleation, sintering, ion exchange, and protein adsorption. These approaches provide access to free-energy landscapes governing degradation and reconstruction processes. By uniting multiple scales, hybrid modeling yields a continuous picture from chemical bonding to bulk mechanical integrity, improving the predictive capability of computational biomaterials design.

4.4. MD in additive manufacturing simulation

Additive manufacturing (AM) in biomaterials should not be viewed as a single process, but rather as a family of strongly non-equilibrium fabrication routes that include laser sintering or melting of inorganic particles, filament-based deposition of thermoplastics, and extrusion- or photopolymerization-based printing of soft hydrogels and bioinks. Across these platforms, the common challenge is that steep local gradients in temperature, stress, shear, solvent content, and composition restructure interfaces at the nanoscale, while the final printed performance is determined by how these transient events are frozen into microstructure [206]. In this regard, MD simulations are especially valuable because they can resolve the atomistic origin of particle neck formation, interlayer healing between deposited filaments, chain orientation under flow, and the stability of hydration and crosslinking environments in cell-compatible hydrogel systems.

In ceramic- and glass-based AM, MD can quantify the earliest stages of particle coalescence by tracking surface diffusion, viscous flow, local densification, and neck-growth kinetics under rapid heating and cooling. These simulations provide mechanistic insight into how particle size, local composition, and thermal history control residual porosity, defect formation, and metastable interfacial structures that later influence mechanical integrity, dissolution, and biological performance. Likewise, in oxide-containing or metallic biomaterial systems, atomistic calculations can identify local stress concentrations, defect generation, and interface stability during repeated thermal cycling, thereby supplying physically meaningful descriptors for process maps that would otherwise rely mainly on empirical optimization [207].

For polymer-based biomaterials, the key AM questions are different but equally well suited to MD. During fused-filament fabrication or extrusion-based deposition, interlayer strength depends on segmental diffusion across weld lines, recovery of chain entanglement, shear-induced orientation, crystallization kinetics, and the retention of residual anisotropy after cooling [208]. MD simulations can therefore clarify how print temperature, cooling rate, molecular weight, copolymer composition, and filler loading influence filament–filament healing and mechanical anisotropy. In soft hydrogel and bioink formulations, atomistic and coarse-grained simulations are additionally valuable for resolving solvent retention, ion-mediated gelation, reversible supramolecular crosslinks, and photo- or chemically induced network formation. These factors directly affect print fidelity, post-print shape stability, nutrient transport, degradation behavior, and ultimately cell compatibility in printed constructs.

The broader significance of MD in AM lies in multiscale integration. Although MD alone cannot represent the full dimensions of a printed scaffold, it can provide constitutive descriptors, such as surface

diffusivity, melt-relaxation times, interfacial adhesion energies, defect formation tendencies, and crosslinking-related structural metrics, that can be transferred into mesoscale, computational-fluid-dynamics, phase-field, or finite-element process models. In this way, MD moves beyond post hoc interpretation and becomes part of a predictive workflow for selecting feedstock chemistry, processing windows, and architectural design parameters in printed biomaterials [209]. Such an approach is particularly promising for multi-material implants, graded scaffolds, and biofabricated hydrogel systems, where final function emerges from the coupling of processing history, local microstructure, and biological response.

5. Validation and experimental correlation

Validation of MD simulations, besides experimental data, is essential to confirm that atomistic models provide physically meaningful and predictive insights into biomaterials behavior. The predictive value of MD depends strongly on how well simulated properties, including structure, energy, mechanical response, and transformation kinetics, correlate with experimental measurements obtained from techniques such as inductively coupled plasma (ICP) spectroscopy, X-ray diffraction (XRD), differential scanning calorimetry (DSC), and nanoindentation. Correct validation enhances confidence in MD results, supports model refinement, and bridges the gap between simulation and practical applications in biomaterials, composites, and additively-manufactured systems. Effective validation in biomaterials should be multi-property, application-specific, and uncertainty-aware rather than based on agreement with a single experimental observable. For structural biomaterials such as glasses and ceramics, validation may require simultaneous consistency in density, coordination numbers, RDF/PDF features, and network-connectivity descriptors. For polymers and hydrogels, more informative benchmarks often include glass transition behavior, chain dimensions, swelling ratio, diffusion coefficients, and stress-strain response. For biointerface problems, adsorption free energies, protein orientation trends, hydration structure, contact-angle behavior, and force-spectroscopy observables may be more meaningful than structural agreement alone. Accordingly, the goal of validation is not merely to reproduce one experiment, but to demonstrate that the same MD framework captures a coherent set of structural, thermodynamic, dynamical, and functional observables relevant to the intended biomaterial application.

5.1. Structural validation

Structural validation typically involves comparing MD-predicted atomic arrangements and short-range order with experimental data obtained from XRD, neutron diffraction, or solid-state nuclear magnetic resonance (NMR). For example, the partial pair distribution functions (PDFs) generated from MD can be compared with those derived from XRD or neutron scattering to confirm the correct representation of coordination environments and bond lengths [210–212]. In BGs, for instance, Si-O and Ca-O coordination numbers derived from MD simulations have demonstrated good agreement with experimental values, confirming the reliability of potential models used to describe silicate networks [213,214]. Similarly, in polymer-based biomaterials, MD simulations of amorphous regions can be validated through the comparison of predicted X-ray scattering profiles or simulated density with experimentally measured densities [215].

Beyond peak positions, structural validation increasingly examines peak widths (indicative of disorder/strain), medium-range features (rings or pores), and topology of amorphous networks, allowing MD to predict not just what bonds exist but how the network classifies, which is critical for hierarchical biomaterials. Furthermore, MD can predict changes in structure under external stimulus (e.g., hydration, temperature, porosity), which experiments can then probe (via in-situ XRD or PDF) to validate dynamic behavior.

5.2. Thermal and energetic validation

Differential scanning calorimetry (DSC) and dynamic mechanical analysis (DMA) provide key thermal parameters such as glass transition temperature (T_g), crystallization onset, melting temperature (T_m), and enthalpic relaxation signatures. These parameters are often estimated in MD simulations through analysis of volume-temperature or enthalpy-temperature curves obtained through heating and cooling cycles [216]. Although MD typically overestimates T_g and T_m due to accelerated cooling rates (on the order of 10^9 – 10^{12} K/s compared to experimental 1–10 K/min), trends in compositional dependence and comparative changes are often well described [217]. In polymeric and glassy biomaterials, such correlations have been applied to modify composition and cross-linking density to achieve desired mechanical and thermal stability. Furthermore, energy distributions from MD can be linked with calorimetric data to validate relaxation dynamics and metastable phase formation.

A remarkably advanced approach is to track simulated enthalpy or volume changes while scanning with various cooling or heating protocols and compare the inflection points or derivative peaks with DSC experiments. This enables calibration of quench-rate dependencies, allowing one to adjust simulation cooling rates or apply empirical corrections. By linking MD and DSC, researchers can infer activation parameters for transitions, validate how network topology evolves with temperature, and evaluate whether MD potentials correctly encode enthalpic landscapes of the material.

5.3. Mechanical property validation

Mechanical validation aims at comparing elastic moduli, hardness, yield behavior, and failure mechanisms predicted from MD with experimental nanoindentation, tensile, or compression data [218]. For example, the Young's modulus and hardness derived from simulated stress-strain curves can be quantitatively compared to nanoindentation results on corresponding materials. In calcium-silicate or phosphate-based glasses, the modulus values predicted via MD typically agree within 10–20% of experimental measurements, signifying that MD captures the primary load-bearing mechanisms, even though discrepancies may arise from the absence of porosity or defects in the atomic model [219]. For polymers and composites, MD simulations incorporating cross-linking, chain-length distributions, and filler architectures have successfully predicted modulus and yield strength consistent with tensile testing data.

In additively manufactured biomaterials or metallic lattice structures, MD can simulate nanoindentation or virtual tensile deformation of small volumes to extract effective mechanical response, which can then be compared with micro-CT guided experimental mechanical testing. This synergy strengthens confidence in simulation predictions and allows for the prediction of local heterogeneity effects (grain boundaries, interfaces) that experiments may not easily resolve. In the specific context of additive manufacturing, mechanical validation must bridge local printed interfaces and bulk scaffold response. MD-predicted interlayer adhesion energies, diffusion lengths across filament welds, nanoparticle neck-growth rates, residual orientation factors, or local defect sensitivities can be compared with experimentally measured crystallinity, micro-CT-informed porosity, results from nanoindentations at interlayer junctions, strength/stiffness of printed single struts etc. This is particularly important when there are anisotropies/heterogeneities in printed biomaterials and agreement should therefore be assessed not only in average modulus or strength, but also in the spatial variability introduced by layer boundaries, unmelted particles, porosity gradients, or compositionally graded architectures. Such hierarchical comparison strengthens confidence in simulation-guided process optimization and highlights additive manufacturing as a particularly powerful use case for multiscale MD validation.

5.4. Limitations and divergences

Despite significant progress, several limitations still affect the accuracy of MD-experiment correlation. Simplifications in force fields, limited simulation time scales, and system sizes may neglect slow structural relaxations, chemical reactions, large-scale heterogeneities, or long-term aging phenomena that occur experimentally. Moreover, experimental uncertainty, sample preparation conditions, and measurement resolution can contribute to noticeable discrepancies. For instance, while MD provides atomistic insight into interfacial bonding or ionic diffusion, it regularly cannot directly reproduce long-term degradation or bioactivity observed experimentally. Hybrid approaches combining MD with mesoscale or continuum models are progressively being explored to overcome such limitations and bridge the gap between nanoscale simulations and macroscale experimental phenomena [220].

For example, force-field transferability remains a major issue: a potential calibrated for one temperature and composition may fail under another, leading to erroneous property predictions. Also, rapid quenching in MD tends to produce higher residual stresses and unrealistically small microstructural features compared with experiments. Additionally, many experiments (e.g., fatigue, long-term ion release, corrosion) operate on timescales inaccessible to MD, so simulation must be complemented by ageing or accelerated-dynamics techniques or treated as predictive rather than definitive.

5.5. Toward quantitative correlation

Future developments in validation are moving toward quantitative, multi-property benchmarking, where structural, thermal, and mechanical predictions are instantaneously compared against a unified experimental dataset. The use of machine learning to optimize force-field parameters based on experimental data is a promising direction that can systematically improve the reliability of MD simulations for biomaterials. Such integrated frameworks hold potential for establishing MD not only as a qualitative interpretive tool but also as a quantitative predictive platform in biomaterials design. For example, combining MD with finite-element or mesoscale models allows property predictions at the macroscale that are directly comparable to experiment, with MD providing defect- and interface-level parameters for continuum modelling. Further, high-throughput MD screening allied with experimental combinatorial documents may enable rapid validation loops, accelerating material discovery and qualification.

6. Gaps, challenges, and future directions

MD simulations have evolved into a robust methodology for exploring atomistic mechanisms in biomaterials. However, as the variety of systems broadens from inorganic bioglasses and crystalline ceramics to complex polymer networks and organic-inorganic composites, the limitations inherent in current MD practices become increasingly apparent. Bridging these gaps is crucial for transforming MD from a primarily exploratory tool into a predictive framework that can effectively inform large-scale biomaterial design. Many of the remaining limitations in biomaterials MD become most visible when simulations are expected to support translational workflows such as additive manufacturing, degradation prediction, scaffold optimization, or bio-interface screening. In these contexts, the key challenge is no longer only to simulate larger systems or longer times, but to preserve chemical realism across heterogeneous interfaces, quantify uncertainty in model predictions, and transfer MD-derived descriptors into higher-scale models without losing mechanistic meaning. The following subsections therefore focus on the major scientific and methodological barriers that must be overcome before MD can function routinely as a predictive design tool in biomaterials engineering.

6.1. Biological realism vs. computational cost

Achieving biologically relevant realism while managing computational costs remains a significant challenge. Biomaterials function within inherently complex environments characterized by physiological ionic strengths, structured hydration layers, variable pH levels, protein corona formation, and dynamic interactions with membranes or extracellular matrices. Accurately modeling these conditions at an atomic level necessitates large system sizes and extended simulation times, often requiring tens to hundreds of nanoseconds for minor biological events and microseconds for more complex processes like protein adsorption, conformational changes, or interfacial wetting transitions.

This complexity creates tension, as increasing biological fidelity complicates simulations, which do not scale well with traditional MD approaches. For instance, simulating the mechanical response of a polymer-ceramic composite scaffold in the presence of solvents and biological macromolecules often involves millions of atoms, rendering such simulations prohibitively expensive even with GPU-accelerated algorithms. To address this issue, researchers are exploring multiscale solutions. Carefully parameterized coarse-grained models can capture mesoscale behavior while significantly reducing computational demands. However, these models frequently overlook critical chemical details vital for biomaterials, such as ion-mediated cross-linking, pH effects on bonding, or the reactive dissolution of bioglasses. Hybrid atomistic-coarse-grained strategies show promise, but they require meticulous calibration to ensure accuracy across different scales. Future developments may involve automated resolution switching, allowing regions with high chemical activity to be treated atomistically while the rest of the material is represented in a simplified manner. Such adaptive multiscale approaches could strike a necessary balance between biological realism and computational feasibility.

6.2. Lack of validated force fields for hybrid organic-inorganic systems

Despite the advancements in MD, a notable gap exists in the availability of validated force fields for hybrid organic-inorganic systems. While MD has benefited from extensive force-field development in traditional areas such as proteins, nucleic acids, and simple inorganic solids, the integration of diverse chemical components in biomaterials presents unique challenges. A single biomaterial may encompass peptides, polymer linkers, metal ions, silicate networks, or calcium phosphate nanocrystals. Most established force fields were not designed to handle these hybrid interfaces, leading to inconsistent predictions when combining components.

The intricacies are particularly evident in inorganic phases. For instance, accurately modeling the dissolution of BGs or ion mobility in doped calcium phosphate systems requires force fields that account for polarization, coordination flexibility, and environment-dependent bonding. Concurrently, polymer matrices necessitate reliable descriptions of chain flexibility, cross-link behavior, and interactions with both water and inorganic fillers. Existing force fields often treat these properties independently, while cross-interaction parameters governing the interactions between organic and inorganic domains remain underdeveloped. Many biomaterial applications also involve conditions where chemical properties change dynamically, such as surface hydroxylation, proton exchange, degradation reactions, and ion release. Classical fixed-charge force fields struggle to represent these phenomena accurately. Reactive MD approaches, such as ReaxFF or machine-learned potentials, present potential solutions but require comprehensive training and careful validation to avoid introducing artifacts.

The future of modeling hybrid biomaterials hinges on the establishment of coherent, transferable, and openly accessible force fields tailored for complex material interfaces. Automated parameterization pipelines, guided by density DFT datasets and genetic optimization methods, are accelerating progress in this area. However, widespread adoption will necessitate community-driven initiatives to systematically

benchmark these force fields and ensure reproducible performance across various materials.

6.3. Need for standardized benchmarks and validation protocols

Benchmarking remains a fundamental practice in computational disciplines, yet the field of biomaterials MD lacks a unified set of reference systems or validation metrics. This absence complicates the assessment of simulation reliability and the comparison of methodologies across studies. The diversity of biomaterial types and their highly context-dependent behaviors exacerbate this issue. A method validated for hydrated silica may not perform adequately for polymer-ceramic complexes or organic-inorganic self-assembled scaffolds.

Standardized benchmarks could take various forms, including reference structural datasets such as fully hydrated silica surfaces, collagen-mineral interfaces, or polymer networks with controlled cross-link densities. Additionally, clearly defined environmental conditions such as ionic strength, temperature, pH, and solvent composition, along with mechanical testing protocols, would facilitate comparisons of elastic moduli, yield points, or fracture pathways predicted by different force fields. Interfacial interaction benchmarks, quantifying adhesion energies or contact angles for peptide-surface or polymer-oxide interactions, would further enhance reproducibility, accelerate method development, and clarify the reliability and limitations of force fields. Furthermore, these benchmarks would support the integration of emerging machine-learning potentials by providing standardized validation datasets for tuning and testing new models.

6.4. Open datasets, transparency, and reproducibility in biomaterials MD

Reproducibility in biomaterials MD remains a significant concern, not due to inherent irreproducibility in MD itself, but because simulation details are often inadequately documented. Simulations in biomaterials are sensitive to various factors, including surface preparation, partial charge assignment, boundary conditions, thermostat/barostat algorithms, and equilibration protocols. Without access to comprehensive input files and metadata, reproducing results even for experts can be extremely challenging.

The broader computational science community is increasingly adopting FAIR data principles (Findable, Accessible, Interoperable, Reusable), yet biomaterials MD lags behind other fields, such as protein modeling or materials informatics, where dataset sharing is more prevalent. The biomaterials community would significantly benefit from public repositories for simulation data, encompassing trajectories, input scripts, and force-field parameters. Standardized metadata formats that capture model construction, simulation workflows, and analysis pipelines would enhance transparency. Additionally, open-source analysis tools ensure consistency in metrics such as diffusion coefficients, radial distribution functions, and mechanical response curves. Shared libraries of surface models, including hydrated glass surfaces or functionalized polymers, would promote consistent starting structures. Increasing transparency and openness would not only improve reproducibility but also create valuable resources for training machine-learning models, paving the way for the next generation of predictive tools in biomaterials design.

6.5. Future directions in modeling novel biomaterials

The biomaterials landscape is rapidly evolving, driven by innovations in regenerative medicine, drug delivery, soft robotics, and bioinspired ceramics. Consequently, MD simulations must adapt alongside these advancements. Key future directions include the development of reactive and environment-aware simulations, which are essential for materials that undergo degradation, mineralization, swelling, or ion exchange. Incorporating pH-sensitive force fields, electrochemical potentials, and environment-triggered events will be

increasingly important as biomaterials are engineered for dynamic biological functions.

Machine-learned interatomic potentials, trained on extensive quantum mechanical datasets, offer the potential for near-DFT accuracy with MD efficiency, particularly for heterogeneous interfaces where classical force fields face limitations. Future modeling efforts are likely to integrate multiphysics and multiscale frameworks, enabling couplings between MD, continuum mechanics, and biochemical network models. This integration could allow for predictive simulations of scaffold deformation under physiological loads, nutrient transport in hydrogels, or growth-material interactions in regenerative systems.

Additionally, the establishment of autonomous simulation workflows, compatible with AI-driven design loops, will facilitate rapid screening of polymer blends, surface functionalizations, or nanoparticle additives. These automated, high-throughput MD pipelines will reduce reliance on intuition-driven trial-and-error approaches, supporting a data-rich exploration of biomaterial design spaces.

7. Conclusion

7.1. Summary of MD contributions to biomaterials science

Over the past several decades, MD simulations have become integral to understanding and designing biomaterials across molecular, nano-, and mesoscale regimes. MD provides unique insights into phenomena that are challenging or impossible to observe experimentally, such as the atomic-level dissolution of BGs, polymer conformational dynamics under mechanical stress, water structuring near ceramic surfaces, and peptide adsorption onto mineral substrates. Through these capabilities, MD has contributed to optimizing implant coatings, designing bio-inspired composites with enhanced toughness, developing polymeric scaffolds with tunable mechanical properties, and refining responsive hydrogels for therapeutic delivery.

By elucidating the interplay among chemistry, structure, and environment, MD simulations have clarified the fundamental mechanisms governing biomaterial performance. Consequently, computational modeling is now recognized not only as a complementary tool but also as a core element of biomaterials research, guiding material synthesis, modification, and application. Fig. 8 summarizes this perspective by showing how MD simulations functionalize the evaluation of biomaterials with the suitable properties into experimentally relevant and application-driven biomaterial functionalities.

7.2. Future outlook: integration with ai, faster simulations, and predictive design

Looking ahead, the forthcoming decade of biomaterials modeling will be characterized by convergence—the integration of molecular simulation, machine learning, data science, and high-performance computing. Several anticipated advancements are poised to reshape the field. AI-enhanced force fields and sampling methods are expected to enable simulations that approach quantum accuracy while maintaining the efficiency required for large systems, addressing longstanding challenges in modeling hybrid materials and dynamic biological environments. GPU acceleration, exascale computing, and advanced parallelization will lower the barriers to simulating million-atom systems over microsecond timescales, allowing researchers to investigate material behavior under realistic physiological conditions.

Predictive, data-driven design pipelines that combine MD-generated descriptors with machine-learning optimization will facilitate the rapid screening of candidate biomaterials. Researchers may soon be able to computationally evaluate thousands of formulations, narrowing the field to the most promising designs instead of relying solely on empirical testing. Improved reproducibility and open data practices will further democratize biomaterials modeling, support community benchmarking, and enhance the reliability of MD-based predictions.

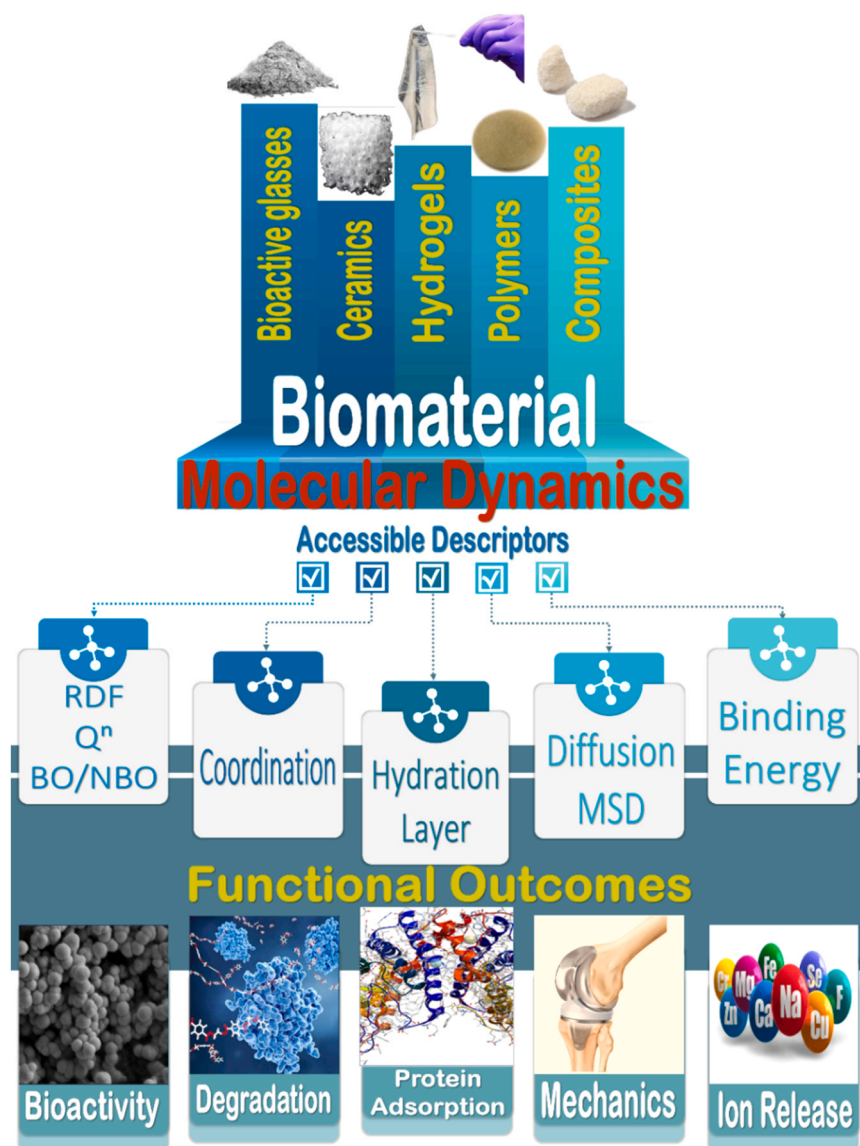


Fig. 8. Schematic overview of MD role in enabling the extraction of descriptors and their related functional outcome in biomaterials. Structural, dynamical, and energetic descriptors provide insights into bioactivity, degradation, protein adsorption, mechanical behavior, and ion release in a variety of biomaterials research, while helping the rational design of advanced and optimized biomaterials.

Together, these developments signal a future where MD is not merely a tool for mechanistic insight but a foundational component of biomaterials innovation. By coupling simulations with experimental validation and AI-driven design, researchers will gain unprecedented control over performance, biocompatibility, and functionality, ultimately accelerating progress in regenerative medicine, tissue engineering, implantable devices, and sustainable biomaterial technologies.

CRediT authorship contribution statement

Ahmet Akif Kızılkurtlu: Writing – original draft, Investigation. **Amirhossein Moghanian:** Writing – original draft, Supervision, Investigation, Conceptualization. **Majid Vaseghi:** Writing – review & editing, Supervision, Investigation. **Ali Akpek:** Writing – review & editing, Investigation. **Francesco Baino:** Writing – review & editing, Supervision, Investigation. **Sirus Safaee:** Writing – original draft, Investigation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

References

- [1] Y. Zhou, Y. Pan, B. Sun, Q. Gao, Mechanical properties of an improved flexible auxetic metamaterial and its application in self-expanding vascular stents, *Eur. J. Mech. -A/Solids* 106 (2024) 105345.
- [2] J. Li, X. Hu, Y. Chen, D. Fan, C. Tan, S. Yang, H. Wu, Y. Wang, Q. An, Z. Xiao, Review of recent progress in vascular stents: from conventional to functional vascular stents, *Chin. Chem. Lett.* 36 (7) (2025) 110492.
- [3] Y. Huang, Y. Xu, X. Chen, J.P. Armstrong, M. Caputo, Q. Qi, B. Hicks, C. Vyas, P. Bartolo, G. Biglino, 3D printing of biodegradable polymer vascular stents to treat cardiovascular diseases: a review, *Addit. Manuf.* (2025) 104984.
- [4] X. Yang, Y. Chen, Q. Wu, H. Yang, G. Duan, Q. Fu, H. He, P. Zhang, J. Ji, S. Jiang, Electrospun anisotropic fiber reinforced composites for artificial heart valves, *J. Mater. Chem. B* 13 (12) (2025) 3918–3929.
- [5] P. Sonthithai, P. Kaewkong, S. Channasanon, S. Tanodekaew, 3D-Printed PEG–PLA/Gelatin hydrogel: characterization toward in vitro chondrocyte redifferentiation, *ACS Biomater. Sci. & Eng.* 11 (4) (2025) 2157–2166.
- [6] L. Xie, F. Gan, Y. Hu, Y. Zheng, J. Lan, Y. Liu, X. Zhou, J. Zheng, X. Zhou, J. Lou, From blood to therapy: the revolutionary application of platelets in cancer-targeted drug delivery, *J. Funct. Biomater.* 16 (1) (2025) 15.
- [7] S. Safaei, S. Abe, M. Nesabi, A. Moghanian, P. Asadi, K. Yoshida, H. Okazaki, K. Harada, I. Watanabe, H. Murata, Multi-layered coatings for improving implant performance of titanium, *Ceram. Int.* (2025).
- [8] F. Bairo, R. Gabrieli, E. Verné, A. Schiavi, M. Schwentenwein, L. D'Andrea, P. Vena, Recent trends in design, manufacturing and challenges of bone-like bioceramic scaffolds, *Ceram. Int.* 51 (10) (2025) 12355–12369.
- [9] M. Montazerian, F. Bairo, E. Fiume, C. Migneco, A. Alaghmandfar, O. Sedighi, A.V. DeCeanne, C.J. Wilkinson, J.C. Mauro, Glass-ceramics in dentistry: fundamentals, technologies, experimental techniques, applications, and open issues, *Prog. Mater. Sci.* 132 (2023) 101023.
- [10] A. Moghanian, M. Zohourfazel, M.H.M. Tajer, A.K. Miri, Comprehensive in vitro studies of novel sol gel-derived Zr_{4+}/Zn^{2+} co-substituted bioactive glass with enhanced biological properties for bone healing, *J. Non-Cryst. Solids* 566 (2021) 120887.
- [11] A. Moghanian, S. Nasiripour, Z. Miri, Z. Hajifathali, S.H. Hosseini, M. Sajjadnejad, R. Aghabarari, N. Nankali, A.K. Miri, M. Tahriri, Structural and in vitro biological evaluation of sol-gel derived multifunctional Ti^{4+}/Sr^{2+} 2 co-doped bioactive glass with enhanced properties for bone healing, *Ceram. Int.* 47 (20) (2021) 29451–29462.
- [12] A. Moghanian, Biocompatibility and dissolution experiments with molecular dynamics structural characterization of melt-derived Mg-substituted silicate bioactive potential glasses, *J. Non-Cryst. Solids* 672 (2026) 123855.
- [13] A. Moghanian, M. Vaseghi, S. Safaei, Molecular dynamics and experimental investigation of MgO/SrO Co-substituted silicate bioactive glasses: structural and dissolution behavior, *Mater. Res. Bull.* (2025) 113882.
- [14] A. Moghanian, R. Farmani, A. Tayebi, N. Kolivand, S. Safaei, Investigating zinc-doped silicate-based ternary glass using computational chemistry: a study on interatomic interaction, dissolution behavior, and antibacterial efficiency, *J. Non-Cryst. Solids* 669 (2025) 123804.
- [15] Q. Fu, F. Bairo, E. Saiz, H. Bai, J.C. Mauro, Nature-inspired hierarchical materials, *J. Am. Ceram. Soc.* 108 (11) (2025) e70156.
- [16] L.L. Hench, H. Paschall, Direct chemical bond of bioactive glass-ceramic materials to bone and muscle, *J. Biomed. Mater. Res.* 7 (3) (1973) 25–42.
- [17] A. Moghanian, S. Nasiripour, A. Koohfar, M. Sajjadnejad, S. Hosseini, M. Taherkhani, Z. Miri, S.H. Hosseini, M. Aminitabar, A. Rashvand, Characterization, in vitro bioactivity and biological studies of sol-gel-derived TiO_2 substituted 58S bioactive glass, *Int. J. Appl. Ceram. Technol.* 18 (5) (2021) 1430–1441.
- [18] A. Pazhoueshgar, S.A.S. Vanini, A. Moghanian, The experimental and numerical study of fracture behavior of 58s bioactive glass/polysulfone composite using the extended finite elements method, *Mater. Res. Express* 6 (9) (2019) 095208.
- [19] A. Moghanian, S. Safaei, A.A. Kizilkurtlu, R. Farmani, Advanced rare-earth doped bioactive glasses: a comprehensive review on structural, biological, and mechanical improvements, *Silicon* (2025) 1–28.
- [20] A. Tilocca, Molecular dynamics simulations of a bioactive glass nanoparticle, *J. Mater. Chem.* 21 (34) (2011) 12660–12667.
- [21] A. Atila, On the relationship between water adsorption and surface chemistry in soda-lime silicate glasses, *ChemPhysChem* 25 (24) (2024) e202400370.
- [22] Z. Zhang, S. Ispas, W. Kob, Structure and vibrational properties of sodium silicate glass surfaces, *J. Chem. Phys.* 153 (12) (2020).
- [23] A. Tilocca, Structural models of bioactive glasses from molecular dynamics simulations, *Proc. R. Soc. A Math. Phys. Eng. Sci.* 465 (2104) (2009) 1003–1027.
- [24] K. Deshmukh, T. Kovarik, T. Krenk, D. Docheva, T. Stich, J. Pola, Recent advances and future perspectives of sol-gel derived porous bioactive glasses: a review, *RSC Adv.* 10 (56) (2020) 33782–33835.
- [25] R. Mathew, B. Svensson, A. Tilocca, M. Edén, Toward a rational design of bioactive glasses with optimal structural features: composition–structure correlations unveiled by solid-state NMR and MD simulations, *J. Phys. Chem. B* 118 (3) (2014) 833–844.
- [26] M. Corno, C. Busco, V. Bolis, S. Tosoni, P. Uglierio, Water adsorption on the stoichiometric (001) and (010) surfaces of hydroxyapatite: a periodic B3LYP study, *Langmuir* 25 (4) (2009) 2188–2198.
- [27] J.-W. Shen, T. Wu, Q. Wang, H.-H. Pan, Molecular simulation of protein adsorption and desorption on hydroxyapatite surfaces, *Biomaterials* 29 (5) (2008) 513–532.
- [28] T. Yoshioka, T. Ikoma, A. Monkawa, T. Tonegawa, D. Chakarov, B. Kasemo, N. Hanagata, M. Tanaka, Protein adsorption on hydroxyapatite nano-crystals with quartz crystal microbalance technique, *Key Eng. Mater.* 361 (2008) 1119–1122.
- [29] J. Zhu, J. Zhu, T.A. Springer, Complete integrin headpiece opening in eight steps, *J. Cell Biol.* 201 (7) (2013) 1053–1068.
- [30] A. Göpferich, Mechanisms of polymer degradation and erosion, *Biomaterials Silver Jubilee Compendium* (1996) 117–128.
- [31] I. Grizzi, H. Garreau, S. Li, M. Vert, Hydrolytic degradation of devices based on poly (DL-lactic acid) size-dependence, *Biomaterials* 16 (4) (1995) 305–311.
- [32] G.L. Siparsky, K.J. Voorhees, F. Miao, Hydrolysis of polylactic acid (PLA) and polycaprolactone (PCL) in aqueous acetonitrile solutions: autocatalysis, *J. Environ. Polym. Degrad.* 6 (1) (1998) 31–41.
- [33] S. Teixeira, K.M. Eblagon, F. Miranda, M.F. R. Pereira, J.L. Figueiredo, Towards controlled degradation of poly (lactic) acid in technical applications, *C 7* (2) (2021) 42.
- [34] Y. Chen, S. Zhou, Q. Li, Mathematical modeling of degradation for bulk-erosive polymers: applications in tissue engineering scaffolds and drug delivery systems, *Acta Biomater.* 7 (3) (2011) 1140–1149.
- [35] T.K. Giri, A. Thakur, A. Alexander, H. Badwaik, D.K. Tripathi, Modified chitosan hydrogels as drug delivery and tissue engineering systems: present status and applications, *Acta Pharm. Sin. B* 2 (5) (2012) 439–449.
- [36] A.M. Alswieleh, N. Cheng, G.J. Leggett, S.P. Armes, Spatial control over cross-linking dictates the pH-responsive behavior of poly (2-(tert-butylamino) ethyl methacrylate) brushes, *Langmuir* 30 (5) (2014) 1391–1400.
- [37] Y. Yang, M. Guo, R. Qian, C. Liu, X. Zong, Y.-Q. Li, W. Li, Binding efficacy and kinetics of chitosan with DNA duplex: The effects of deacetylation degree and nucleotide sequences, *Carbohydr. Polym.* 169 (2017) 451–457.
- [38] L. Li, Z. Liu, R. Qi, Molecular dynamics simulations in hydrogel research and its applications in energy utilization: a review, *Energy Rev.* 3 (3) (2024) 100072.
- [39] X. Yang, B.L. Dargaville, D.W. Hutmacher, Elucidating the molecular mechanisms for the interaction of water with polyethylene glycol-based hydrogels: influence of ionic strength and gel network structure, *Polymers* 13 (6) (2021) 845.
- [40] J. Wu, S. Chen, Investigation of the hydration of nonfouling material poly (ethylene glycol) by low-field nuclear magnetic resonance, *Langmuir* 28 (4) (2012) 2137–2144.
- [41] C. Perrino, S. Lee, N.D. Spencer, Quantitative comparison of the hydration capacity of surface-bound dextran and polyethylene glycol, *Langmuir* 40 (27) (2024) 14130–14140.
- [42] G.T. Grant, E.R. Morris, D.A. Rees, P.J. Smith, D. Thom, Biological interactions between polysaccharides and divalent cations: the egg-box model, *FEBS Lett.* 32 (1) (1973) 195–198.
- [43] I. Braccini, S. Pérez, Molecular basis of Ca^{2+} -induced gelation in alginates and pectins: the egg-box model revisited, *Biomacromolecules* 2 (4) (2001) 1089–1096.
- [44] K. Draget, G.S. Bræk, O. Smidsrød, Alginic acid gels: the effect of alginate chemical composition and molecular weight, *Carbohydr. Polym.* 25 (1) (1994) 31–38.
- [45] Y.Å. Mørch, I. Donati, B.L. Strand, G. Skjåk-Bræk, Eff. Ca^{2+} Ba^{2+} Sr^{2+} Alginate Microbe (2006).
- [46] Q. Chen, H. Chen, L. Zhu, J. Zheng, Fundamentals of double network hydrogels, *J. Mater. Chem. B* 3 (18) (2015) 3654–3676.
- [47] R. Zhu, D. Zhu, Z. Zheng, X. Wang, Tough double network hydrogels with rapid self-reinforcement and low hysteresis based on highly entangled networks, *Nat. Commun.* 15 (1) (2024) 1344.
- [48] J.O. Akindoyo, M.D. Beg, S. Ghazali, H.P. Heim, M. Feldmann, Impact modified PLA-hydroxyapatite composites-Thermo-mechanical properties, *Composites Part A Applied Science Manufacturing* 107 (2018) 326–333.
- [49] J.S. Watai, P.S. Calvão, T.R. Rigolin, S.Hd.P. Bettini, A.M.C. Souza, Retardation effect of nanohydroxyapatite on the hydrolytic degradation of poly (lactic acid), *Polym. Eng. Sci.* 60 (9) (2020) 2152–2162.
- [50] H.S. Wijerathne, D. Yan, B. Zeng, Y. Xie, H. Hu, M.N. Wickramaratne, Y. Han, Effect of nano-hydroxyapatite on protein adsorption and cell adhesion of poly (lactic acid)/nano-hydroxyapatite composite microspheres, *SN Appl. Sci.* 2 (4) (2020) 722.
- [51] P. Gkeka, Molecular dynamics studies of peptide-membrane interactions: insights from coarse-grained models, University of Edinburgh, 2010.
- [52] K. Matsuzaki, Why and how are peptide-lipid interactions utilized for self defence? *Biochem. Soc. Trans.* 29 (4) (2001) 598–601.
- [53] Y. Lyu, G. Narsimhan, Antimicrobial peptides—membrane interactions. One Hundred Years of Colloid Symposia: Looking Back and Looking Forward, ACS Publications, 2023, pp. 381–430.
- [54] A.C. Van Duin, S. Dasgupta, F. Lorient, W.A. Goddard, ReaxFF: a reactive force field for hydrocarbons, *J. Phys. Chem. A* 105 (41) (2001) 9396–9409.
- [55] M. Karplus, J. Kuriyan, Molecular dynamics and protein function, *Proc. Natl. Acad. Sci.* 102 (19) (2005) 6679–6685.
- [56] T. Nagy, J. Yosa Reyes, M. Meuwly, Multisurface adiabatic reactive molecular dynamics, *J. Chem. Theory Comput.* 10 (4) (2014) 1366–1375.
- [57] N. Dasgupta, D.E. Yilmaz, A. Van Duin, Simulations of the biodegradation of citrate-based polymers for artificial scaffolds using accelerated reactive molecular dynamics, *J. Phys. Chem. B* 124 (25) (2020) 5311–5322.
- [58] A. Vashisth, C. Ashraf, W. Zhang, C.E. Bakis, A.C. Van Duin, Accelerated ReaxFF simulations for describing the reactive cross-linking of polymers, *J. Phys. Chem. A* 122 (32) (2018) 6633–6642.
- [59] J.R. Gissinger, B.D. Jensen, K.E. Wise, Molecular modeling of reactive systems with REACTER, *Comput. Phys. Commun.* 304 (2024) 109287.

- [60] S.W. Wendji, R. Piotrowski, C. Massobrio, M. Boero, C. Tugène, F. Shuaib, D. Hamani, P.-M. Geffroy, P. Thomas, A. Bouzid, Enhanced structural description of sodium vanadium phosphate glasses: a combined experimental and molecular dynamics study, *J. Non-Cryst. Solids* 655 (2025) 123420.
- [61] F. Wang, F. Yang, J. Zhao, P. Lv, H. Sun, A. Ejaz, T. Wang, Molecular dynamics simulation of dynamic stretching of borosilicate glass, *J. Non-Cryst. Solids* 655 (2025) 123475.
- [62] Y. Noguchi, M. Shimizu, S. Sukenaga, R. Endo, T. Nishi, Y. Shimotsuma, K. Miura, Dominant factors of thermal conduction in alkali silicate glasses and melts: a molecular dynamics study, *J. Am. Ceram. Soc.* 108 (4) (2025) e20288.
- [63] J. Du, A. Cormack, The medium range structure of sodium silicate glasses: a molecular dynamics simulation, *J. Non-Cryst. Solids* 349 (2004) 66–79.
- [64] J. Du, A.N. Cormack, Molecular dynamics simulation of the structure and hydroxylation of silica glass surfaces, *J. Am. Ceram. Soc.* 88 (9) (2005) 2532–2539.
- [65] M. Ren, L. Deng, J. Du, Surface structures of sodium borosilicate glasses from molecular dynamics simulations, *J. Am. Ceram. Soc.* 100 (6) (2017) 2516–2524.
- [66] G. Greaves, S. Sen, Inorganic glasses, glass-forming liquids and amorphizing solids, *Adv. Phys.* 56 (1) (2007) 1–166.
- [67] L.L. Hench, The story of Bioglass®, *Journal materials science materials medicine* 17 (11) (2006) 967–978.
- [68] J.R. Jones, Review of bioactive glass: from Hench to hybrids, *Acta Biomater.* 9 (1) (2013) 4457–4486.
- [69] B. Stevansson, Y. Yu, M. Edén, Structure–composition trends in multicomponent borosilicate-based glasses deduced from molecular dynamics simulations with improved B–O and P–O force fields, *Phys. Chem. Chem. Phys.* 20 (12) (2018) 8192–8209.
- [70] E.A. Abou Neel, D.M. Pickup, S.P. Valappil, R.J. Newport, J.C. Knowles, Bioactive functional materials: a perspective on phosphate-based glasses, *J. Mater. Chem.* 19 (6) (2009) 690–701.
- [71] U. Hoppe, R. Kranold, D. Stachel, A. Barz, A.C. Hannonb, Variation in PO bonding in phosphate glasses—a neutron diffraction study, *Z. F. üR. Naturforsch. A* 55 (3–4) (2000) 369–380.
- [72] P. Stoch, A. Stoch, M. Ciecinska, I. Krakowiak, M. Sitarz, Structure of phosphate and iron-phosphate glasses by DFT calculations and FTIR/Raman spectroscopy, *J. Non-Cryst. Solids* 450 (2016) 48–60.
- [73] A. Pedone, X. Chen, R.G. Hill, N. Karpukhina, Molecular dynamics investigation of halide-containing phospho-silicate bioactive glasses, *J. Phys. Chem. B* 122 (11) (2018) 2940–2948.
- [74] B. Milborne, L. Murrell, I. Cardillo-Zallo, J. Titman, L. Briggs, C. Scotchford, A. Thompson, R. Layfield, I. Ahmed, Developing porous ortho-and pyrophosphate-containing glass microspheres; structural and cytocompatibility characterisation, *Bioengineering* 9 (11) (2022) 611.
- [75] Z. Fallah, J.K. Christie, Development of a ReaxFF reactive force field for ternary phosphate-based bioactive glasses, *J. Chem. Phys.* 160 (18) (2024).
- [76] A. Moghanian, S. Safaei, A. Pazhouheshgar, R. Farmani, A. Tayebi, A. Rajabpour, Integrating Molecular Dynamics and Polynomial Regression for Predicting Structure–Property Relationships in Silicate-based Melt-Quench Derived Binary Bioactive Glasses, *Ceramics International*, 2025.
- [77] A. Moghanian, R. Farmani, N. Kolivand, A. Tayebi, S. Safaei, Investigation on structure–property relationships of MgO–SrO containing silicate-based bioactive glasses: an experimental and molecular dynamics simulation study, *Methods* (2025).
- [78] Y. Ouldhnini, A. Atila, S. Ouaskit, A. Hasnaoui, Density-diffusion relationship in soda-lime phosphosilicate, *J. Non-Cryst. Solids* 590 (2022) 121665.
- [79] A. Moghanian, S. Safaei, N. Kolivand, R. Farmani, A. Tayebi, Structural, dissolution, and dynamic properties of binary (1-x) SiO₂-xCaO melt-derived bioactive glasses using computational simulation analysis, *J. Mater. Sci.* (2025) 1–16.
- [80] A. Atila, Y. Ouldhnini, S. Ouaskit, A. Hasnaoui, Atomistic insights into the mixed-alkali effect in phosphosilicate glasses, *Phys. Rev. B* 105 (13) (2022) 134101.
- [81] W. Wang, R. Christensen, B. Curtis, S.W. Martin, J. Kieffer, A new model linking elastic properties and ionic conductivity of mixed network former glasses, *Phys. Chem. Chem. Phys.* 20 (3) (2018) 1629–1641.
- [82] K. Yang, X. Xu, B. Yang, B. Cook, H. Ramos, N.A. Krishnan, M.M. Smedskjaer, C. Hoover, M. Bauchy, Predicting the Young's modulus of silicate glasses using high-throughput molecular dynamics simulations and machine learning, *Sci. Rep.* 9 (1) (2019) 8739.
- [83] Y. Ouldhnini, A. Atila, S. Ouaskit, A. Hasnaoui, Atomistic insights into the structure and elasticity of densified 45S5 bioactive glasses, *Phys. Chem. Chem. Phys.* 23 (28) (2021) 15292–15301.
- [84] Y. Zhao, J. Du, X. Cao, C. Zhang, G. Xu, X. Qiao, Y. Liu, S. Peng, G. Han, A modified random network model for P 2 O 5–Na 2 O–Al 2 O 3–SiO 2 glass studied by molecular dynamics simulations, *RSC Adv.* 11 (12) (2021) 7025–7036.
- [85] A. Tilocca, A.N. Cormack, Modeling the water–bioglass interface by ab initio molecular dynamics simulations, *ACS Appl. Mater. Interfaces* 1 (6) (2009) 1324–1333.
- [86] L. Deng, K. Miyatani, M. Suehara, S.-i Amma, M. Ono, S. Urata, J. Du, Ion-exchange mechanisms and interfacial reaction kinetics during aqueous corrosion of sodium silicate glasses, *npj Mater. Degrad.* 5 (1) (2021) 15.
- [87] T.S. Mahadevan, J. Du, Hydration and reaction mechanisms on sodium silicate glass surfaces from molecular dynamics simulations with reactive force fields, *J. Am. Ceram. Soc.* 103 (6) (2020) 3676–3690.
- [88] E. Lopez-Fontal, S. Gin, Insights into calcium phosphate formation induced by the dissolution of 45S5 bioactive glass, *ACS Biomater. Sci. Eng.* 11 (2) (2025) 875–890.
- [89] A.M. Abraham, S. Venkatesan, A review on application of biomaterials for medical and dental implants, *Proc. Inst. Mech. Eng. Part L J. Mater. Des. Appl.* 237 (2) (2023) 249–273.
- [90] P. Sharma, V. Mishra, S. Murab, Unlocking osseointegration: surface engineering strategies for enhanced dental implant integration, *ACS Biomater. Sci. & Eng.* 11 (1) (2024) 67–94.
- [91] D. Jones, J.E. Allen, Y. Yang, W.F. Drew Bennett, M. Gokhale, N. Moshiri, T. S. Rosing, Accelerators for classical molecular dynamics simulations of biomolecules, *J. Chem. Theory Comput.* 18 (7) (2022) 4047–4069.
- [92] D. Di Tommaso, M. Prakash, T. Lemaire, M. Lewerenz, N.H. De Leeuw, S. Naili, Molecular dynamics simulations of hydroxyapatite nanopores in contact with electrolyte solutions: the effect of nanoconfinement and solvated ions on the surface reactivity and the structural, dynamical, and vibrational properties of water, *Crystals* 7 (2) (2017) 57.
- [93] X. Dong, Q. Wang, T. Wu, H. Pan, Understanding adsorption-desorption dynamics of BMP-2 on hydroxyapatite (001) surface, *Biophys. J.* 93 (3) (2007) 750–759.
- [94] X. Chen, Q. Wang, J. Shen, H. Pan, T. Wu, Adsorption of leucine-rich amelogenin protein on hydroxyapatite (001) surface through—COO-Claws, *J. Phys. Chem. C* 111 (3) (2007) 1284–1290.
- [95] S.E. Hoff, J. Liu, H. Heinz, Binding mechanism and binding free energy of amino acids and citrate to hydroxyapatite surfaces as a function of crystallographic facet, pH, and electrolytes, *J. Colloid Interface Sci.* 605 (2022) 685–700.
- [96] Z.-s Zhang, H.-h Pan, R.-k Tang, Molecular dynamics simulations of the adsorption of amino acids on the hydroxyapatite {100}-water interface, *Front. Mater. Sci. China* 2 (3) (2008) 239–245.
- [97] C. Hong, X. Wu, J. Huang, H. Dai, Biomimetic fusion: platyper's dual vision for predicting protein–surface interactions, *Mater. Horiz.* 11 (15) (2024) 3528–3538.
- [98] R. Sabirianov, A. Rubinstein, F. Namavar, Enhanced initial protein adsorption on engineered nanostructured cubic zirconia, *Phys. Chem. Chem. Phys.* 13 (14) (2011) 6597–6609.
- [99] B.J. Williams-Nonan, A. Kamboukos, N. Todorova, I. Yarovsky, Self-assembling peptide biomaterials: insights from spontaneous and enhanced sampling molecular dynamics simulations, *Chem. Phys. Rev.* 4 (2) (2023).
- [100] J. Yang, M. Youssef, B. Yildiz, Structure, kinetics, and thermodynamics of water and its ions at the interface with monoclinic ZrO₂ resolved via ab initio molecular dynamics, *J. Phys. Chem. C* 125 (28) (2021) 15233–15242.
- [101] G. Yan, G. Chen, Z. Peng, Z. Shen, X. Tang, Y. Sun, X. Zeng, L. Lin, The cross-linking mechanism and applications of catechol–metal polymer materials, *Adv. Mater. Interfaces* 8 (19) (2021) 2100239.
- [102] I.-C. Yeh, J.L. Lenhart, J.A. Orllicki, B.C. Rinderspacher, Molecular dynamics simulation study of adsorption of bioinspired oligomers on alumina surfaces, *J. Phys. Chem. B* 123 (32) (2019) 7024–7035.
- [103] L. Shao, J. Ma, J.L. Prelesnik, Y. Zhou, M. Nguyen, M. Zhao, S.A. Jenekhe, S. V. Kalinin, A.L. Ferguson, J. Pfandtner, Hierarchical materials from high information content macromolecular building blocks: construction, dynamic interventions, and prediction, *Chem. Rev.* 122 (24) (2022) 17397–17478.
- [104] M.A. González, Force fields and molecular dynamics simulations, *École thématique De la Soci. été Française De la Neutron.* 12 (2011) 169–200.
- [105] H. Elkhenany, M.W. Soliman, D. Atta, N. El-Badri, Innovative marine-sourced hydroxyapatite, Chitosan, collagen, and gelatin for eco-friendly bone and cartilage regeneration, *J. Biomed. Mater. Res. Part A* 113 (1) (2025) e37833.
- [106] G. Gorczyca, R. Tylingo, P. Szweda, E. Augustin, M. Sadowska, S. Milewski, Preparation and characterization of genipin cross-linked porous chitosan–collagen–gelatin scaffolds using chitosan–CO₂ solution, *Carbohydr. Polym.* 102 (2014) 901–911.
- [107] K. Guillén-Carvajal, B. Valdez-Salas, E. Beltrán-Partida, J. Salomón-Carlos, N. Cheng, Chitosan, gelatin, and collagen hydrogels for bone regeneration, *Polymers* 15 (13) (2023) 2762.
- [108] R. Tylingo, G. Gorczyca, S. Mania, P. Szweda, S. Milewski, Preparation and characterization of porous scaffolds from chitosan–collagen–gelatin composite, *React. Funct. Polym.* 103 (2016) 131–140.
- [109] C. Tang, K. Zhou, Y. Zhu, W. Zhang, Y. Xie, Z. Wang, H. Zhou, T. Yang, Q. Zhang, B. Xu, Collagen and its derivatives: From structure and properties to their applications in food industry, *Food Hydrocoll.* 131 (2022) 107748.
- [110] S.A. Hulgán, J.D. Hartgerink, Recent advances in collagen mimetic peptide structure and design, *Biomacromolecules* 23 (4) (2022) 1475–1489.
- [111] J. Xiao, Collagen mimetic peptides and their biophysical characterization, Springer, 2024.
- [112] B. Jayachandran, T.N. Parvin, M.M. Alam, K. Chanda, B. Mm, Insights on chemical crosslinking strategies for proteins, *Molecules* 27 (23) (2022) 8124.
- [113] J. Hartmann, M. Zacharias, Mechanism of collagen folding propagation studied by Molecular Dynamics simulations, *PLOS Comput. Biol.* 17 (6) (2021) e1009079.
- [114] Y.-X. Tsai, N.-E. Chang, K. Reuter, H.-T. Chang, T.-J. Yang, S. von Bülow, V. Sehrawat, N. Zerrouki, M. Tuffery, M. Gecht, Rapid simulation of glycoprotein structures by grafting and steric exclusion of glycan conformer libraries, *Cell* 187 (5) (2024) 1296–1311. e26.
- [115] K. Piekarska, M. Sikora, M. Owczarek, J. Jóźwik-Pruska, M. Wiśniewska-Wrona, Chitin and chitosan as polymers of the future—obtaining, modification, life cycle assessment and main directions of application, *Polymers* 15 (4) (2023) 793.
- [116] C. Avitabile, M.T. Cerasa, A. D'Aniello, M. Saviano, M. Moccia, Recent cutting-edge technologies for the delivery of peptide nucleic acid, *Chem. A Eur. J.* 31 (34) (2025) e202500469.
- [117] J. Jacob, S. Loganathan, S. Thomas, Chitin-and chitosan-based biocomposites for food packaging applications, CRC Press, 2020.

- [118] M. Hudek, K. Johnston, K. Kubiak-Ossowska, V.A. Ferro, P.A. Mulheran, Molecular dynamics study of chitosan adsorption at a silica surface, *J. Phys. Chem. C* 128 (50) (2024) 21531–21538.
- [119] F.A. Tettey-Engmann, Innovative Hybrid Scaffold Design Using 3D Bioprinting and Electrospinning for Accelerated Wound Healing Applications, North Carolina Agricultural and Technical State University, 2024.
- [120] F. Razmimanesh, S. Amjad-Iranagh, H. Modarress, Molecular dynamics simulation study of chitosan and gemcitabine as a drug delivery system, *J. Mol. Model.* 21 (7) (2015) 165.
- [121] Z. Shariatnia, A. Mazloom-Jalali, Chitosan nanocomposite drug delivery systems designed for the ifosfamide anticancer drug using molecular dynamics simulations, *J. Mol. Liq.* 273 (2019) 346–367.
- [122] S. Moradi, E. Hosseini, M. Abdoli, S. Khani, M. Shahlaei, Comparative molecular dynamic simulation study on the use of chitosan for temperature stabilization of interferon α II, *Carbohydr. Polym.* 203 (2019) 52–59.
- [123] X. Jia, X. Fan, C. Chen, Q. Lu, H. Zhou, Y. Zhao, X. Wang, S. Han, L. Ouyang, H. Yan, Chemical and structural engineering of gelatin-based delivery systems for therapeutic applications: a review, *Biomacromolecules* 25 (2) (2024) 564–589.
- [124] R.M. Hathout, A.A. Metwally, T.J. Woodman, J.G. Hardy, Prediction of drug loading in the gelatin matrix using computational methods, *ACS Omega* 5 (3) (2020) 1549–1556.
- [125] J. Guo, S. Zhu, B. Liu, M. Zheng, H. Chen, J. Pang, Rheological behavior and molecular dynamics simulation of κ -carrageenan/casein under simulated gastrointestinal electrolyte conditions, *Food Hydrocoll.* 136 (2023) 108240.
- [126] S. Das, J.T. Jegadeesan, B. Basu, Gelatin methacryloyl (GelMA)-based biomaterial inks: process science for 3D/4D printing and current status, *Biomacromolecules* 25 (4) (2024) 2156–2221.
- [127] J. Xiong, K. Li, C. Lin, H. Shen, C. Lu, Molecular interaction mechanism and mechanical performance of lignin/gelatin composites by molecular dynamics simulation study, *Langmuir* 41 (12) (2025) 8369–8379.
- [128] A. Zaupa, A.T. Neffe, B.F. Pierce, A. Lendlein, D. Hofmann, A molecular dynamic analysis of gelatin as an amorphous material: Prediction of mechanical properties of gelatin systems, *Int. J. Artif. Organs* 34 (2) (2011) 139–151.
- [129] R. Liang, L. Xu, C. Fan, L. Cao, X. Guo, Structural characteristics and antioxidant mechanism of donkey-hide gelatin peptides by molecular dynamics simulation, *Molecules* 28 (24) (2023) 7975.
- [130] M. Akbar, A. Yaqoob, A. Ahmad, R. Luque, Sodium Alginate Overv. Sodium Alginate-Based Nanomater. Wastewater Treat. (2023) 1–17.
- [131] C. Iurciuc-Tincu, L. Ochiuz, M. Popa, L. Atanase, Crosslinked Marine Polysaccharides for Delivery of Therapeutics, *Marine Biomaterials: Therapeutic Potential*, Springer, 2022, pp. 41–79.
- [132] A.O. Makarova, S.R. Derkach, T. Khair, M.A. Kazantseva, Y.F. Zuev, O.S. Zueva, Ion-induced polysaccharide gelation: Peculiarities of alginate egg-box association with different divalent cations, *Polymers* 15 (5) (2023) 1243.
- [133] R. Garduño-Juárez, D.O. Tovar-Anaya, J.M. Perez-Aguilar, L.F. Lozano-Aguirre Beltran, R.A. Zubillaga, M.A. Alvarez-Perez, E. Villarreal-Ramirez, Molecular dynamic simulations for biopolymers with biomedical applications, *Polymers* 16 (13) (2024) 1864.
- [134] A.A. Agles, I.C. Bourg, Structure–thermodynamic relationship of a polysaccharide gel (alginate) as a function of water content and counterion type (Na vs Ca), *J. Phys. Chem. B* 127 (8) (2023) 1828–1841.
- [135] X. Liu, X. Qin, Y. Wang, J. Zhong, Physicochemical properties and formation mechanism of whey protein isolate-sodium alginate complexes: Experimental and computational study, *Food Hydrocoll.* 131 (2022) 107786.
- [136] B. Boukoussa, A. Mokhtar, A. El Guerdaoui, M. Hachemaoui, H. Ouachtak, S. Abdelkrim, A.Ait Addi, S. Babou, B. Boudina, A. Bengueddach, Adsorption behavior of cationic dye on mesoporous silica SBA-15 carried by calcium alginate beads: experimental and molecular dynamics study, *J. Mol. Liq.* 333 (2021) 115976.
- [137] A. Mahmood, D. Patel, B. Hickson, J. DesRochers, X. Hu, Recent progress in biopolymer-based hydrogel materials for biomedical applications, *Int. J. Mol. Sci.* 23 (3) (2022) 1415.
- [138] X. Zhang, X. Dai, H. Wan, Z. Jiao, W. Chen, W. Wei, J. Li, K. Liu, Z. Yang, L.T. Yan, Interfaces between nanoparticle and biomacromolecular network: dynamic behaviors, confinement and entropy, *Adv. Funct. Mater.* 34 (32) (2024) 2314134.
- [139] R. Kumari, S. Shekhar, Effect of crosslinking on thermodynamics interactions and network parameters of terpolymeric hydrogels, *J. Macromol. Sci. Part B* 62 (12) (2023) 689–717.
- [140] R.D. Guha, O. Idolor, K. Berkowitz, M. Pasquinelli, L.R. Grace, Exploring secondary interactions and the role of temperature in moisture-contaminated polymer networks through molecular simulations, *Soft Matter* 17 (10) (2021) 2942–2956.
- [141] J. Landsgeßel, D. Beyer, P. Hebbeker, P. Košovan, C. Holm, The pH-dependent swelling of weak polyelectrolyte hydrogels modeled at different levels of resolution, *Macromolecules* 55 (8) (2022) 3176–3188.
- [142] A.J. Wright, From Microscale Network Topology to Macroscale Mechanics of Folded protein Hydrogels, University of Leeds, 2021.
- [143] M.J. Webber, M.W. Tibbitt, Dynamic and reconfigurable materials from reversible network interactions, *Nat. Rev. Mater.* 7 (7) (2022) 541–556.
- [144] A. Ghosh, M. Ziatdinov, O. Dyck, B.G. Sumpter, S.V. Kalinin, Bridging microscopy with molecular dynamics and quantum simulations: an atomAI based pipeline, *npj Comput. Mater.* 8 (1) (2022) 74.
- [145] S.Y. Joshi, S.A. Deshmukh, A review of advancements in coarse-grained molecular dynamics simulations, *Mol. Simul.* 47 (10–11) (2021) 786–803.
- [146] G. Giannetti, F. Matsumura, F. Caporaletti, D. Micha, G.H. Koenderink, I.M. Ilie, M. Bonn, S. Woutersen, G. Giubertoni, Water and collagen: a mystery yet to unfold, *Biomacromolecules* 26 (5) (2025) 2784–2799.
- [147] Z. Zhu, X. Luo, S.J. Paddison, Coarse-grained modeling of ion-containing polymers, *Chem. Rev.* 122 (12) (2022) 10710–10745.
- [148] A. Scacchi, M. Vuorte, M. Sammalkorpi, Multiscale modelling of biopolymers, *Adv. Phys. X* 9 (1) (2024) 2358196.
- [149] M. Hussain, S.M. Khan, M. Shafiq, N. Abbas, U. Sajjad, K. Hamid, Advances in biodegradable materials: Degradation mechanisms, mechanical properties, and biocompatibility for orthopedic applications, *Heliyon* 10 (12) (2024).
- [150] X. Dai, Y. Chen, Computational biomaterials: computational simulations for biomedicine, *Adv. Mater.* 35 (7) (2023) 2204798.
- [151] G.T. El-Bassouini, S.M. Mounier, A.M. El-Shamy, Advances in surface modifications of titanium and its alloys: implications for biomedical and pharmaceutical applications, *Multiscale Multidiscip. Model. Exp. Des.* 8 (5) (2025) 265.
- [152] R. Misra, Enabling manufacturing of multi-axial forging-induced ultrafine-grained strong and ductile magnesium alloys: a perspective of process-structure-property paradigm, *Mater. Technol.* 38 (1) (2023) 2189769.
- [153] D. Xu, X. Wang, Y. Lu, Heterogeneous-structured refractory high-entropy alloys: a review of state-of-the-art developments and trends, *Adv. Funct. Mater.* 34 (49) (2024) 2408941.
- [154] M. Gong, J. Graham, V. Taupin, L. Capolungo, The effects of stress, temperature and facet structure on growth of {101 2} twins in Mg: a molecular dynamics and phase field study, *Acta Mater.* 208 (2021) 116603.
- [155] J. Barberi, S. Spriano, Titanium and protein adsorption: an overview of mechanisms and effects of surface features, *Materials* 14 (7) (2021) 1590.
- [156] M.A. Maruf, M. Noor-A-Alam, W. Haider, I. Shabib, Enhancing controlled and uniform degradation of Fe by incorporating Mg and Zn aimed for bio-degradable material applications, *Mater. Chem. Phys.* 285 (2022) 126171.
- [157] H. Wang, H. Yuan, J. Wang, E. Zhang, M. Bai, Y. Sun, J. Wang, S. Zhu, Y. Zheng, S. Guan, Influence of the second phase on protein adsorption on biodegradable Mg alloys' surfaces: comparative experimental and molecular dynamics simulation studies, *Acta Biomater.* 129 (2021) 323–332.
- [158] S.N. Sahoo, S. Mandal, T. Adhikary, V.K. Ramesh, P. Mukherjee, S. Aich, I. Samanta, S.K. Nandi, M. Roy, Synergistic Improvement of Antibacterial and Osteogenic Differentiation of Thermomechanically Processed Mg–Zr–Sr–Ce Alloy: insights into the role of precipitate evolution supported by AIMD simulation study, *ACS Appl. Bio Mater.* 7 (5) (2024) 2762–2780.
- [159] S. Kaviani, M. Izadyar, M. Khavani, M.R. Housaindokht, A combined molecular dynamics and quantum mechanics study on the interaction of Fe3+ and human serum albumin relevant to iron overload disease, *J. Mol. Liq.* 317 (2020) 113933.
- [160] R. Feng, M. Chen, L. Xie, Y. Bao, Y. Ge, Research progress on prediction models of plastic deformation and ductile fracture of titanium alloy, *Phys. Met. Metallogr.* 124 (13) (2023) 1524–1546.
- [161] S. Gowthaman, Study on the influence of temperature and strain rate on the tensile behavior of Ti–5Mo–5Cu alloy biomaterial alloy: a molecular dynamics simulation, *Funct. Compos. Struct.* 5 (3) (2023) 035006.
- [162] J. Sampath, A. Kullman, R. Gebhart, G. Drobny, J. Pfandtner, Molecular recognition and specificity of biomolecules to titanium dioxide from molecular dynamics simulations, *npj Comput. Mater.* 6 (1) (2020) 34.
- [163] P. Sadhukhan, S. Banerjee, H. Kumar, B. Maji, M. Ghosh, A study on Ti–Nb alloys with different composition for orthopedic application using md simulations and experiments, *Adv. Theory Simul.* (2025) 2500137.
- [164] Z.-Q. Zhang, H.-Y. Wang, L. Wang, X.-B. Chen, S.-K. Guan, C.-G. Lin, R.-C. Zeng, Protein conformation and electric attraction adsorption mechanisms on anodized magnesium alloy by molecular dynamics simulations, *J. Magnes. Alloy.* 10 (11) (2022) 3143–3155.
- [165] N. Li, Z. Hao, L. Xu, M. Tang, L. Wei, L. Wang, A review of molecular dynamics simulation of different Ti–Al-based alloys, *Metals* 14 (9) (2024) 1018.
- [166] A. Malinowski, M. Śmiechowski, Solvent exchange around aqueous Zn (II) from ab initio molecular dynamics simulations, *Liquids* 2 (3) (2022) 243–257.
- [167] A.A. Riet, J.A. Van Orman, D.J. Lacks, A molecular dynamics study of grain boundary diffusion in MgO, *Geochim. Et. Cosmochim. Acta* 292 (2021) 203–216.
- [168] E. Liasas, K. Kubiak-Ossowska, R.A. Black, O.R. Thomas, Z.J. Zhang, P. A. Mulheran, Adsorption of fibronectin fragment on surfaces using fully atomistic molecular dynamics simulations, *Int. J. Mol. Sci.* 19 (11) (2018) 3321.
- [169] K. Kubiak-Ossowska, P.A. Mulheran, W. Nowak, Fibronectin module FNIII9 adsorption at contrasting solid model surfaces studied by atomistic molecular dynamics, *J. Phys. Chem. B* 118 (33) (2014) 9900–9908.
- [170] L. Hao, T. Li, L. Wang, X. Shi, Y. Fan, C. Du, Y. Wang, Mechanistic insights into the adsorption and bioactivity of fibronectin on surfaces with varying chemistries by a combination of experimental strategies and molecular simulations, *Bioact. Mater.* 6 (10) (2021) 3125–3135.
- [171] M.M. Castellanos, C.M. Colina, Molecular dynamics simulations of human serum albumin and role of disulfide bonds, *J. Phys. Chem. B* 117 (40) (2013) 11895–11905.
- [172] M. Kooravand, S. Asadpour, H. Haddadi, S. Farhadian, An insight into the interaction between malachite green oxalate with human serum albumin: molecular dynamic simulation and spectroscopic approaches, *J. Hazard. Mater.* 407 (2021) 124878.
- [173] M.A. Abdelaziz, M. Shaldam, R.A. El-Domany, F. Belal, Multi-Spectroscopic, thermodynamic and molecular dynamic simulation studies for investigation of interaction of dapagliflozin with bovine serum albumin, *Spectrochim. Acta Part A Mol. Biomol. Spectrosc.* 264 (2022) 120298.

- [174] H. Huang, W. Feng, Y. Chen, Two-dimensional biomaterials: material science, biological effect and biomedical engineering applications, *Chem. Soc. Rev.* 50 (20) (2021) 11381–11485.
- [175] D. Macuglia, Free-energy calculations in condensed matter: from early challenges to the advent of umbrella sampling, *Arch. Hist. Exact. Sci.* 78 (5) (2024) 479–522.
- [176] H. Ohshima, DLVO theory of colloid stability. *Interface Science and Technology*, Elsevier, 2024, pp. 217–244.
- [177] D. Zerbib, Bacterial Cell Envelopes from Monoderms to Diderms: Composition, Architecture, and Origin. *Handbook of Electroporation*, Springer, 2025, pp. 1–27.
- [178] E.E. Bastounis, P. Radhakrishnan, C.K. Prinz, J.A. Theriot, Mechanical forces govern interactions of host cells with intracellular bacterial pathogens, *Microbiol. Mol. Biol. Rev.* 86 (2) (2022) e00094–20.
- [179] C.K. Ellison, G.B. Whitfield, Y.V. Brun, Type IV Pili: dynamic bacterial nanomachines, *FEMS Microbiol. Rev.* 46 (2) (2022) fuab053.
- [180] A. Sukumaran, M.A. Mansour, Cytotoxicity-Understanding Cellular Damage and Response: Understanding Cellular Damage and Response, *BoD-Books on Demand* 2023.
- [181] F. Vianello, A. Cecconello, M. Magro, Toward the specificity of bare nanomaterial surfaces for protein corona formation, *Int. J. Mol. Sci.* 22 (14) (2021) 7625.
- [182] I. Tvaroška, S. Kozmon, J. Kónia, Molecular modeling insights into the structure and behavior of integrins: a review, *Cells* 12 (2) (2023) 324.
- [183] L. Zhang, Z. Zhang, M. Negahban, A. Jérusalem, Molecular dynamics simulation of cell membrane pore sealing, *Extrem. Mech. Lett.* 27 (2019) 83–93.
- [184] G. Grasso, S. Muscat, M. Rebella, U. Morbiducci, A. Audenino, A. Danani, M. A. Deriu, Cell penetrating peptide modulation of membrane biomechanics by Molecular dynamics, *J. Biomech.* 73 (2018) 137–144.
- [185] S. Mirshahi, B. Vahedi, S.O. Yazdani, M. Golab, A. Sazgarnia, Calculating transmembrane voltage on the electric pulse-affected cancerous cell membrane: using molecular dynamics and finite element simulations, *J. Mol. Model.* 30 (7) (2024) 221.
- [186] S. Kaviani, Molecular dynamics and machine learning framework for predicting ion transport and mechanical properties of ionic liquid@polyvinylidene fluoride gel polymer electrolyte, *J. Ind. Eng. Chem.* (2025).
- [187] M. Khajavian, S. Kaviani, I. Piyanzina, D.A. Tayurskii, O.V. Nedopekin, M. Sillanpää, V. Vatanpour, Single and binary heavy metal adsorption using alginate acid structure: Experimental and density functional theory investigations, *J. Water Process Eng.* 54 (2023) 103996.
- [188] A.P. Bartók, M.C. Payne, R. Kondor, G. Csányi, Gaussian approximation potentials: The accuracy of quantum mechanics, without the electrons, *Phys. Rev. Lett.* 104 (13) (2010) 136403.
- [189] L. Zhang, J. Han, H. Wang, R. Car, E. W. Deep potential molecular dynamics: a scalable model with the accuracy of quantum mechanics, *Phys. Rev. Lett.* 120 (14) (2018) 143001.
- [190] K. Schütt, P.-J. Kindermans, H.E. Saucedo Felix, S. Chmiela, A. Tkatchenko, K.-R. Müller, Schnet: a continuous-filter convolutional neural network for modeling quantum interactions, *Adv. Neural Inf. Process. Syst.* 30 (2017).
- [191] S. Batzner, A. Musaelian, L. Sun, M. Geiger, J.P. Mailoa, M. Kornbluth, N. Molinari, T.E. Smidt, B. Kozinsky, E (3)-equivariant graph neural networks for data-efficient and accurate interatomic potentials, *Nat. Commun.* 13 (1) (2022) 2453.
- [192] A.V. Shapeev, Moment tensor potentials: a class of systematically improvable interatomic potentials, *Multiscale Model. Simul.* 14 (3) (2016) 1153–1173.
- [193] V.L. Deringer, G. Csányi, Machine learning based interatomic potential for amorphous carbon, *Phys. Rev. B* 95 (9) (2017) 094203.
- [194] D.M. Anstine, O. Isayev, Machine learning interatomic potentials and long-range physics, *J. Phys. Chem. A* 127 (11) (2023) 2417–2431.
- [195] Y. Mishin, Machine-learning interatomic potentials for materials science, *Acta Mater.* 214 (2021) 116980.
- [196] Z. Chen, D. Li, J. Liu, K. Gao, Application of gaussian processes and transfer learning to prediction and analysis of polymer properties, *Comput. Mater. Sci.* 216 (2023) 111859.
- [197] P. Huang, Z. Yang, Y. Liu, P. Yu, B. Liu, M. Xu, M. Avdeev, S. Shi, A database of composition-processing-performance parameters for over 400 inorganic-polymer composite solid-state electrolytes, *Adv. Theory Simul.* (2025) e01125.
- [198] J. Schmidt, M.R. Marques, S. Botti, M.A. Marques, Recent advances and applications of machine learning in solid-state materials science, *npj Comput. Mater.* 5 (1) (2019) 83.
- [199] J. Wang, Y. Wang, Y. Chen, Inverse design of materials by machine learning, *Materials* 15 (5) (2022) 1811.
- [200] M.J. Buehler, *Atomistic Modeling of Materials Failure*, Springer, 2008.
- [201] M. Karplus, J.A. McCammon, Molecular dynamics simulations of biomolecules, *Nat. Struct. Biol.* 9 (9) (2002) 646–652.
- [202] A. Laio, M. Parrinello, Escaping free-energy minima, *Proc. Natl. Acad. Sci.* 99 (20) (2002) 12562–12566.
- [203] A. Laio, F.L. Gervasio, Metadynamics: a method to simulate rare events and reconstruct the free energy in biophysics, chemistry and material science, *Rep. Prog. Phys.* 71 (12) (2008) 126601.
- [204] A.F. Voter, Hyperdynamics: accelerated molecular dynamics of infrequent events, *Phys. Rev. Lett.* 78 (20) (1997) 3908.
- [205] D.E. Kleiman, H. Nadeem, D. Shukla, Adaptive sampling methods for molecular dynamics in the era of machine learning, *J. Phys. Chem. B* 127 (50) (2023) 10669–10681.
- [206] M. Rahman, K.S. Islam, T.M. Dip, M.F.M. Chowdhury, S.R. Debnath, S.M. M. Hasan, M.S. Sakib, T. Saha, R. Padhye, S. Houshyar, A review on nanomaterial-based additive manufacturing: dynamics in properties, prospects, and challenges, *Prog. Addit. Manuf.* 9 (4) (2024) 1197–1224.
- [207] B. Efremenko, Y. Chabak, I. Petryshynets, T. Zhao, V. Efremenko, K. Wu, T. Xia, M. Džupon, S. Arshad, Evaluation of the suitability of high-temperature post-processing annealing for property enhancement in LPBF 316L Steel: a comprehensive mechanical and corrosion assessment, *Metals* 15 (6) (2025) 684.
- [208] J. Lee. Understanding the Processing-Structure-Property Relationship in Abs: Focus on Interlayer Bonding in Fused Filament Fabrication (FFF), University of Massachusetts Lowell, 2025.
- [209] R.A. Patel, M.A. Webb, Data-driven design of polymer-based biomaterials: high-throughput simulation, experimentation, and machine learning, *ACS Appl. Bio Mater.* 7 (2) (2023) 510–527.
- [210] L. Qian, H. Zhao, Nanoindentation of soft biological materials, *Micromachines* 9 (12) (2018) 654.
- [211] X. Long, R. Dong, Y. Su, C. Chang, Critical review of nanoindentation-based numerical methods for evaluating elastoplastic material properties, *Coatings* 13 (8) (2023) 1334.
- [212] K.K. Bejagam, C.N. Iverson, B.L. Marrone, G. Pilania, Molecular dynamics simulations for glass transition temperature predictions of polyhydroxyalkanoate biopolymers, *Phys. Chem. Chem. Phys.* 22 (32) (2020) 17880–17889.
- [213] K.-H. Lin, L. Paterson, F. May, D. Andrienko, Glass transition temperature prediction of disordered molecular solids, *npj Comput. Mater.* 7 (1) (2021) 179.
- [214] Z. Tang, S. Okazaki, All-atomistic molecular dynamics study of the glass transition of amorphous polymers, *Polymer* 254 (2022) 125044.
- [215] D. Xu, T. Harvey, E. Begiristain, C. Domínguez, L. Sánchez-Abella, M. Browne, R. B. Cook, Measuring the elastic modulus of soft biomaterials using nanoindentation, *J. Mech. Behav. Biomed. Mater.* 133 (2022) 105329.
- [216] O. Pshyk, O. Nakonechna, E. Coy, Current view of nanoindentation: recent developments and application in material characterization, *ACS Appl. Mater. Interfaces* (2025).
- [217] T. Jin, C.W. Coley, A. Alexander-Katz, Molecular signatures of the glass transition in polymers, *Phys. Rev. E* 106 (1) (2022) 014506.
- [218] J.L. Suter, W.A. Muller, M. Vassaux, A. Anastasiou, M. Simmons, D. Tilbrook, P. V. Coveney, Rapid, accurate and reproducible prediction of the glass transition temperature using ensemble-based molecular dynamics simulation, *J. Chem. Theory Comput.* 21 (3) (2025) 1405–1421.
- [219] O. Adjaoud, K. Albe, Nanoindentation of nanoglasses tested by molecular dynamics simulations: influence of structural relaxation and chemical segregation on the mechanical response, *Front. Mater.* 8 (2021) 664220.
- [220] G. Wu, M. Gotthardt, M. Gollasch, Assessment of nanoindentation in stiffness measurement of soft biomaterials: kidney, liver, spleen and uterus, *Sci. Rep.* 10 (1) (2020) 18784.